MIDDLE YEARS PROGRAMME
YEAR 3 MATHEMATICS

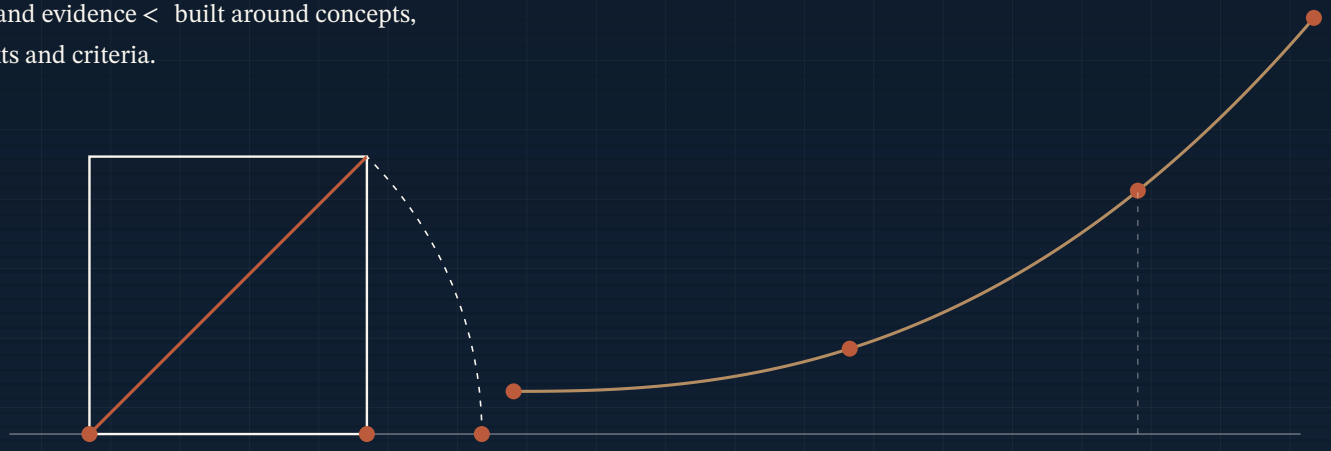
FIRST EDITION
2026

MYP

MATHEMATICS 3

3 *Student Coursebook*

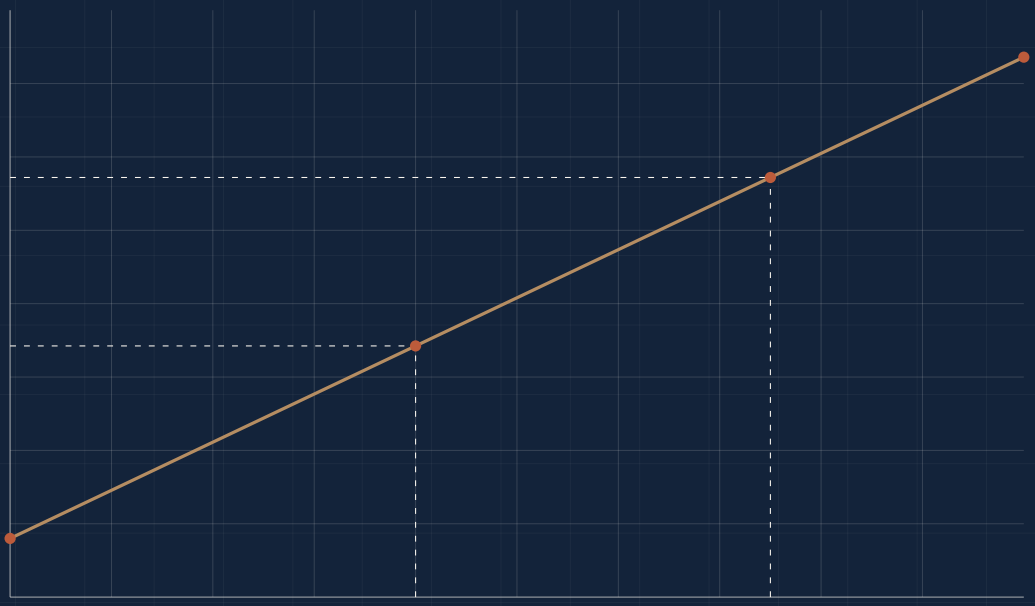
An inquiry-led course in number, algebra, proof, space and evidence < built around concepts, contexts and criteria.



Real numbers <Powers and scientific notation <Functions and algebra
Linear relationships <Equations and systems <Transformations and congruence
Similarity and trigonometry <Solids and measurement <Bivariate data <Probability

SADIKOU MOUKODDIM

STUDENT EDITION
MYP YEAR 3



Every number on the line *is either a fraction or it is not.*

The whole course begins with that distinction.

MYP MATHEMATICS

YEAR 3

Mathematics

3 *Student Coursebook*

An inquiry-led course in number, algebraic reasoning, linear relationships, proof, space, measurement and evidence < designed around the concepts, contexts and assessment criteria of the Middle Years Programme.

SADIKOU MOUKODDIM

Mathematics teacher author

IMPROVED FIRST EDITION

2026

Independent educational publication

MYP Mathematics 3 ≠ Student Coursebook

Improved first edition ?2026

Author: SADIKOU MOUKODDIM.

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> ABOUT THIS EDITION
>
> This is an **improved first edition**, not a second edition. The course content, chapter sequence, topics, learning objectives and level of difficulty are those of the original first edition. What has changed is the quality of the typesetting, the clarity of the explanations, the completeness of the worked solutions, the provision of graded support, core and extension pathways, and the integration of original diagrams and contextual photographs. Each change is recorded in the design and preservation reports that accompany this file.

Production

Typeset in Playfair Display (display), STIX Two Text (running text and inline mathematics), STIX Two Math (display mathematics) and Inter (labels, tables and captions). All four families are licensed under the SIL Open Font License 1.1. Diagrams were drawn for this edition and are original work. Photographs are credited individually at the point of use and in the image register that accompanies this file.

A note on the mathematics

Every worked example, answer and diagram in this book has been checked independently. Answers to practice sets are given in full at the back, with the reasoning rather than only the final value, because an answer without a reason teaches nothing that can be reused.

How to use this coursebook

Year 3 is where several familiar ideas become exact. The number line gains numbers that cannot be written as fractions; algebra gains functions with domains; geometry gains proof. This book is organised so that each of those steps is taken in the same order: a question first, then evidence, then explanation.

THE SEVEN STAGES OF EVERY CHAPTER

One architecture, ten times

Every chapter follows the same seven stages. The band at the top of a page tells you which stage you are in, so you always know whether you are being asked to notice, to explain, to practise or to reflect.

	Stage	What happens
1	Engage	a context, problem or photograph that raises the question
2	Explore	an investigation before the formal teaching
3	Explain	rigorous, accessible mathematical development
4	Worked	fully reasoned step-by-step solutions
5	Practice	progressive exercises: fluency, reasoning, problem solving
6	Apply	substantial investigation and real-life application
7	Reflect	summary, vocabulary, misconceptions and revision

The icons and panels of a page

Every marked element in the book carries a label and a colour. The colour is the quickest way to know what you are being shown: teal is a definition, sand is a key idea or a learning skill, blush is a mistake worth avoiding, and a terracotta rule marks an inquiry task.

The first page of every part

Each part opens on a navy plate that names the part, lists the chapters it contains and states what the part establishes. Turn the page and the chapter itself opens on a photograph: the context the inquiry is set in, carried at full width so that it can be studied rather than glanced at.



The photographic opening of a chapter: a drawn construction on paper, the instrument and the argument in one frame.
Original photograph made for this edition.

Exercise tags

F FLUENCY	Short, deliberate practice to make a method accurate and efficient.
R REASONING	Explain, compare, justify, find a counterexample or diagnose an error.
P PROBLEM SOLVING	Multi-step and unfamiliar contexts that need selection and planning.
E EXTENSION	Deeper structure, generalisation or a more open task.

Panels

DEFINITION	A statement worth remembering, in a mist-blue panel.
KEY IDEA	A method connected to the larger principle behind it.
COMMON ERROR	A mistake worth anticipating, with the reason it happens.
ATL	A learning skill made explicit, with a way to practise it.
WORKED EXAMPLE	A full solution in numbered steps, with a check at the end.
CRITERION STRIP	The MYP criterion and strands developed by the section.

The parts of a page

Element	What it does
Section heading	A numbered teaching section, with the assessment criterion and related concepts it addresses.
Worked example	A full solution in numbered steps, with a check at the end. The navy bar gives the reference number and the topic.
Practice	Graded exercises. F = fluency, R = reasoning, P = problem solving, E = extension. Support, core and extension ranges are always given.
Investigation	A Criterion B task: you gather evidence and form a conjecture before you are told the rule.
Application	A Criterion D task: mathematics used in a real situation, with a decision to justify.
Criterion task	A longer written task that is marked against a descriptor strip, so you can see what the next band requires.

ATL SELF-MANAGEMENT

How to use the practice sets

Do the support items until you no longer need the worked example in front of you. Then do the core items in one sitting, without notes. Only then attempt the extension items < they are not harder versions of the same thing, they are questions that require a decision about method.

PAGE FURNITURE

Running heads, folios and rules

A verso (left-hand) page carries the part title at the outer edge and the chapter title at the inner edge; a recto (right-hand) page reverses them. The folio, in terracotta, sits at the outer corner. Reference numbers such as **1.4.2** mean chapter 1, section 1.4, worked example 2, so any cross-reference can be found without the contents page.

CONVENTIONS USED IN THIS BOOK

Angles are in degrees unless stated. The decimal point is a point. A repeating decimal is shown with a bar over the repeating block where typography allows, and with an ellipsis otherwise. Answers are given to three significant figures unless an exact form is asked for.

The MYP framework in this course

The Middle Years Programme organises mathematics around concepts rather than topics. A concept is a big idea that recurs: form, relationships, quantity, equivalence, systems, change. Topics are how the ideas are met.

Key and related concepts

Each chapter names one **key concept** and two or three **related concepts**. The key concept is the lens; the related concepts are the specific mathematical ideas that the lens brings into focus. Together they determine the kind of question the chapter asks.

Key concept	Chapters	What it asks
Form	1, 2, 3, 6	What is this, and how is it structured? What forms can the same idea take?
Relationships	4, 5, 7	How are these quantities connected? What stays true as one changes?
Quantity	8	How much? How precisely can we know, and how does error behave?
Systems	9, 10	How do the parts of a whole interact? What does the evidence support?

Global contexts

Every chapter is set in one of the six global contexts. The context is not decoration: it decides which questions are worth asking, and it supplies the data that the mathematics is used on.

Global context	Chapters	Exploration
Orientation in space and time	1, 6	How cultures represented and measured quantity; how space is described.
Scientific and technical innovation	2, 8	Scale, precision, and the design of objects that must work.
Fairness and development	3, 5	How resources, costs and constraints are shared and decided.
Globalisation and sustainability	4, 9, 10	Patterns in public data, and claims made from evidence.
Identities and relationships	7	Similarity, ratio and the mathematics of comparison.

Statements of inquiry

A statement of inquiry is one sentence that joins a concept, a context and a purpose. The statement for each chapter appears on its unit planner page and is written to be arguable: you should be able to say what evidence would support it.

> FORM OF THE SENTENCE
>
> **In order for a student to□ , students must□**

Every inquiry question in this book is phrased through that construction. The factual questions establish vocabulary; the conceptual questions ask why the mathematics works; the debatable questions ask what should be done with it. A chapter is not finished when the factual questions are answered.

Approaches to learning

ATL category	What it looks like in this course
Communication	Writing a chain of reasoning that another person can check; using notation consistently; choosing an appropriate form of answer.
Thinking	Testing a conjecture with cases; looking for a counterexample; deciding whether evidence is enough for a proof.
Research	Reading a claim, finding its source, checking whether the data support the conclusion.
Self-management	Planning a multi-step problem; keeping track of units and precision; reviewing errors rather than only marks.
Collaboration	Comparing methods, defending a choice, and revising an answer in the light of another person's argument.

Assessment criteria in practice

Mathematics in the MYP is assessed against four criteria. They are not four kinds of question; they are four kinds of evidence. The same problem can produce evidence for more than one criterion, depending on what you are asked to do with it.

Criterion	The evidence looked for	Typical command words
A — Knowing and understanding	Selecting and applying the right mathematics; recognising patterns in familiar situations; using correct notation and terminology.	State, calculate, classify, convert, simplify.
B — Investigating patterns	Choosing and using a representation; describing a pattern; testing it; justifying a general rule.	Investigate, generalise, verify, justify.
C — Communicating	Using notation and language appropriately for the audience; organising a line of reasoning; interpreting a result in context.	Explain, prove, describe, report.
D — Applying mathematics in real-life contexts	Identifying the relevant variables; building a model; checking whether the model is adequate; justifying a decision.	Model, evaluate, decide, recommend.

How to read the descriptor strips

Each criterion task closes with a four-band strip. The bands are written so that you can see what distinguishes them. The difference between a 3<4 and a 5<6 is almost never more correct arithmetic; it is a more complete justification, or a clearer statement of what the result means.

Criterion	What strong work does that adequate work does not
A	Chooses a method and says why it is appropriate, rather than applying the most recent one.
B	States the general rule symbolically, tests it on a case that was not used to find it, and explains why the pattern must continue.
C	Defines terms before using them, keeps notation consistent, and interprets the final result in the language of the original problem.
D	Names the assumptions in the model and says how sensitive the conclusion is to them.

> **A WORD ABOUT MARKS**
>
> A correct answer with no reasoning earns less than an incorrect answer with a visible, valid method. This is deliberate: the criterion is about what you know and can do, and a method you cannot explain is a method you cannot yet rely on.

Course map and progression

The sequence is one coherent course, not ten separate topics. Each part builds the language the next part uses.

	Chapter	The idea that organises it
1	The real number system	The real number system contains both rational and irrational numbers<
2	Powers, roots and scientific notation	In order for a student to compare quantities across many orders of magnitude
3	Functions, expressions and algebraic reasoning	In order for a student to generalise from particular cases
4	Linear relationships	In order for a student to compare competing rates and decide between them on evidence
5	Equations, inequalities and systems	In order for a student to express and satisfy several constraints at once
6	Transformations and congruence	In order for a student to determine whether two figures are the same shape
7	Similarity and trigonometry	In order for a student to determine a length or angle they cannot measure directly
8	Solids, measurement and accuracy	In order for a student to measure and describe a three-dimensional object reliably
9	Bivariate data and correlation	In order for a student to evaluate a claim that two quantities are related
10	Probability and risk	In order for a student to make a defensible decision under uncertainty

What each part contributes

Part	Chapters	What it establishes
I Number and algebraic structure	1<3	The number system, the laws of indices, and the language of functions and expressions. Everything later assumes these.
II Relationships and equations	4<5	Linear relationships as models, and the algebra of solving. The bridge from expression to prediction.
III Geometry, proof and space	6<8	Congruence, similarity and trigonometry; then three-dimensional measurement, where precision becomes an issue.
IV Data, chance and evidence	9<10	Bivariate data and probability, ending with the critical reading of claims made from evidence.

> READING THE MAP
>
> If a chapter feels difficult, the reason is usually two chapters back. The progression table on each unit planner page names the prerequisite ideas explicitly; use it before you re-read the whole of the previous chapter.

Notation, reasoning and habits

Mathematical notation is a form of compression. A symbol that both you and your reader understand saves a sentence; a symbol that only you understand costs a mark.

Symbol	Meaning	Example
$\in$, $\notin$	is an element of, is not an element of	$\sqrt{2} \in \mathbb{Q}$
$\subseteq$	is a subset of	$\mathbb{N} \subseteq \mathbb{Z}$
$\Rightarrow$	implies	n even $\Rightarrow$ n^2 even
$\Leftrightarrow$	if and only if	$a < b \Leftrightarrow a^2 < b^2$ for $a, b > 0$
$\therefore$	therefore	p^2 even $\therefore$ p even
$\neq$	is not equal to	$q \neq 0$
$\approx$	is approximately equal to	$\pi \approx 3.142$
$\cong$	is congruent to	$\triangle ABC \cong \triangle DEF$
$\sim$	is similar to	$\triangle ABC \sim \triangle DEF$
$\lvert x \rvert$	the absolute value of x	$\lvert -3 \rvert = 3$
$\angle$	angle	$\angle ABC = 90^\circ$
$\perp$, $\parallel$	perpendicular to, parallel to	$AB \perp BC$

Six habits this course builds

1. **State the rule.** When you use a result, name it: ‹the sum of angles on a straight line is 180°›, not just ‹it's 180›
2. **Show the substitution.** Write the formula, then the numbers, then the result. A reader should be able to find your error without asking you.
3. **Keep the units.** A number without a unit is a number without a meaning. Carry units through the calculation and check that the answer's unit makes sense.
4. **Say what the answer means.** A speed of 4.2 is not an answer; 4.2 m/s is, and ‹she walks about 4 m/s, which is a brisk walk›is a better one.
5. **Distinguish exact from approximate.** $\sqrt{2}$ is exact; 1.414 is a rounded value. Say which one you are giving.
6. **Check one case.** Substitute your answer back, or test the formula on a case you can compute by hand.

ATL THINKING

When you are stuck

Write down what you know, then what you want. Then look for a relation that contains both. If you cannot find one, try a small case with numbers: a worked small case frequently shows the shape of the general method.

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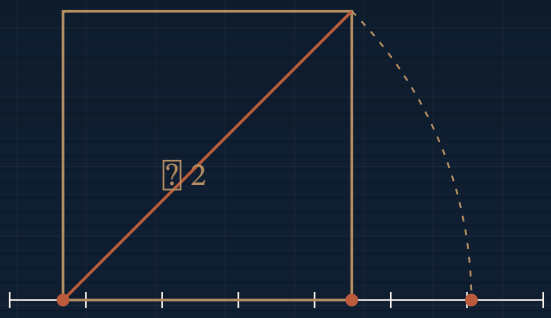
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I

PART I

Number and algebraic structure

The number line is not a list of labels; it is a system with gaps, extensions and rules. This part extends the numbers you know, compresses repeated multiplication into powers, and shows how a function organises a relationship between two quantities. The aim is not to memorise more rules, but to see why the rules are forced by the structure.



01 The real number system

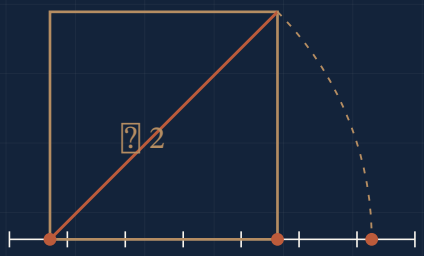
02 Powers, roots and scientific notation

03 Functions, expressions and algebraic reasoning



1

NUMBER, STRUCTURE AND PROOF



The real number system

A number line drawn with a ruler looks continuous, yet the numbers that fill it are of strikingly different kinds. Some can be written as exact fractions; some cannot be written as fractions at all. Understanding that difference changes what ⟨exact⟨ means.

KEY CONCEPT ≣FORM | RELATED ≣QUANTITY | RELATED ≣EQUIVALENCE | CONTEXT ≣ORIENTATION IN SPACE AND TIME

INQUIRY

If every point on a line corresponds to a number, how can we describe the numbers that lie between the fractions < and how can we prove that some of them can never be written as fractions?

ATL ≣CRITICAL THINKING IN ACTION

Consider the claim ⟨every number can be written as a fraction⟨. Test it on integers, terminating decimals and repeating decimals. Then try ?2. When a single counterexample appears, decide whether the claim is false, or whether the definition of ⟨number⟨ needs to be widened. In this chapter you do both: you reject the universal claim and extend the number system.

MYP UNIT PLANNER THE WHOLE CHAPTER IN ONE VIEW

Framing the inquiry

Unit 1 approximately 14 hours of class time ATL focus: critical thinking

KEY AND RELATED CONCEPTS

Form. A number has more than one possible written form: fraction, terminating decimal, repeating decimal, surd or approximation. The form reveals which operations are exact and which are limited.

Quantity. The real number line assigns a position to every decimal, including those with infinitely many digits. Order and distance can be described without listing every digit.

Equivalence. Different decimals, fractions and radical expressions can name the same real number. Recognising equivalence is central to simplifying and to proving.

GLOBAL CONTEXT AND EXPLORATION

Orientation in space and time. How different cultures represented and approximated quantities, and why the discovery of incommensurable lengths challenged the idea that every length could be expressed as a whole-number ratio. Greek mathematicians initially assumed that any two lengths could be compared using a common unit. The diagonal of a square exposed a length with no such unit. That discovery did not break mathematics; it forced a broader conception of number. Today irrational numbers appear in geometry, music theory, engineering tolerances and computer graphics.

STATEMENT OF INQUIRY

The real number system contains both rational and irrational numbers; precise definitions and logical proof let us distinguish exact forms from approximations and reason about quantities that cannot be written as fractions.

FACTUAL

<What is a rational number?

<How can a repeating decimal be written as a fraction?

<What does the square-root symbol mean?

CONCEPTUAL

<Why is the square root of 2 not rational, and how does a proof by contradiction establish that claim?

DEBATABLE

<When engineers use 1.414 for 2, are they working with the number itself or with a different, rational number?

Criterion	Strands	Evidence in this chapter
A Knowing and understanding	i, ii, iii	Classify numbers; convert between forms; simplify surds; order values; solve interval and tolerance problems.
B Investigating patterns	i, ii, iii	Investigate decimal expansions and successive rational approximations to 2; describe and verify the pattern.
C Communicating	i, ii, iii, iv	Write a structured proof; use set, interval and radical notation; explain the meaning of an approximation.
D Applying mathematics	i, ii, iii, iv, v	Interpret a manufacturing tolerance as an interval and justify accept or reject decisions.

Each teaching section closes with a graded practice set. The investigation addresses Criterion B; the application addresses Criterion D; the criterion task asks for extended written communication (Criterion C) about knowledge and understanding (Criterion A).

Learning intentions and success criteria

The objectives below are the whole of Criterion A for this chapter. The success criteria beside them are written as statements you can check against your own work, not as a summary of the teaching.

YOU ARE LEARNING TO

- **Understand** the relationships among natural, integer, rational and real numbers.
- **Know** how to recognise terminating and repeating decimal expansions.
- **Be able to** convert between fractions, decimals and simple surd forms.
- **Be able to** order real numbers, including irrational values, using exact reasoning or careful approximation.
- **Be able to** interpret and use intervals, absolute value and tolerance.
- **Be able to** construct a short proof, including a proof by contradiction.

SUCCESS CRITERIA ≣ I CAN

- ☐ I can classify a number as natural, integer, rational or irrational and justify the classification.
- ☐ I can decide whether a fraction terminates or repeats without doing a long division.
- ☐ I can convert a simple repeating decimal to a fraction.
- ☐ I can simplify a surd and rationalise a one-term denominator.
- ☐ I can place values such as √5, √ and √7/3 in order on a number line.
- ☐ I can write a tolerance statement as an inequality and as an absolute-value statement.
- ☐ I can explain why √2 is irrational using a valid chain of reasoning.

PREREQUISITE CHECK ≣ 12 MINUTES, NO CALCULATOR

1. Simplify 18/24.
2. Write 0.35 as a fraction in lowest terms.
3. Calculate 7^2 and $(-7)^2$.
4. Which is greater, 4/3 or 7/5? Show a reason.
5. Calculate 2.5 × 4.
6. Write the first five square numbers.

Answers are in the chapter answer section. If more than two are incorrect, review fraction equivalence and square numbers before continuing.



A drawn construction, made with compass and straightedge on paper. Every construction in this chapter assumes two things at once: that lengths can be compared, and that the comparison can be written down as a ratio.
Original photograph made for this edition.

Explore: the square that will not close

You are going to meet a length that no fraction can express. Before anything is explained, try to measure it, and see for yourself what goes wrong.

YOUR TASK BEFORE YOU ARE TAUGHT

1. Draw a square of side 10 cm on squared paper. Draw its diagonal and measure it to the nearest millimetre. Record the measurement in a table.
2. Divide your measurement by 10 to get the diagonal of a unit square. Compare your value with those of three classmates. Record the range of the class's answers.
3. Test the fractions 7/5, 17/12 and 41/29 by squaring them. Which is closest to 2? Extend the pattern of denominators if you can see it.
4. Copy and complete a table with columns for the fraction p/q, its square, and the difference from 2. Keep the differences exact, as fractions.
5. Write a conjecture: can a fraction ever square to exactly 2? State the evidence you have for it and the evidence you would need to be certain.
6. Describe one thing your measurements cannot settle, and one thing the exact fractions can.

Keep your table. The proof in section 1.5 will settle what your measurements could only suggest, and you will be asked to compare the two kinds of evidence at the end of the chapter.

BEFORE YEAR 3	YEAR 3 THIS CHAPTER	LATER
▪ Integers, fractions, decimals, percentages.	▪ Rational and irrational numbers.	▪ Real exponents and logarithms.
▪ Square numbers and simple roots.	▪ Decimal expansions and surds.	▪ Limits of sequences.
▪ Ordering positive numbers.	▪ Intervals, absolute value.	▪ Rigorous ε–δ arguments.
	▪ Proof by contradiction.	▪ Completeness of the real numbers.

Where this chapter sits in the sequence.

Inquiry pathway for this chapter

	Stage	What you do
1	Context	Ancient measurement and the diagonal of a square expose a length with no common unit.
2	Question	Which numbers can be written as fractions, and which cannot?
3	Concept	Rational form, decimal expansion, irrationality, order, interval.
4	Investigation	Approximate $\sqrt{2}$ by rationals and examine the error.
5	Communication	Write a proof by contradiction; interpret a tolerance interval.
6	Reflection	What does "exact" mean when the decimal expansion never ends?

1.1 Rational numbers and decimal form

CRITERION A, II • FRACTIONS
TERMINATING AND REPEATING DECIMALS

A rational number is a quotient of two integers. That simple definition unifies whole numbers, fractions, terminating decimals and repeating decimals: they are different surface forms of the same underlying structure.

Every integer is rational: $5 = 5/1$ and $-12 = -12/1$. Every terminating decimal is rational because it can be written over a power of 10: $0.375 = 375/1000 = 3/8$. A repeating decimal is also rational, although that is less obvious. The repetition is the clue: it allows a finite algebraic calculation to recover an exact fraction.

DEFINITION

A rational number is a number that can be written in the form p/q, where p and q are integers and $q \neq 0$. The set of rational numbers is written $\mathbb{Q}$.

Not every rational number has a terminating decimal. In lowest terms, the denominator decides. If the denominator has only the prime factors 2 and 5, the decimal terminates, because multiplying numerator and denominator by a suitable power of 10 gives an equivalent fraction with denominator 10^n. If the denominator has any other prime factor, the decimal repeats forever.

Terminating	Repeating
$3/8 = 0.375$ $7/40 = 0.175$ $9/25 = 0.36$ Denominators $8 = 2^3$, $40 = 2^3 \times 5$, $25 = 5^2$ use only 2 and 5.	$5/6 = 0.8333\overline{3}$ $2/11 = 0.1818\overline{18}$ $7/9 = 0.7777\overline{7}$ Denominators $6 = 2 \times 3$, 11, $9 = 3^2$ carry another prime factor.

The denominator, in lowest terms, decides the form of the decimal.

Why the denominator decides

Writing a/b over a power of ten means finding n with $10^n = k\,b$. Because $10 = 2 \cdot 5$, this is possible exactly when every prime factor of b is 2 or 5. Otherwise the long division can never reach a remainder of zero; and because there are only $b - 1$ possible remainders, one of them must eventually repeat < and with it, the whole block of digits.

KEY IDEA

Density

Between any two distinct rational numbers there is another rational number. If $a < b$, the midpoint $(a+b)/2$ is rational and lies strictly between them. Repeating the process shows there are infinitely many rationals in any interval, however small. This property is called **density**.

WORKED EXAMPLE 1.1.1 FROM REPEATING DECIMAL TO

PROBLEM

Write 0.363636< as a fraction in lowest terms.

1. Let $x = 0.363636<$. Two digits repeat, so the block length is 2.
2. Multiply by 10^2: $100x = 36.363636<$.
3. Subtract the first equation from the second: $100x - x = 36.363636< - 0.363636<$, so $99x = 36$.
4. Hence $x = 36/99$.
5. Divide numerator and denominator by 9: $x = 4/11$.

ANSWER 4/11

$4 \cdot 11 = 0.363636<$, which matches the original decimal.

COMMON ERROR

Treating the block as if it ended

Writing 0.36< as 36/100 treats a repeating decimal as if it terminated. The value $36/100 = 0.36$ is slightly smaller than $0.363636<$. When digits repeat, the denominator must be 9, 99, 999 or a related value < one 9 for each digit in the block < never a power of 10.

PRACTICE 1.1 RATIONAL FORM AND DECIMAL EXPANSION

SUPPORT 1–3 CORE 1–6 EXTENSION 7–8

Where a decimal repeats, write at least six digits and mark the repeating block, for example 0.16< with the 6 repeating. Items marked **F** build fluency, **R** ask for reasoning, **P** are problem solving and **E** extend.

1. Classify each number as natural, integer or rational: (a) -8 (b) 3/4 (c) 0.25 (d) 15. F
2. Write each fraction as a terminating decimal: (a) 7/20 (b) 13/8 (c) 9/50. F
3. Write each fraction as a repeating decimal, writing at least six digits and identifying the repeating block: (a) 2/3 (b) 5/11 (c) 7/12. F
4. Without dividing, decide whether each fraction terminates or repeats: (a) 3/24 (b) 4/40 (c) 11/30 (d) 9/64. Explain your method. R
5. Convert each repeating decimal to a fraction in lowest terms: (a) 0.555< (b) 0.2727< (c) 0.41666< . F
6. Find a rational number strictly between 1/4 and 1/3. Then find a second rational between your answer and 1/4. R
7. Explain why 0.999< is equal to 1, not just less than 1< Use the same algebraic method as Worked example 1.1.1. R
8. A student claims that every fraction with a denominator less than 10 has a terminating decimal. Find a counterexample and explain why the claim fails. E

1.2 Irrational numbers and surds

CRITERION A ▤ I, III • REAL NUMBERS • PROOF

Density is seductive: between any two rationals there is another. It is tempting to conclude that the rationals fill the line. They do not. The diagonal of a unit square has a length that is not any fraction.

A number is **irrational** when it cannot be written as a quotient of two integers. Equivalently, its decimal expansion neither terminates nor repeats. This second description is a consequence, not a definition: a fraction always produces a terminating or repeating decimal, so a decimal that does neither cannot come from a fraction.

> **DEFINITION**
>
> **A surd** is an irrational root left in exact form, such as $\sqrt{2}$, $\sqrt{7}$ or $3\sqrt{5}$. Writing the root rather than a decimal keeps the value exact. The number under the radical sign is the **radicand**; the number in front is the coefficient.

The discovery that $\sqrt{2}$ is not a fraction is one of the turning points of mathematics. It was made geometrically: in a square of side 1, the diagonal squared equals $1^2 + 1^2 = 2$, so the diagonal is $\sqrt{2}$. No whole-number ratio of lengths can express it, so the Greek assumption that all lengths are commensurable had to be abandoned.

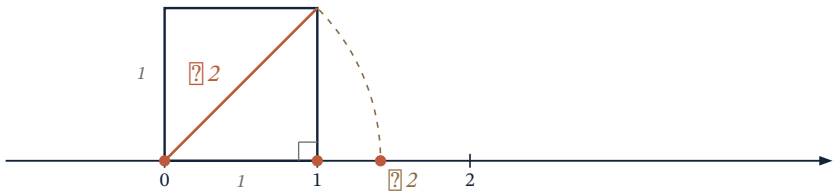


Figure 1.1 The side of the square is 1, so the diagonal is $\sqrt{2}$. Swinging that diagonal onto the number line locates $\sqrt{2}$ between 1 and 2 ▤ and proves that no fraction can sit exactly there.

THEOREM

Irrationality of $\sqrt{2}$

$\sqrt{2}$ is irrational. Suppose, for contradiction, that $\sqrt{2} = p/q$ where p and q are integers with no common factor. Squaring gives $p^2 = 2q^2$, so p^2 is even, and therefore p is even: write $p = 2k$. Then $4k^2 = 2q^2$, so $q^2 = 2k^2$ and q is even too. But then p and q share the factor 2, contradicting the assumption. Hence no such p and q exist, and $\sqrt{2}$ is irrational.

Reading a proof by contradiction

The proof has three moves, and every proof by contradiction uses the same three. First, **assume the opposite** of what you want to prove. Second, **reason validly** from that assumption. Third, **arrive at an impossibility** < here, two numbers that were supposed to share no factor but do. The only escape is that the assumption was false.

- A proof must work for *every* case; one example never proves a universal statement.
- A single counterexample *disproves* a universal statement.
- A statement, once proved, becomes available as a reason in later proofs < which is why proofs are worth writing down.

WORKED EXAMPLE 1.2.1 SIMPLIFYING A SURD

PROBLEM

Write $\sqrt{72}$ in the form $k\sqrt{m}$ with k and m integers and m as small as possible.

1. Factor the radicand, looking for square factors: $72 = 36 \cdot 2$.
2. Use the rule $\sqrt{(ab)} = \sqrt{a}\,\sqrt{b}$, valid for $a, b \geq 0$: $\sqrt{72} = \sqrt{36}\,\sqrt{2}$.
3. Evaluate the square root of the square factor: $\sqrt{36} = 6$.
4. Hence $\sqrt{72} = 6\sqrt{2}$.

ANSWER $6\sqrt{2}$

$6\sqrt{2} \approx 6 \times 1.4142 = 8.485$, and $\sqrt{72} \approx 8.485$.

WORKED EXAMPLE 1.2.2 RATIONALISING A DENOMINATOR

PROBLEM

Write $6/\sqrt{3}$ with a rational denominator.

1. Multiplying numerator and denominator by the same nonzero quantity does not change a fraction.
2. Multiply by $\sqrt{3}/\sqrt{3}$: $6/\sqrt{3} = (6\sqrt{3})/(\sqrt{3}\cdot\sqrt{3})$.
3. Simplify the denominator: $\sqrt{3}\cdot\sqrt{3} = 3$.
4. Hence $6/\sqrt{3} = 6\sqrt{3}/3 = 2\sqrt{3}$.

ANSWER $2\sqrt{3}$

$2\sqrt{3} \approx 3.464$ and $6 \approx 1.732 \approx 3.464$.

COMMON ERROR

Splitting a root over a sum

$\sqrt{(a+b)}$ is **not** $\sqrt{a}+\sqrt{b}$. Test it: $\sqrt{(9+16)} = \sqrt{25} = 5$, but $\sqrt{9}+\sqrt{16} = 3+4 = 7$. The rule that splits a root works for products, never for sums.

PRACTICE 1.2 IRRATIONAL NUMBERS AND SURDS

SUPPORT 1–4 CORE 1–7 EXTENSION 8–9

1. State whether each number is rational or irrational: (a) $\sqrt{9}$ (b) $\sqrt{10}$ (c) 0.454545< (d) $\sqrt{}$ (e) -7. F
2. Simplify: (a) $\sqrt{50}$ (b) $\sqrt{48}$ (c) $\sqrt{200}$ (d) $\sqrt{(12/3)}$. F
3. Write with a rational denominator: (a) $4/\sqrt{2}$ (b) $10/\sqrt{5}$ (c) $\sqrt{18}/\sqrt{6}$. F
4. Between which two consecutive integers does each value lie? (a) $\sqrt{40}$ (b) $\sqrt{75}$ (c) $-\sqrt{20}$. F
5. Explain why $\sqrt{3}\cdot\sqrt{3} = 3$ but $\sqrt{3}+\sqrt{3} \neq 3$. Use the definitions of root and product. R
6. Prove that $\sqrt{5}$ is irrational, adapting the proof for $\sqrt{2}$. State clearly where you use the fact that 5 is prime. P
7. A student writes $\sqrt{8} = 2\sqrt{2}$ and $\sqrt{2} \approx 1.41$, then concludes $\sqrt{8} = 2.82$ exactly. Explain what is wrong with the word <exactly< R
8. Show that $\sqrt{2}+\sqrt{8} = 3\sqrt{2}$, and decide whether $\sqrt{2}+\sqrt{3}$ can be written as a single surd. Justify your answer. E

1.3 Order, absolute value and intervals

CRITERION A V, V • QUANTITY • DISTANCE

Ordering irrational numbers looks difficult because their decimals never stop. It is not: compare squares, or compare exact surd forms, and the difficulty disappears.

To compare $\sqrt{5}$ and 2.2, square both: $(\sqrt{5})^2 = 5$ and $2.2^2 = 4.84$. Since $5 > 4.84$ and both numbers are positive, $\sqrt{5} > 2.2$. Squaring is safe here only because both numbers are positive; squaring can reverse nothing among positives, but it destroys sign information in general.

KEY IDEA

Comparing surds

For positive a and b: $a < b$ if and only if $a^2 < b^2$. This converts every comparison of surds into a comparison of integers.

Absolute value as distance

The absolute value $|x|$ is the distance from x to zero on the number line. The more useful reading is $|x - a|$, the distance from x to a. Distance language turns inequalities into statements about position rather than about sign.

DEFINITION

$|x| = x$ when $x \geq 0$, and $|x| = -x$ when $x < 0$. In general $|x - a|$ is the distance between x and a on the number line.

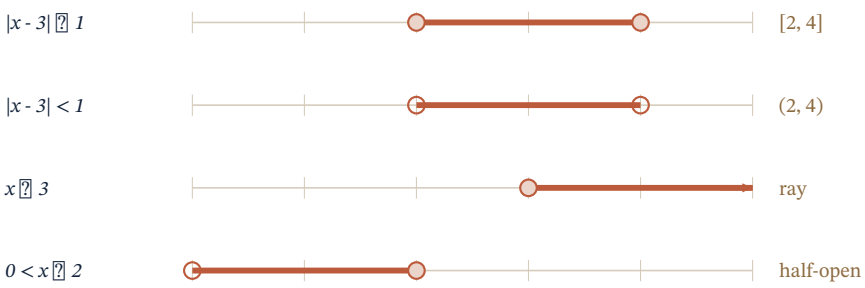


Figure 1.2 Four ways of writing the same set: as a distance statement, an inequality, a number-line picture and an interval.

An **interval** is a connected stretch of the number line. Square brackets include an endpoint, round brackets exclude it, and the symbols $\pm\infty$ never take a square bracket because infinity is not a number that can be reached.

Interval	Inequality	Distance form	Meaning
$[2, 5]$	$2 \le x \le 5$	$\lvert x - 3.5\rvert \le 1.5$	includes both ends
$(2, 5)$	$2 < x < 5$	$<$	excludes both ends
$[2, \infty)$	$x \ge 2$	$<$	ray, unbounded above
$(a - r, a + r)$	$a - r < x < a + r$	$\lvert x - a\rvert < r$	open interval of radius r

WORKED EXAMPLE 1.3.1 ORDERING A MIXED SET

PROBLEM

Arrange $\sqrt{7}$, 2.6, 8/3 and π in ascending order.

1. Compare squares where possible: $(\sqrt{7})^2 = 7$ and $2.6^2 = 6.76$, so $\sqrt{7} > 2.6$.
2. Compare the fractions as decimals to three places: $8/3 = 2.667$ and $\pi \approx 3.142$.
3. Now compare 2.6 with $8/3 = 2.667$: $2.6 < 2.667$.
4. Compare $\sqrt{7} \approx 2.646$ with $8/3 \approx 2.667$: $\sqrt{7} < 8/3$ because $7 < (8/3)^2 = 64/9 \approx 7.11$.

ANSWER $2.6 < \sqrt{7} < 8/3 < \pi$

Numerically: $2.6 < 2.646 < 2.667 < 3.142$.

WORKED EXAMPLE 1.3.2 DISTANCE AS AN ABSOLUTE VALUE

PROBLEM

Describe the set of points within 0.4 of 3.5 as an inequality, an interval and a distance statement.

1. Within 0.4 of 3.5 means the distance from x to 3.5 is at most 0.4: $|x - 3.5| \le 0.4$.
2. Unpack the absolute value: $-0.4 \le x - 3.5 \le 0.4$.
3. Add 3.5 throughout: $3.1 \le x \le 3.9$.
4. As an interval: $[3.1, 3.9]$.

ANSWER $|x - 3.5| \le 0.4$, that is $3.1 \le x \le 3.9$, that is $[3.1, 3.9]$.

COMMON ERROR

Dropping one side of an absolute-value inequality

Writing $|x - 3| < 2$ as $x < 5$ loses half the solution. The absolute value describes distance in **both** directions, so the solution is the whole interval $1 < x < 5$.

PRACTICE 1.3 ORDER, DISTANCE AND INTERVALS

SUPPORT 1–4 CORE 1–7 EXTENSION 8–9

1. Insert $<$ or $>$: (a) $\sqrt{6}$ ___ 2.4 (b) $\sqrt{11}$ ___ 3.3 (c) $\sqrt{2}/2$ ___ 0.7. F
2. Write each set in interval notation: (a) $-1 \le x < 4$ (b) $x > 6$ (c) $x \le -2$. F
3. Solve and sketch: (a) $|x - 4| = 3$ (b) $|x + 1| < 2$ (c) $|x - 5| \le 1$. F
4. Write $2.9 < x < 3.1$ as an absolute-value inequality. F
5. Order $\sqrt{10}$, 3.1, 22/7 and π from smallest to largest, showing the comparison for each neighbouring pair. R
6. Explain why squaring is a valid way to compare two positive surds but not a valid way to compare -3 and -5. R
7. A machine must produce rods of length 25 cm to within 0.5 mm. Write the acceptable interval in millimetres and as an absolute-value inequality. P
8. Find all integers n with $|n - 7| < 3$. Explain how the interval and the integer restriction interact. E

1.4 Working with surds

CRITERION A II • EQUIVALENCE • SIMPLIFICATION

Surd arithmetic is algebra with one extra rule: like surds combine, unlike surds do not. Knowing which is which is the whole skill.

Two surds are **like** when they have the same radicand after simplification. Like surds add and subtract exactly as like terms do: $3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$. Unlike surds do not combine: $\sqrt{2} + \sqrt{3}$ stays as it is, because no exact simplification exists.

KEY IDEA

The one rule that matters

Products and quotients of roots can always be combined: $\sqrt{a}\sqrt{b} = \sqrt{(ab)}$ and $\sqrt{a}/\sqrt{b} = \sqrt{(a/b)}$, for $a, b \geq 0$ with $b \neq 0$. Sums and differences of unlike roots cannot.

WORKED EXAMPLE 1.4.1 COLLECTING LIKE SURDS

PROBLEM

Simplify $\sqrt{50} + \sqrt{18} - \sqrt{8}$.

1. Simplify each surd on its own: $\sqrt{50} = 5\sqrt{2}$, $\sqrt{18} = 3\sqrt{2}$, $\sqrt{8} = 2\sqrt{2}$.
2. Now all three terms are like surds in $\sqrt{2}$.
3. Collect the coefficients: $5 + 3 - 2 = 6$.
4. Hence the sum is $6\sqrt{2}$.

ANSWER $6\sqrt{2}$

$5\sqrt{2} \approx 7.07$, $3\sqrt{2} \approx 4.24$, $2\sqrt{2} \approx 2.83$; $7.07 + 4.24 - 2.83 = 8.48$, and $6\sqrt{2} \approx 8.49$.

WORKED EXAMPLE 1.4.2 EXPANDING A PRODUCT WITH

PROBLEM

Expand and simplify $(2+\sqrt{3})(4-\sqrt{3})$.

1. Expand as for any binomial product: $2\sqrt{}4=8$, $2\sqrt{}(-\sqrt{}3)=-2\sqrt{}3$, $\sqrt{}3\sqrt{}4=4\sqrt{}3$, $\sqrt{}3\sqrt{}(-\sqrt{}3)=-3$.
2. Collect like terms: $8-3=5$ and $-2\sqrt{}3+4\sqrt{}3=2\sqrt{}3$.
3. Hence the product is $5+2\sqrt{}3$.

ANSWER $5+2\sqrt{}3$

The surd term did not vanish because the two middle terms had opposite signs; only a conjugate product cancels it.

A **conjugate** pair is $a+\sqrt{}b$ and $a-\sqrt{}b$. Their product, a^2-b, contains no surd at all. This is why multiplying by the conjugate is the standard way to rationalise a two-term denominator.

WORKED EXAMPLE 1.4.3 RATIONALISING WITH A CONJUGATE

PROBLEM

Write $1/(3-\sqrt{}5)$ with a rational denominator.

1. The conjugate of $3-\sqrt{}5$ is $3+\sqrt{}5$.
2. Multiply numerator and denominator by it: $(3+\sqrt{}5)/((3-\sqrt{}5)(3+\sqrt{}5))$.
3. Evaluate the denominator as a difference of squares: $3^2-5=9-5=4$.
4. Hence the expression equals $(3+\sqrt{}5)/4$.

ANSWER $(3+\sqrt{}5)/4$

$(3+2.236)/4=1.309$ and $1/(3-2.236)=1.309$.

COMMON ERROR

Roots do not distribute over sums

$\sqrt{}(a^2+b^2)$ is not $a+b$. The square root undoes a square, not a sum. For example $\sqrt{}(3^2+4^2)=5$, not 7.

NOTE

Why exact forms matter

Surds are not an obstacle to be removed; they are the exact answer. √2 is a better answer than 1.414, because it is the number rather than an approximation to it. Rationalising a denominator does not make a value more exact < it makes it easier to compare and to add.

PRACTICE 1.4 SURD ARITHMETIC

SUPPORT 1–4 CORE 1–7 EXTENSION 8–9

1. Simplify: (a) √12 + √27 (b) √45 - √20 (c) √32 + √8 - √18. F
2. Expand and simplify: (a) (√5 + 1)(√5 - 1) (b) (2√3)² (c) (1 + √2)(3 - √2). F
3. Rationalise: (a) 1/(2 + √3) (b) 5/(√7 - √2) (c) (√6 + √2)/(√6 - √2). F
4. Write √8/√2 in two ways and explain why both give the same value. R
5. Explain why √2 × √18 is an integer, but √2 + √18 is not. R
6. A rectangle has sides (3 + √2) cm and (3 - √2) cm. Find its area and its perimeter exactly. P
7. Show that 1/(√2 + 1) = √2 - 1 by rationalising the denominator. Then show that 1/(√2 - 1) = √2 + 1, and explain what the two results say about reciprocals of surd expressions. P
8. Find integers a and b such that $(3 + \sqrt{2})^2 = a + b\sqrt{2}$. Then find $(3 + \sqrt{2})^3$ in the same form. E

1.5 Sets, logic and proof

CRITERION C, III • REASONING • NOTATION

Mathematics is written in a notation that compresses argument. Learning the notation is not decoration: it is what allows another person to check your reasoning without asking you what you meant.

The number systems nest inside one another. Every natural number is an integer, every integer is rational, and every rational is real. Writing the nesting with set notation makes the relationships visible in a single line.

DEFINITION

, where is the set of natural numbers, the integers, the rationals and the real numbers. The symbol means ‹is a subset of› means ‹is an element of› means ‹is not an element of›

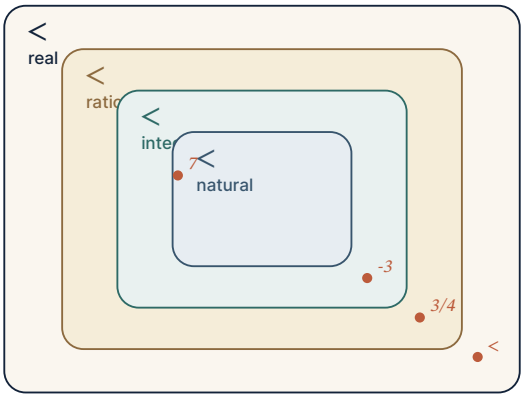


Figure 1.3 The nested number systems, with an example in each region. The irrational numbers are with the rationals removed.

Quantifiers and counterexamples

A universal statement claims something about every case: ‹for all x› An existential statement claims at least one: ‹there exists x› The two are destroyed in different ways. One counterexample destroys a universal claim; one example establishes an existential claim. Confusing the two is the most

common error in student proofs.

Statement	Type	How to prove it	How to disprove it
Every prime is odd	universal	show it for all primes	one even prime (2)
Some prime is even	existential	exhibit one example (2)	<
2 is irrational	universal over forms	proof by contradiction	<

WORKED EXAMPLE 1.5.1 A DIRECT PROOF

PROBLEM

Prove that the sum of any two even integers is even.

1. Write the objects in the form the definition allows: let the integers be $2a$ and $2b$, where a and b are integers. This is what <even< means.
2. Add them: $2a + 2b$.
3. Factor out 2: $2a + 2b = 2(a + b)$.
4. Since $a + b$ is an integer, the sum is twice an integer, and so is even.
5. Because a and b were arbitrary, the result holds for all even integers.

ANSWER The sum of two even integers is even.

The proof works because the *form* $2n$ was used, not particular numbers.

WORKED EXAMPLE 1.5.2 A COUNTEREXAMPLE, PROPERLY

PROBLEM

A student claims: <if n^2 is divisible by 4 then n is divisible by 4<. Disprove the claim.

1. Look for a value where the hypothesis holds but the conclusion fails.
2. Try $n = 6$: $n^2 = 36$, which is divisible by 4, since $36 = 4$ 9.
3. But 6 is not divisible by 4.
4. One counterexample is enough to make a universal claim false, so the claim is disproved.

ANSWER $n = 6$: 36 is divisible by 4 but 6 is not.

Note what the counterexample does *not* do: it does not show the claim is false for every n, only that it is not universally true.

COMMON ERROR

Proof by example

Proving a universal statement with a single example. Showing that $3^2 + 4^2 = 5^2$ does not prove Pythagoras' theorem; it exhibits one right triangle. A proof must reason about the general case.

ATL

ATL ≠Thinking skills

Critical thinking. When you meet a claim, ask three questions: is it universal or existential, what would make it false, and what evidence would settle it? Writing the answers down is the beginning of a proof.

PRACTICE 1.5 CLASSIFICATION, COUNTEREXAMPLE AND PROOF

SUPPORT 1–3 CORE 1–6 EXTENSION 7–9

1. Insert □ or □: (a) -3 □ (b) 0 □ (c) □4 □ (d) □3 □. F
2. State whether each statement is universal or existential, and say what would disprove it: (a) all squares are rectangles; (b) some squares are rectangles. R
3. Disprove: <every odd number is prime< F
4. Prove that the product of two odd integers is odd. P
5. Explain why □2 □ □2 = 2 does **not** show that the product of two irrational numbers is rational in general. Find a counterexample to that general claim. R
6. Prove that if n is odd then n^2 is odd. Use the form $n = 2k + 1$. P
7. Prove that there is no largest rational number less than 1. E
8. A proof begins <Assume □3 = p/q where p and q have no common factor< Explain why that assumption is allowed and why <no common factor<is needed. R

An investigation is not a longer exercise. It is a short piece of mathematics you do yourself: you gather evidence, look for a pattern, say what you think is true, and then test it until you can defend it.

INVESTIGATION

Chasing $\sqrt{2}$ with fractions

Long before calculators, mathematicians approximated square roots using only fractions and arithmetic. A Babylonian method starts with a guess x_1, then repeatedly computes $x_{n+1} = (x_n + 2/x_n)/2$. The method is a feedback process: if x_n is too large, then $2/x_n$ is too small, and the average moves closer to $\sqrt{2}$.

n	x_n (8 decimal places)	x_n^2	$\lvert\sqrt{2} - x_n\rvert$
1	1.50000000	2.25000000	0.08578644
2	1.41666667	2.00694445	0.00245311
3	1.41421569	2.00000601	0.00000212
4	1.41421356	2.00000000	0.00000000

1. **Compute.** Verify the value of x_2 from $x_1 = 1.5$. Then calculate x_3 yourself and compare with the table.
2. **Describe.** How does the error change from one row to the next? Is the change approximately the same each time, or does the error shrink much faster?
3. **Explain.** Why does the method require x_n and $2/x_n$ to lie on opposite sides of $\sqrt{2}$? Use the fact that their product is 2.
4. **Generalise.** Adapt the method to approximate $\sqrt{7}$. Start with $x_1 = 3$ and compute three iterations.
5. **Verify.** Use your generalised formula to approximate $\sqrt{10}$. Compare with a calculator and describe the size of the error after three iterations.
6. **Conclude.** The sequence produces rational numbers forever. Explain why no term can ever equal $\sqrt{2}$, even though the terms get arbitrarily close. What does this show about the completeness of the rationals on the number line?

Report your table, your formula for the general case, and a conclusion of three or four sentences. A strong conclusion distinguishes evidence from proof.

APPLICATION

Tolerance is an interval

A bicycle workshop produces seat posts with a nominal diameter of 27.2 mm. A post that is too wide will not fit; one that is too narrow will slip. The specification allows a deviation of at most 0.05 mm in either direction.

Measured diameter (mm)	Device resolution (mm)	Recorded value	Decision
27.24	0.01	27.24	accept
27.16	0.01	27.16	accept
27.252	0.01	27.25	borderline
27.148	0.01	27.15	borderline

1. Write the acceptable interval using inequalities and using absolute value.
2. Which recorded values in the table are inside the interval? Which are borderline, and why?
3. The measuring device reads to 0.01 mm. Explain how rounding to the nearest 0.01 mm can move a value across the boundary of the interval.
4. Recommend an accept-or-reject decision for each borderline part, and state the cost of each kind of error: a rejected good post, and an accepted bad one.
5. The workshop wants to guarantee that no accepted post lies outside the true interval. Suggest a tighter internal limit and justify it.

CRITERION C COMMUNICATING

Why ≠exact≠ is a claim about form

Write a short article for the school mathematics magazine. Your reader knows fractions and decimals but has never met an irrational number.

1. Explain, with one worked example, why every repeating decimal is a fraction.
2. Explain why $\sqrt{2}$ cannot be written as a fraction. Set the proof out as a numbered chain of reasoning, so that a reader can check each step.
3. A reader objects: <but 1.41421356 is $\sqrt{2}$, my calculator says so< Answer the objection, using the language of approximation and accuracy.
4. Explain the difference between an exact answer and a rounded answer, and say when each is the appropriate form to report.

1–2 Uses some correct notation; explains one idea, with errors in reasoning or structure.

3–4 Uses correct notation and a logical sequence; explains the conversion and the irrationality argument adequately.

5–6 Explains completely and concisely, using notation consistently; the proof is valid and the objection is answered precisely.

7–8 As above, plus a clear distinction between exact and approximate forms, and a judgement about when each should be reported.

Chapter summary and review

Key ideas

- A rational number can be written as p/q with integers p, q and $q \neq 0$.
- A fraction terminates in decimal form exactly when the denominator, in lowest terms, has no prime factor other than 2 or 5.
- A repeating decimal converts to a fraction by subtracting two multiples of the decimal.
- Irrational numbers are real numbers that are not rational; their decimals neither terminate nor repeat.
- $\sqrt{2}$ is irrational, and the proof is by contradiction: assume the opposite and derive an impossibility.
- Surds multiply and divide using radical rules; only like surds add or subtract.
- Absolute value measures distance; $|x - a| \leq r$ describes an interval centred at a with radius r.

Vocabulary

rational number
A number of the form p/q with integers p, q and $q \neq 0$.

irrational number
A real number that is not rational.

surd
An irrational root, such as $\sqrt{2}$, left in exact form.

radicand
The number under a radical sign.

terminating decimal
A decimal that ends after finitely many digits.

repeating decimal
A decimal with a block of digits that repeats forever.

absolute value
The distance of a number from zero.

interval
A connected set of real numbers described by endpoints.

counterexample
A case that makes a universal statement false.

proof by contradiction
A proof that assumes the negation and derives a contradiction.

density
Having another element between any two distinct elements.

rationalise
Rewrite a quotient so no surd remains in the denominator.

What students often think	What is actually true
0.999< is just less than 1.	It equals 1: $x = 0.999<$ gives $10x - x = 9$, so $x = 1$.
$\sqrt{}(a+b) = \sqrt{}a + \sqrt{}b$.	Roots split over products, not sums: $\sqrt{}(9+16) = 5 \neq 7$.
Every number on the line is a fraction, because fractions are dense.	Density does not fill the line; $\sqrt{}2$, π and infinitely many others are not fractions.
A rounded decimal is an exact answer.	A rounded decimal is an approximation; the surd or fraction is the exact form.

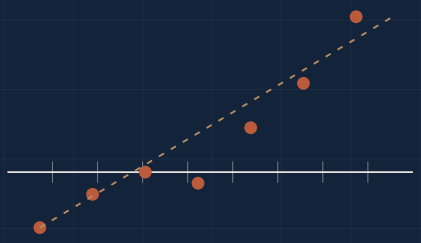
Misconceptions worth confronting directly.

Revision checklist

- ☐ I can classify numbers using ℕ ℤ ℚ ℝ ℂ ℙ ℍ 𝕆.
- ☐ I can convert a repeating decimal to a fraction and check it.
- ☐ I can simplify a surd and rationalise a one-term denominator.
- ☐ I can order a mixed set of rational and irrational values.
- ☐ I can express a tolerance as an interval and make an accept-or-reject decision.
- ☐ I can distinguish a counterexample from a proof.
- ☐ I can reproduce the proof that $\sqrt{}2$ is irrational, justifying each step.
- ☐ I can write a short argument using set, interval and radical notation.

> **REFLECT**
>
> Rate each checklist item **secure**, **developing** or **not yet**. For every developing item, name the practice question that would test it, and attempt that question again in a week without looking at the answer.





2

NUMBER, STRUCTURE AND PROOF

Powers, roots and scientific notation

Repeated multiplication has its own arithmetic. Once the rules are written down, a number of any size can be held in a few symbols — which is why an astronomer and a chemist can compare an orbit with an atom without writing out the zeros.

KEY CONCEPT ▪ FORM | RELATED ▪ QUANTITY | RELATED ▪ REPRESENTATION

CONTEXT ▪ SCIENTIFIC AND TECHNICAL INNOVATION

INQUIRY

Why does a compact notation for very large and very small numbers change what we are able to reason about — and when should a number be reported to three significant figures rather than exactly?

ATL ▪ RESEARCH AND THINKING SKILLS

Find two published quantities, one far larger than the other, both written in scientific notation. Convert each to an ordinary decimal to see how unwieldy it is, then compare them by dividing. Decide what the ratio tells you that neither number alone does.

MYP UNIT PLANNER ≣THE WHOLE CHAPTER IN ONE VIEW

Framing the inquiry

Unit 2 ?approximately 14 hours of class time ?ATL focus: research and thinking

KEY AND RELATED CONCEPTS

Form. A power is a compressed product; a root is the inverse of a power; scientific notation is a standard form that makes magnitude visible. The form chosen changes what is easy to see.

Quantity. Orders of magnitude express comparison rather than exact size. Two quantities differing by 10^3 are not <a bit<different; one is a thousand times the other.

Representation. The same number can be written as a power, a root, a decimal, a fraction or in scientific notation. Each representation makes a different property visible.

GLOBAL CONTEXT AND EXPLORATION

Scientific and technical innovation. Every measuring instrument has a resolution, and every reported number carries an implicit claim about how precisely it is known. The notation of powers and significant figures is what allows engineers, chemists and astronomers to state that claim honestly. A scale model of the solar system, a drug dose in micrograms and the tolerance on an aircraft part are all written in this language.

STATEMENT OF INQUIRY

In order for a student to compare quantities across many orders of magnitude, students must be able to move fluently between powers, roots, decimals and scientific notation, and must report a result with a precision that the measurement can support.

FACTUAL

<What does a negative exponent mean?

<How is a number written in scientific notation?

<What is a significant figure?

CONCEPTUAL

<Why do the exponent laws have to hold for negative and fractional exponents if the notation is to stay consistent?

DEBATABLE

<Is a reported value of 3.10 m more honest than 3.1 m, or merely more precise-looking?

Criterion	Strands	Evidence in this chapter
A ?Knowing and understanding	i, ii, iii	Apply the exponent laws; evaluate roots; convert between forms; round to a given number of significant figures.
B ?Investigating patterns	i, ii, iii	Investigate how repeated folding grows without bound; describe the pattern and generalise it as a power.
C ?Communicating	i, ii, iv	Report values with appropriate precision; justify a rounding decision; explain the meaning of an order of magnitude.
D ?Applying mathematics	i, ii, iii, v	Build and evaluate a scale model; convert units of area and volume; check whether a reported precision is credible.

Each teaching section closes with a graded practice set. The investigation addresses Criterion B; the application addresses Criterion D; the criterion task asks for extended written communication (Criterion C) about knowledge and understanding (Criterion A).

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** a power as repeated multiplication and a root as its inverse.
- **Know** and apply the exponent laws, including zero, negative and fractional exponents.
- **Be able to** write and interpret numbers in scientific notation.
- **Be able to** estimate using orders of magnitude and check a result for reasonableness.
- **Be able to** round to a given number of significant figures and state the precision of a measurement.
- **Understand** how error behaves when quantities are multiplied or divided.

SUCCESS CRITERIA I CAN

- ☐ I can simplify expressions using the exponent laws and say which law I used.
- ☐ I can evaluate $16^{3/4}$ and $27^{-2/3}$ without a calculator.
- ☐ I can write 0.000 042 and 5 600 000 in scientific notation, and back again.
- ☐ I can estimate $(4.9 \times 10^{7}) \times (2.1 \times 10^{3})$ before computing it.
- ☐ I can state the number of significant figures in a reported value and round correctly.
- ☐ I can explain why 3.10 m and 3.1 m make different claims about precision.

PREREQUISITE CHECK 10 MINUTES, NO CALCULATOR

1. Calculate 3^4.
2. Calculate $\sqrt{81}$ and $\sqrt{(1/4)}$.
3. Write 10 000 as a power of 10.
4. Simplify $x^2 \times x^3$.
5. Round 4.678 to two decimal places.
6. How many millimetres are there in 3.5 m?

Answers are in the chapter answer section. If more than two are incorrect, review index notation and place value before continuing.



A micrometer and a measurement log. The instrument fixes the number of digits that can honestly be reported; the log records what was actually read, not what was later calculated.
Original photograph made for this edition.

Explore: how many digits does a number need?

Take one large quantity and one very small one and write each out in full. Then count the characters. The count is the reason scientific notation exists.

YOUR TASK ▤ BEFORE YOU ARE TAUGHT

1. Choose a quantity larger than a million (a distance, a population, a file size) and write it in full with all its zeros. Count the digits.
2. Choose a quantity smaller than one thousandth (a particle size, a thickness, a concentration) and write it in full. Count the zeros between the point and the first digit.
3. Record both in a table: ordinary form, number of characters, number of significant figures you actually know.
4. Multiply your large quantity by your small one without a calculator, estimating first. Then check with a calculator and compare.
5. Write a rule of your own for how many places the point must move to write a number in the form $a \times 10^{n}$ with $1 \leq a < 10$.
6. Test your rule on 0.000 4 and on 4 000 000. If it fails on either, revise it and say what you changed.

You have just written the definition in section 2.3 for yourself. The sections that follow give the notation a name and make it reliable.

BEFORE YEAR 3	YEAR 3 THIS CHAPTER	LATER
▪ Powers with whole-number exponents. ▪ Square and cube roots of perfect squares and cubes. ▪ Multiplying and dividing by powers of 10.	▪ Exponent laws, including negative and fractional exponents. ▪ Scientific notation and orders of magnitude. ▪ Significant figures and precision.	▪ Logarithms as the inverse of exponentials. ▪ Exponential functions and their graphs. ▪ Error propagation and uncertainty analysis.

Where this chapter sits in the sequence.

Inquiry pathway for this chapter

	Stage	What you do
1	Context	A folded sheet of paper and a scale model of the solar system.
2	Question	How can a short notation hold a number too long to write?
3	Concept	Power, root, exponent law, order of magnitude, significant figure.
4	Investigation	Fold a sheet forty-two times and find where it reaches.
5	Communication	Report a measurement with an honest precision.
6	Reflection	When is a more precise number a less truthful one?

2.1 Integer exponents and the exponent laws

CRITERION A, II • POWERS • STRUCTURE

A power is repeated multiplication written compactly. The compactness is useful, but the exponent laws are what make the notation powerful: they turn multiplication into addition.

In a^n, the number a is the **base** and n is the **exponent** or index. For a positive integer n, a^n means n copies of a multiplied together: $2^5 = 32$. Two powers with the same base can be multiplied by adding the exponents, because the total number of factors is simply the sum.

DEFINITION

For any real base a and positive integers m and n: $a^m \times a^n = a^{m+n}$, $a^m \div a^n = a^{m-n}$, $(a^m)^n = a^{mn}$, $(ab)^n = a^n b^n$, and $(a/b)^n = a^n/b^n$ for $b \neq 0$.

KEY IDEA

Why the laws force the definitions

The laws are not arbitrary rules to memorise; each one is a statement about counting factors. $a^m \times a^n$ has $m + n$ factors of a, so it must equal a^{m+n}. If the laws are to keep working, they force the meanings of a^0 and a^{-n} as well.

Zero and negative exponents

Dividing a^3 by a^3 should give 1, and the subtraction law gives $a^{3-3} = a^0$. So a^0 must be 1 for any nonzero a. Similarly $a^2 \times a^5$ is $1/a^3$ by direct cancellation, while the same law gives a^{-3}; so a negative exponent means a reciprocal. Nothing is being invented here < the definitions are the only ones that keep the arithmetic consistent.

WORKED EXAMPLE 2.1.1 APPLYING THE LAWS

PROBLEM

Simplify: (a) $x^5 \cdot x^3$ (b) $y^9 \div y^4$ (c) $(2z^3)^4$ (d) $(6a^5)/(3a^2)$.

1. (a) Same base, multiplication: add the exponents, $5 + 3 = 8$, giving x^8.
2. (b) Same base, division: subtract, $9 - 4 = 5$, giving y^5.
3. (c) The exponent applies to both factors: $2^4(z^3)^4 = 16z^{12}$.
4. (d) Divide the coefficients and subtract the exponents: $(6/3)a^{5-2} = 2a^3$.

ANSWER (a) x^8 (b) y^5 (c) $16z^{12}$ (d) $2a^3$

Test (a) with $x = 2$: $2^5 \cdot 2^3 = 32 \cdot 8 = 256 = 2^8$, and it works.

WORKED EXAMPLE 2.1.2 ZERO AND NEGATIVE EXPONENTS

PROBLEM

Evaluate: (a) 7^0 (b) 2^{-4} (c) $(3/4)^{-2}$ (d) $5x^0$ when $x = 3$.

1. (a) Any nonzero base to the power zero is 1: $7^0 = 1$.
2. (b) A negative exponent gives a reciprocal: $2^{-4} = 1/2^4 = 1/16$.
3. (c) Invert the fraction and square: $(3/4)^{-2} = (4/3)^2 = 16/9$.
4. (d) The exponent applies only to x, not to 5: $5x^0 = 5 \cdot 1 = 5$.

ANSWER (a) 1 (b) 1/16 (c) 16/9 (d) 5

In (d), $(5x)^0$ would be 1 < the bracket matters.

COMMON ERROR

Mixing up the two laws

$2^3 \cdot 2^4 = 2^{12}$ and $(x^2)^3 = x^5$ are both wrong. Multiplication of like bases **adds** exponents; a power of a power **multiplies** them. Check with numbers: $2^3 \cdot 2^4 = 8 \cdot 16 = 128 = 2^7$, not 4096.

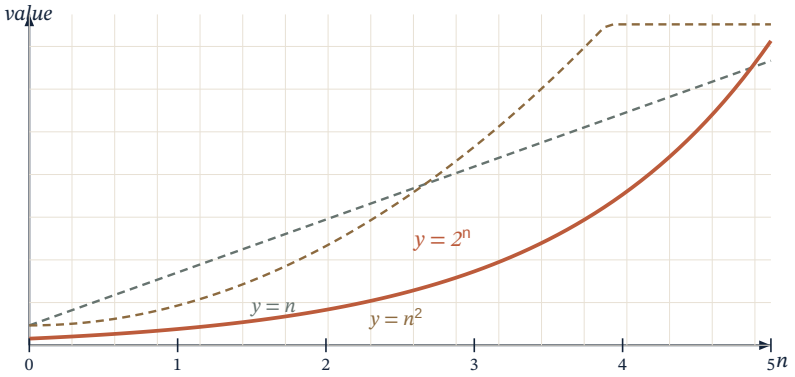


Figure 2.1 Linear, quadratic and exponential growth compared. The exponent controls how fast a quantity changes: each increase of 1 in the exponent doubles the value, which is why exponential graphs overtake every polynomial.

PRACTICE 2.1 ▤ EXPONENT LAWS

SUPPORT 1▤4 ▤ CORE 1▤7 ▤ EXTENSION 8▤0

1. Simplify: (a) $a^4 \cdot a^6$ (b) $b^7 \div b^2$ (c) $(c^3)^5$ (d) $(2d)^3$. F
2. Evaluate: (a) 3^0 (b) 5^{-2} (c) $(2/5)^{-3}$ (d) 10^{-1}. F
3. Simplify: (a) $(8x^6)/(2x^2)$ (b) $(3a^2b)^2$ (c) $(x^4y^3)/(x^2y)$. F
4. Write as a single power of 2: (a) 8 (b) 1/16 (c) $(2^5 \cdot 2^3)/2^4$. F
5. Explain, using the subtraction law, why $a^0 = 1$ for every nonzero a. R
6. A student writes $(3x)^2 = 3x^2$. Show, with a numerical substitution, that this is wrong, and give the correct expansion. R
7. Simplify and state which law you used at each step: $(2m^3n^2)^2 \cdot (3m\,n^4)$. P
8. Show that $(a^m)^n = (a^n)^m$ and explain what this means about the order of applying powers. E
9. If $2^x = 32$ and $3^y = 81$, find $x + y$. Then find $2^x \cdot 3^y$. P

2.2 Roots and rational exponents

CRITERION A ▤ I • INVERSE OPERATIONS • EQUIVALENCE

A root undoes a power. Writing a root as a fractional exponent is not a trick: it is the only notation that keeps the exponent laws true for every rational exponent.

The square root of a is the non-negative number whose square is a. The cube root is the number whose cube is a, and unlike the square root it exists for negative numbers: $\sqrt[3]{-8} = -2$ because $(-2)^3 = -8$. In general the nth root $\sqrt[n]{a}$ is the number that must be raised to the power n to give a.

DEFINITION

$\sqrt[n]{a} = a^{1/n}$, and $\sqrt[n]{a^m} = a^{m/n}$. For an even root the radicand must be non-negative if the result is to be real; for an odd root it may be negative.

KEY IDEA

The notation is forced

Why $a^{1/2}$ must be $\sqrt{a}$: apply the multiplication law to $a^{1/2} \cdot a^{1/2}$. The exponents add to 1, so the product is $a^1 = a$. The only non-negative number whose square is a is $\sqrt{a}$, so $a^{1/2}$ has no choice.

fraction	decimal	behaviour
3/8	0.375	*terminates*
7/40	0.175	*terminates*
5/6	0.8333<	*repeats*
2/11	0.1818<	*repeats*
7/9	0.7777<	*repeats*

Figure 2.2 Powers, roots and decimals as three representations of the same quantity ▤ and the behaviour that distinguishes them.

WORKED EXAMPLE 2.2.1 EVALUATING A RATIONAL EXPONENT

PROBLEM

Evaluate $16^{3/4}$ without a calculator.

1. Write the exponent as a root and a power: $16^{3/4} = (16^{1/4})^3$.
2. Take the root first, because the numbers stay small: $16^{1/4} = 2$, since $2^4 = 16$.
3. Cube the result: $2^3 = 8$.

ANSWER 8

The other order works too, but is harder: $16^3 = 4096$ and $\sqrt[4]{4096} = 8$.

WORKED EXAMPLE 2.2.2 A NEGATIVE RATIONAL EXPONENT

PROBLEM

Evaluate $27^{-2/3}$.

1. The negative sign means reciprocal: $27^{-2/3} = 1/27^{2/3}$.
2. Take the cube root: $27^{1/3} = 3$.
3. Square it: $3^2 = 9$.
4. So the value is $1/9$.

ANSWER $1/9$

$27^{2/3} = 9$, so its reciprocal is $1/9$.

COMMON ERROR

Forgetting the absolute value

$\sqrt{(a^2)}$ is not a when a is negative: $\sqrt{((-5)^2)} = \sqrt{25} = 5$, not -5. The square root returns the **non-negative** root, so $\sqrt{(a^2)} = |a|$.

COMMON ERROR

Even roots of negative numbers

An even root of a negative number has no real value: $\sqrt{(-9)}$ is not -3, because $(-3)^2 = 9$, not -9. There is no real number whose square is negative.

PRACTICE 2.2 ▤ ROOTS AND RATIONAL EXPONENTS

SUPPORT 1▤4 ▤ CORE 1▤7 ▤ EXTENSION 8▤9

1. Evaluate: (a) $25^{1/2}$ (b) $8^{1/3}$ (c) $81^{1/4}$ (d) $(-27)^{1/3}$. F
2. Evaluate: (a) $8^{2/3}$ (b) $25^{3/2}$ (c) $16^{-1/2}$ (d) $64^{-2/3}$. F
3. Write as a single power: (a) $□a□ \lhd□ a$ (b) $□(a^3)/□a$ (c) $(□a)^6$. F
4. State whether each is real, and give the value if it is: (a) $□(-4)$ (b) $\lhd□(-125)$ (c) $(-16)^{1/4}$. F
5. Explain why $□(a^2) = |a|$ rather than a. Use $a = -3$ as an example. R
6. Show that $8^{2/3} = (8^{1/3})^2 = (8^2)^{1/3}$, and explain why the two orders of operation agree. R
7. A cube has volume 125 cm< Find its edge length, and write your working using a rational exponent. P
8. Solve $x^{3/2} = 27$ for positive x, showing each step. P
9. Decide whether $□2□ \lhd□ 2$ can be written as a single power of 2. Justify your answer. E

2.3 Scientific notation

CRITERION A ▤ II • REPRESENTATION • MAGNITUDE

A number written in scientific notation has two parts: a number between 1 and 10, and a power of ten. The first part carries the digits; the second carries the size.

> **DEFINITION**
>
> A number is in **scientific notation** when it is written as $a \times 10^{n}$, where $1 \leq a < 10$ and n is an integer.

For numbers greater than 1 the exponent is positive and equals the number of places the decimal point moves to the left: $5\,600\,000 = 5.6 \times 10^{6}$. For numbers between 0 and 1 the exponent is negative: $0.000\,042 = 4.2 \times 10^{-5}$. The exponent counts the size, and the mantissa carries the digits.

Ordinary form	Scientific notation	Order of magnitude
5 600 000	5.6×10^{6}	millions
0.000 042	4.2×10^{-5}	hundred-thousandths
384 400 000	3.844×10^{8}	hundreds of millions
0.000 000 001	1×10^{-9}	billionths

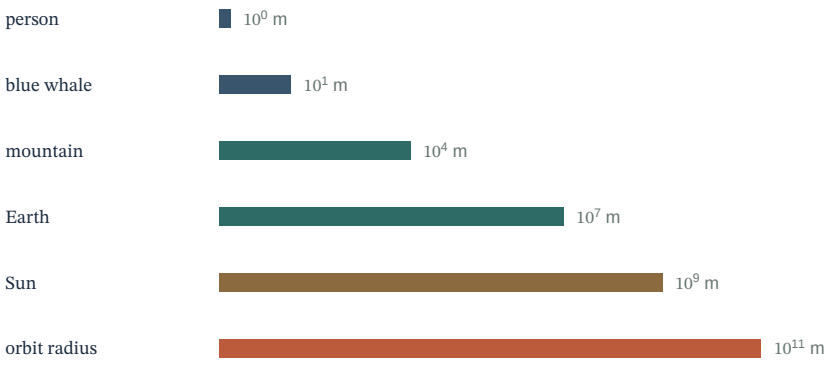


Figure 2.3 Six quantities on one logarithmic scale. Reading a logarithmic axis, each step is a multiplication by ten, so equal distances are equal ratios ▤ not equal differences.

WORKED EXAMPLE 2.3.1 MULTIPLYING IN SCIENTIFIC

PROBLEM

Calculate $(3.2 \times 10^5) \times (4.5 \times 10^{-8})$, giving the answer in scientific notation.

1. Multiply the mantissas: $3.2 \times 4.5 = 14.4$.
2. Add the exponents: $10^5 \times 10^{-8} = 10^{-3}$.
3. Combine: 14.4×10^{-3}.
4. The mantissa is not between 1 and 10, so adjust: $14.4 \times 10^{-3} = 1.44 \times 10^{-2}$.

ANSWER 1.44×10^{-2}, that is 0.0144

Estimate first: about $3 \times 4.5 = 13.5$ and $10^{5-8} = 10^{-3}$, so about 1.4×10^{-2}.

WORKED EXAMPLE 2.3.2 DIVIDING AND INTERPRETING

PROBLEM

The Earth is about 1.5×10^8 km from the Sun, and the Moon is about 3.8×10^5 km from the Earth. Roughly how many times further is the Sun than the Moon?

1. Form the ratio: $(1.5 \times 10^8) \div (3.8 \times 10^5)$.
2. Divide the mantissas: $1.5 \div 3.8 \approx 0.395$.
3. Subtract the exponents: $10^{8-5} = 10^3$.
4. Combine and adjust: $0.395 \times 10^3 = 3.95 \times 10^2$.

ANSWER about 4.0×10^2, that is about 400 times

$4 \times 3.8 \times 10^5 = 1.52 \times 10^6$ km, close to the Moon's 150 million km figure.

COMMON ERROR

A mantissa outside the range

0.42×10^{-3} is **not** in scientific notation, because 0.42 is less than 1. Move the point one place right and reduce the exponent by one: 4.2×10^{-4}.

PRACTICE 2.3 SCIENTIFIC NOTATION

SUPPORT 1–4 CORE 1–7 EXTENSION 8–9

1. Write in scientific notation: (a) 47 000 (b) 0.000 81 (c) 6 300 000 (d) 0.000 000 5. F
2. Write as an ordinary number: (a) 2.7×10^4 (b) 9.03×10^{-3} (c) 1.0×10^6. F

PRACTICE 2.3 ▤ SCIENTIFIC NOTATION (CONTINUED)

SUPPORT 1▤4 ▤ CORE 1▤7 ▤ EXTENSION 8▤9

1. Calculate, giving your answer in scientific notation: (a) $(2 \times 10^3)(4 \times 10^5)$ (b) $(9 \times 10^{-2}) \div (3 \times 10^4)$. F
2. Correct each so that it is in scientific notation: (a) 12.4×10^4 (b) 0.37×10^{-2}. F
3. Estimate $(5.9 \times 10^6) \div (2.1 \times 10^2)$ by rounding the mantissas to one significant figure, then compute it exactly and compare. R
4. A grain of rice has mass about 2×10^{-5} kg. Estimate the mass of rice needed to feed 600 students if each eats 90 g. P
5. Explain why an exponent of -4 corresponds to a number between 0.0001 and 0.001, rather than between 0.001 and 0.01. R
6. The mass of a proton is 1.67×10^{-27} kg. How many protons have a total mass of 1 kg? Give your answer to two significant figures. E

2.4 Orders of magnitude and estimation

CRITERION A V • QUANTITY • APPROXIMATION

An order of magnitude is not a vague word. It is the exponent in scientific notation, rounded to the nearest whole number. Comparing orders of magnitude answers the question <how many times bigger?<before any digits are needed.

Two quantities differ by one order of magnitude when one is about ten times the other. The radius of the Earth is 6.4×10^3 km and the radius of the Sun is 7.0×10^5 km: the exponents differ by 2, so the Sun is about $10^2 = 100$ times wider. Exact radius ratio: $7.0 \times 10^5 \div 6.4 \times 10^3 \approx 109$.

KEY IDEA

Estimation is a claim, not a guess

An estimate is only useful if it is honest about its accuracy. Rounding every input to one significant figure gives an answer good to roughly one significant figure < not to six. Writing $\approx 4 \times 10^5$ is a stronger statement than writing = 412 733, because the first does not pretend to a precision the inputs never had.

WORKED EXAMPLE 2.4.1 ESTIMATING BEFORE COMPUTING

PROBLEM

Estimate $(6.2 \times 10^4) \times (3.9 \times 10^{-2})$ and say how reliable the estimate is.

1. Round each mantissa to one significant figure: $6.2 \to 6$ and $3.9 \to 4$.
2. Multiply the rounded mantissas: $6 \times 4 = 24$.
3. Add the exponents: $10^{4-2} = 10^2$.
4. Combine and adjust: $24 \times 10^2 = 2.4 \times 10^3$.

ANSWER $\approx 2.4 \times 10^3$, reliable to one significant figure

Exact value: $6.2 \times 3.9 \times 10^2 = 2.4 \times 10^3$, so the estimate happens to be close < but that is luck, and the method does not guarantee it.

WORKED EXAMPLE 2.4.2 A FERMI ESTIMATE

PROBLEM

Estimate how many breaths a person takes in a lifetime. State each assumption.

1. Assumptions: a lifetime of 80 years, 12 breaths per minute, 16 waking hours per day.
2. Breaths per day: $12 \times 60 \times 16 \times 1.2 \times 10^4$.
3. Days in a lifetime: $80 \times 365 \times 3 \times 10^4$.
4. Multiply: $1.2 \times 10^4 \times 3 \times 10^4 = 3.6 \times 10^8$.

ANSWER of the order of 10^8 breaths, about 400 million

Any answer between 10^8 and 10^9 is defensible; the assumptions, not the digits, are the substance of the answer.

NOTE

Why estimate

Order-of-magnitude answers are not sloppy answers. They are the answers that survive being wrong about a detail, and they are what tells you whether a calculator result is plausible.

PRACTICE 2.4 ORDERS OF MAGNITUDE AND ESTIMATION

SUPPORT 1–3 CORE 1–6 EXTENSION 7–8

1. Round each to one significant figure and give its order of magnitude: (a) 8 400 (b) 0.031 (c) 5.9×10^7. F
2. Estimate, stating your method: (a) 29×41 (b) $(6.1 \times 10^5) \times (1.9 \times 10^2)$ (c) 0.0048×0.0021. F
3. How many times greater is 7×10^6 than 2×10^4? State the answer to one significant figure. F
4. A calculator gives 123 456.789 for a question whose inputs were each given to two significant figures. Write an appropriate final answer and explain your reasoning. R
5. Estimate the number of grains of rice in a 1 kg bag. State your assumptions and give the order of magnitude. P
6. Estimate the mass of the air in your classroom. State each assumption and the resulting order of magnitude. P
7. Explain why “twice as large” and “one order of magnitude larger” are very different claims. R
8. Two students estimate the same quantity: one gets 3×10^6 and the other 8×10^5. Are the estimates consistent? Justify your answer by comparing orders of magnitude. E

2.5 Significant figures and precision

CRITERION A • PRECISION • COMMUNICATION

Every measurement ends somewhere. The last digit you report is a claim about how finely you could measure, and a number reported with more digits than the instrument can support is a false claim.

The **significant figures** of a number are the digits that carry information about its size. All nonzero digits are significant; zeros between them are significant; leading zeros never are; trailing zeros are significant only if the value is written in a way that shows they are intended.

Value	Significant figures	Reason
0.004 07	3	leading zeros are not significant
1 500	2, 3 or 4	ambiguous without notation < write 1.5×10^3 to be clear
1.500	4	trailing zeros after a point are significant
20.06	4	zeros between significant digits count

KEY IDEA

One rounding, at the end

Round only at the end. Carrying full precision through a calculation and rounding the final answer avoids accumulated error; rounding at every step can change the last digit of the answer, and sometimes the first.

WORKED EXAMPLE 2.5.1 ROUNDING TO SIGNIFICANT FIGURES

PROBLEM

Round 0.045 678 to (a) two, (b) three and (c) four significant figures.

1. Identify the first significant figure: it is the 4 (the leading zeros do not count).
2. (a) Two figures: keep 4 and 5, and look at the next digit, 6, which rounds up: 0.046.
3. (b) Three figures: keep 4, 5 and 6; the next digit is 7, so round up: 0.0457.
4. (c) Four figures: keep 4, 5, 6, 7; the next digit is 8, so round up: 0.045 68.

ANSWER (a) 0.046 (b) 0.0457 (c) 0.045 68

WORKED EXAMPLE 2.5.2 PRECISION IN A CALCULATION

PROBLEM

A rectangle measures 4.2 cm by 7.35 cm. Calculate its area and state an appropriate precision.

1. Multiply: 4.2 × 7.35 = 30.87.
2. The least precise input, 4.2 cm, has two significant figures.
3. A product cannot be known more precisely than its least precise factor.
4. Round the result to two significant figures.

ANSWER about 31 cm<

If the length is really between 4.15 and 4.25 cm, the area lies between 30.5 and 31.2 cm<< the second digit is already uncertain.

COMMON ERROR

Precision is not the same as decimal places

Reporting 30.87 cm<from a measurement of 4.2 cm × 7.35 cm. The arithmetic is right but the precision claim is false: the result cannot be known to four significant figures when one input is known to two.

PRACTICE 2.5 ▤ SIGNIFICANT FIGURES AND PRECISION

SUPPORT 1▤4 ▤ CORE 1▤7 ▤ EXTENSION 8▤9

1. State the number of significant figures: (a) 0.0032 (b) 5.040 (c) 1 200 (written without a point) (d) 6.02×10^{23}. F
2. Round to two significant figures: (a) 0.087 64 (b) 45.9 (c) 9.997 (d) 0.005 06. F
3. Round to three significant figures: (a) 12 345 (b) 0.123 45 (c) 1.0055. F
4. A pencil measures 17.4 cm and another 16.2 cm. Calculate the total length and state it to an appropriate precision. F
5. Explain why writing 1 500 as 1.5×10^3 removes an ambiguity that the plain number has. R
6. A cyclist covers 42.0 km in 1.6 hours. Calculate the average speed and state an appropriate number of significant figures, with a reason. P
7. A student calculates $6.02 \times 10^{23} \times 2$ and writes 1.204×10^{24}. Is this appropriate? Justify your answer. R
8. A rectangle is measured as 3.4 cm by 5.6 cm, and then 3.40 cm by 5.60 cm with a better instrument. Calculate both areas and state, with reasons, which result you would report for each set of measurements. E

An investigation is not a longer exercise. It is a short piece of mathematics you do yourself: you gather evidence, look for a pattern, say what you think is true, and then test it until you can defend it.

INVESTIGATION

Forty-two folds to the Moon

A sheet of paper 0.1 mm thick is folded in half repeatedly. After one fold the thickness is 0.2 mm; after two, 0.4 mm. The thickness doubles each time, which is what a power of 2 describes.

Folds n	Thickness (mm)	Thickness (m)	Nearest everyday object
1	0.2	0.000 2	a thick page
5	3.2	0.003 2	a grain of rice
10	102.4	0.10	a hand span
20	104 857.6	104.9	a 30-storey building
30	1.07×10^{8}	107 374	above the atmosphere
42	4.40×10^{11}	4.40×10^{8}	<

1. **Compute.** Write the thickness after n folds as a formula in the form $a \times 2^{n}$. Check it against the table for $n = 1$ and $n = 10$.
2. **Describe.** Plot the thickness against n on a graph. Why does the curve look almost flat for the first ten folds?
3. **Explain.** Each new fold doubles the previous thickness. Show that this is equivalent to multiplying by 2^{n} rather than adding n times 0.1 mm.
4. **Extend.** The Moon is about 3.84×10^{8} m away. Use your formula to find the smallest whole number of folds for which the thickness exceeds this distance.
5. **Test.** Check your answer from the previous question by computing the thickness at that number of folds and at one fewer.
6. **Conclude.** The formula is a model, and it fails for large n because paper cannot be folded that often. Explain why the failure does not make the mathematics wrong, and state what the model is actually predicting.

Present your formula, your graph, the algebra for the crossing point, and a conclusion that distinguishes what the model predicts from what is physically possible.

APPLICATION

A scale model of the solar system

A school wants a scale model along a corridor 40 m long. The real distance from the Sun to Neptune is about 4.5×10^9 km. The model must fit the corridor and must still have visible planets.

Quantity	Real value	Model value on a $1 : 1.125 \times 10^{11}$ scale
Sun's diameter	1.39×10^6 km	12.4 mm
Earth's diameter	1.27×10^4 km	0.11 mm
Earth–Sun distance	1.50×10^8 km	1.33 m
Neptune–Sun distance	4.50×10^9 km	40.0 m

1. Convert the scale into a form you can use: 1 m on the model represents how many kilometres?
2. Check the model diameter of the Sun in the table, and explain the arithmetic.
3. The Earth is 0.11 mm across on this scale. Discuss the practical difficulty and suggest one way the model could be made usable.
4. Choose a different scale that makes the Earth at least 2 mm across. How long would the corridor need to be?
5. Recommend a scale for the school, and justify your recommendation in terms of what the model is meant to teach. State one thing the model misrepresents.

CRITERION C COMMUNICATING

A public briefing on very small particles

You are asked to write one page for a museum panel explaining how scientists describe the very small. Your reader is a curious adult with no mathematics beyond school arithmetic.

1. Explain scientific notation using one quantity from this chapter, showing the ordinary form and the notation side by side.
2. Explain why 10^{-9} is a convenient unit for a nanometre, and how many nanometres make a metre.
3. Explain the difference between precision and accuracy using a measurement you describe yourself.
4. A colleague wants the panel to say ‹the particle is 0.000 000 025 m across‹ Improve the sentence, and explain to your colleague what is wrong with the original.

1–2 Uses some correct notation; the explanations are present but imprecise or contain errors.

3–4 Uses notation correctly and explains the conversions adequately for the audience.

5–6 Explains clearly and concisely, choosing forms that suit the reader, and distinguishes precision from accuracy correctly.

7–8 As above, and justifies the choice of form for each quantity, anticipating a reader's misunderstanding.

Chapter summary and review

Key ideas

- a^n means n factors of a; the exponent laws follow from counting factors.
- $a^0 = 1$ and $a^{-n} = 1/a^n$ are forced by the laws, not invented.
- $a^{1/n} = \sqrt[n]{a}$ and $a^{m/n} = \sqrt[n]{(a^m)}$.
- Scientific notation is $a \times 10^n$ with $1 \leq a < 10$.
- An order of magnitude is the exponent rounded to a whole number.
- Significant figures report the precision of a measurement; the last digit reported is a claim.
- A product or quotient cannot be more precise than its least precise input.

Vocabulary

base
In a^n, the number a that is multiplied repeatedly.

exponent
The number n in a^n, counting the factors.

reciprocal
The result of dividing 1 by a number; $1/a$.

surd
An irrational root left in exact form.

scientific notation
A number written as $a \times 10^n$ with $1 \leq a < 10$.

order of magnitude
The exponent in scientific notation, expressing size as a power of ten.

mantissa
The a part of $a \times 10^n$.

significant figures
The digits of a number that carry information about its precision.

precision
The fineness of a measurement, shown by its significant figures.

estimate
A value obtained by rounding inputs to convenient sizes.

What students often think	What is actually true
$2^3 \times 2^4 = 2^{12}$.	Add exponents when multiplying: 2^7.
$(x^2)^3 = x^5$.	Multiply exponents for a power of a power: x^6.
$3x^2$ means $(3x)^2$.	The exponent applies to x only: $3x^2$ is three times x squared.

Misconceptions worth confronting directly.

What students often think	What is actually true
A negative exponent makes the answer negative.	It makes a reciprocal: $2^{-3} = 1/8$, which is positive.
More decimal places means a better measurement.	Precision is limited by the instrument, not by the number of digits typed.

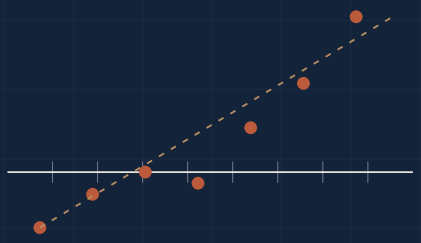
Revision checklist

- ☐ I can apply each exponent law and name it.
- ☐ I can evaluate expressions with zero, negative and fractional exponents.
- ☐ I can convert between ordinary form and scientific notation in both directions.
- ☐ I can compare two quantities by order of magnitude.
- ☐ I can estimate before computing, and check a result for plausibility.
- ☐ I can state the number of significant figures and round correctly.
- ☐ I can decide the appropriate precision for a calculated result.
- ☐ I can explain why precision and decimal places are different ideas.

REFLECT

Choose the three checklist items you are least confident about. For each, write one question of your own that would test it, swap with a partner, and answer each other's questions without notes.





3

NUMBER, STRUCTURE AND PROOF

Functions, expressions and algebraic reasoning

Algebra is compressed reasoning. A letter stands for a number you have not chosen yet, and an expression is a sentence about every number at once. Learning to read that sentence < and to rewrite it without changing its meaning < is the work of this chapter.

KEY CONCEPT ≡ FORM

RELATED ≡ RELATIONSHIPS

RELATED ≡ EQUIVALENCE

CONTEXT ≡ FAIRNESS AND DEVELOPMENT

> **INQUIRY**
>
> How can one expression describe an infinite family of numbers < and how do we know when two expressions mean the same thing?

ATL ≡ THINKING AND COMMUNICATION SKILLS

Write two expressions that you believe are equal for every value of the variable. Test them on five values, including a negative and a fraction. Then decide whether your evidence is a proof, and if not, what would be.

MYP UNIT PLANNER ≣THE WHOLE CHAPTER IN ONE VIEW

Framing the inquiry

Unit 3 □approximately 15 hours of class time □ATL focus: thinking and communication

KEY AND RELATED CONCEPTS

Form. An expression has a structure < terms, factors, a degree < that determines which operations simplify it and which do not. Recognising the structure is what makes factorising possible.

Relationships. A function pairs each input with exactly one output. The domain and range describe the pairing, and the notation $f(x)$ makes the pairing explicit.

Equivalence. Two expressions are equivalent when they give the same value for every allowed input. Expanding and factorising are ways of moving between equivalent forms.

GLOBAL CONTEXT AND EXPLORATION

Fairness and development. A formula is a model of a decision: a price that depends on distance, a wage that depends on hours, a cost that depends on the number of people. Writing the model and rearranging it lets a community compare options and argue about them with numbers rather than impressions. Rearranging a formula to make one variable the subject is exactly what is needed to answer <how many hours until this is worth it?<

STATEMENT OF INQUIRY

In order for a student to generalise from particular cases, students must be able to read the structure of an expression, use function notation to describe a relationship, and rearrange a formula without changing its meaning.

FACTUAL

<What is a variable?

<What does $f(3)$ mean?

<What is the difference between an expression and an equation?

CONCEPTUAL

<Why is expanding $(x + 3)^2$ as $x^2 + 9$ wrong, and what does the correct expansion show about structure?

DEBATABLE

<If a model fits the data we have, when are we justified in using it to predict beyond them?

Criterion	Strands	Evidence in this chapter
A □Knowing and understanding	i, ii, iii	Substitute, expand, factorise, rearrange; evaluate a function and state its domain and range.
B □Investigating patterns	i, ii, iii, iv	Investigate a growing pattern, express it algebraically, and justify the general term.
C □Communicating	i, ii, iii	Use function notation correctly; explain why two expressions are equivalent; present a generalisation.
D □Applying mathematics	i, ii, iii, v	Build a cost model from a context, rearrange it, and justify a decision.

Each teaching section closes with a graded practice set. The investigation addresses Criterion B; the application addresses Criterion D; the criterion task asks for extended written communication (Criterion C) about knowledge and understanding (Criterion A).

Learning intentions and success criteria

YOU ARE LEARNING TO

- **Understand** the difference between a term, a factor and a coefficient, and use the vocabulary precisely.
- **Know** how to expand products of binomials and factorise common factors and simple quadratics.
- **Be able to** use function notation, and to state the domain and range of a function from its rule or graph.
- **Be able to** rearrange a formula to make a chosen variable the subject.
- **Be able to** find and justify the general term of a linear sequence.
- **Understand** that testing cases supports a conjecture but does not prove it.

SUCCESS CRITERIA I CAN

- ☐ I can expand $(2x - 5)(x + 3)$ and factorise $x^2 + 2x - 15$.
- ☐ I can evaluate $f(x) = 3x - 4$ at $x = -2$ and solve $f(x) = 11$.
- ☐ I can state the domain of $f(x) = 1/(x - 2)$ and explain the restriction.
- ☐ I can make r the subject of $A = \pi r^2$.
- ☐ I can find the nth term of 5, 9, 13, 17 and test it.
- ☐ I can explain why three matching cases do not prove a general rule.

PREREQUISITE CHECK 12 MINUTES, NO CALCULATOR

1. Simplify $3x + 5x - 2x$.
2. Expand $4(2x - 3)$.
3. Evaluate $2x + 1$ when $x = -3$.
4. Solve $5x - 7 = 18$.
5. Write the next three terms of 2, 5, 8, 11.
6. Calculate $(-2)^2$ and -2^2.

Answers are in the chapter answer section. If more than two are incorrect, review collecting like terms, substitution and solving linear equations before continuing.



Wooden blocks stacked in a repeating step. Each step adds the same height to the one before, so the total height is a linear function of the number of blocks ▤ the physical version of the general term $2n + 1$.
Original photograph made for this edition.

Explore: a pattern you can build

Before any algebra, build a growing pattern with counters or matches and try to describe it in words. Then try to write your description as a formula.

YOUR TASK ▤ BEFORE YOU ARE TAUGHT

1. Build the first four figures of a pattern in which each figure adds the same number of counters to the one before. Record the totals in a table against the figure number.
2. Describe your pattern in one sentence, without using algebra.
3. Write the total for figure n as a formula. Test it on figure 5 by building it.
4. Swap patterns with a partner. Work out their formula from their table only, then check against their figures.
5. Find a value of n for which your formula gives a total larger than 100, and explain how you found it without building the figure.
6. Write down one thing your formula tells you that the table does not.

Section 3.5 gives the formal language for what you have just done, and asks what makes a pattern a result rather than a coincidence.

BEFORE YEAR 3	YEAR 3 THIS CHAPTER	LATER
▪ Using letters for unknown numbers. ▪ Simplifying expressions by collecting like terms. ▪ Solving one-step and two-step linear equations.	▪ Function notation, domain and range. ▪ Expanding, factorising, rearranging formulas. ▪ Sequences, generalisation and justification.	▪ Quadratic functions and their graphs. ▪ Inverse and composite functions. ▪ Mathematical induction and proof by exhaustion.

Where this chapter sits in the sequence.

Inquiry pathway for this chapter

	Stage	What you do
1	Context	A community hall is painted by a team paid per hour; the cost formula must be rearranged to answer a question.
2	Question	How can one expression stand for infinitely many cases?
3	Concept	Term, factor, coefficient, function, domain, equivalence.
4	Investigation	Matchstick triangles: find and justify the general rule.
5	Communication	Show that two expressions are equivalent, and prove a general term.
6	Reflection	What would make you believe a pattern continues forever?

3.1 Variables, expressions and substitution

CRITERION A≣, II • VARIABLES • STRUCTURE

A variable is not an unknown. It is a placeholder that may take many values, and an expression is a rule for what to do with whatever value arrives.

In the expression $3x^2 - 5x + 2$, the expression is made of three **terms**: $3x^2$, $-5x$ and 2. In the first term the **coefficient** is 3 and the variable part is x^2. The **constant term** is 2. Reading the structure correctly is what allows you to decide which operations are legal: like terms combine, unlike terms do not.

DEFINITION

A **term** is a product of numbers and variables. A **coefficient** is the numerical factor of a term. **Like terms** have identical variable parts. An **expression** is a sum of terms and has no equals sign; an **equation** states that two expressions are equal.

KEY IDEA

Testing a claim

Substitution is the test of equivalence. If two expressions are claimed to be equivalent, they must give the same value for every allowed value of the variable < including negative values and fractions, where sign and order errors appear.

WORKED EXAMPLE 3.1.1 READING THE STRUCTURE

PROBLEM

For the expression $4x^3 - 7x + 6$, state (a) the number of terms, (b) the coefficient of x, (c) the constant term, and (d) the value when $x = -2$.

1. (a) The terms are $4x^3$, $-7x$ and 6: three terms.
2. (b) The coefficient of x is -7, including the sign.
3. (c) The constant term is 6.
4. (d) Substitute carefully: $4(-2)^3 - 7(-2) + 6 = 4(-8) + 14 + 6$.
5. Evaluate: $-32 + 14 + 6 = -12$.

ANSWER (a) 3 (b) -7 (c) 6 (d) -12

$(-2)^3 = -8$, not 8: an odd power keeps the sign.

WORKED EXAMPLE 3.1.2 SUBSTITUTING A FRACTION

PROBLEM

Evaluate $6x^2 - 2x + 1$ when $x = 1/2$.

1. Square first: $(1/2)^2 = 1/4$.
2. Multiply: $6 \times 1/4 = 3/2$.
3. The middle term: $-2 \times 1/2 = -1$.
4. Add: $3/2 - 1 + 1 = 3/2$.

ANSWER 3/2 or 1.5

Keep fractions as fractions until the end; converting to decimals too early invites rounding error.

COMMON ERROR

The bracket that changes the sign

-2^2 and $(-2)^2$ are different: the first is $-(2^2) = -4$ because the exponent binds to 2 only; the second is $(-2) \times (-2) = 4$ because the bracket makes the sign part of the base.

PRACTICE 3.1 EXPRESSIONS AND SUBSTITUTION

SUPPORT 1–4 CORE 1–6 EXTENSION 7–9

1. For $5a^2 - 3a + 8$, state the number of terms, the coefficient of a and the constant term. F
2. Evaluate $3x^2 + 2x - 1$ when (a) $x = 2$ (b) $x = -1$ (c) $x = 1/3$. F

PRACTICE 3.1 ≣EXPRESSIONS AND SUBSTITUTION (CONTINUED)

SUPPORT 1≣4 ≣CORE 1≣6 ≣EXTENSION 7≣9

1. Simplify: (a) $7y - 3y + 2y$ (b) $4m + 3n - m + 2n$ (c) $5p^2 + 2p - 3p^2$. F
2. Evaluate $-x^2$ and $(-x)^2$ when $x = 3$, and explain the difference. F
3. Explain why $3x$ and x^3 are not like terms, using a numerical example with $x = 2$. R
4. A taxi fare is $2.50 + 1.80k$ where k is the number of kilometres. Find the fare for 7 km and state what the constant term represents. P
5. Find a value of x for which $x^2 > x^3$ and a value for which $x^2 < x^3$. What does this show about assuming that higher powers are always larger? E
6. A student says $2x + 3$ is always odd. Test it for $x = 1, 2, 3$ and for $x = 1/2$, then decide whether the claim is true. R

3.2 Functions, domain and range

CRITERION A I • RELATIONSHIPS • NOTATION

A function is a rule with a promise attached: one output for each input. The notation $f(x)$ is the promise made visible.

Writing $f(x) = 3x + 1$ names the rule and the input at the same time. $f(4) = 13$ says: apply the rule to 4 and the output is 13. The **domain** is the set of inputs the rule accepts; the **range** is the set of outputs it produces. Some rules accept every real number; others contain an operation that fails for some inputs.

DEFINITION

A **function** assigns exactly one output to each input in its **domain**. $f(a)$ denotes the output of f at input a. The **range** is the set of all outputs.

Rule	Domain restriction	Why
$f(x) = 2x + 5$	all real x	adding and multiplying always work
$g(x) = 1/(x - 3)$	$x \neq 3$	division by zero is undefined
$h(x) = \sqrt{(x - 1)}$	$x \geq 1$	no real square root of a negative
$k(x) = x^2$	all real x	squaring always works

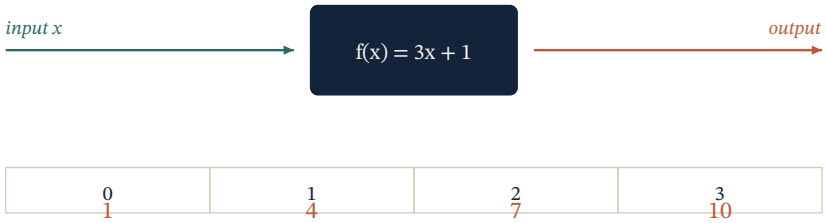


Figure 3.1 A function as a machine and as a table. The rule is the same object seen twice: as an operation and as a list of input–output pairs.

WORKED EXAMPLE 3.2.1 EVALUATING AND SOLVING

PROBLEM

For $f(x) = 3x - 4$, find (a) $f(5)$ and (b) the value of x for which $f(x) = 11$.

1. (a) Substitute: $f(5) = 3(5) - 4 = 15 - 4 = 11$.
2. (b) Set the output equal to 11: $3x - 4 = 11$.
3. Add 4: $3x = 15$.
4. Divide by 3: $x = 5$.

ANSWER (a) 11 (b) $x = 5$

The two parts are inverse questions. Solving $f(x) = k$ asks what input produces output k.

WORKED EXAMPLE 3.2.2 FINDING A DOMAIN

PROBLEM

State the domain of $g(x) = (x + 1)/(x^2 - 9)$.

1. The only operation that can fail is division.
2. Set the denominator equal to zero: $x^2 - 9 = 0$.
3. Factor: $(x - 3)(x + 3) = 0$, so $x = 3$ or $x = -3$.
4. These two values must be excluded; every other real number is allowed.

ANSWER all real x except 3 and -3

Check by substitution: $g(3)$ would require dividing by zero.

COMMON ERROR

Losing the endpoint

Writing the domain of $\sqrt{}(x - 1)$ as $x > 1$. The value $x = 1$ is allowed, because $\sqrt{}0 = 0$. The correct condition is $x \geq 1$.

PRACTICE 3.2 FUNCTIONS, DOMAIN AND RANGE

SUPPORT 1–4 CORE 1–6 EXTENSION 7–9

1. For $f(x) = 2x + 7$, find (a) $f(0)$ (b) $f(-3)$ (c) $f(1/2)$. F
2. For $g(x) = x^2 - 4$, find (a) $g(3)$ (b) $g(-3)$ (c) the values of x for which $g(x) = 0$. F
3. State the domain: (a) $f(x) = 5x - 2$ (b) $g(x) = 1/(x + 4)$ (c) $h(x) = \sqrt{}(x + 2)$. F
4. For $f(x) = 4x - 1$, solve $f(x) = 15$ and $f(x) = -1$. F

PRACTICE 3.2 FUNCTIONS, DOMAIN AND RANGE (CONTINUED)

SUPPORT 1–4 CORE 1–6 EXTENSION 7–9

1. Explain why a function cannot have two different outputs for the same input, and describe what a graph would look like if it did. R
2. The cost of hiring a hall is $C(n) = 120 + 15n$ for n people. Find $C(40)$ and solve $C(n) = 420$, interpreting each answer. P
3. Find the range of $f(x) = x^2 + 1$ for $-2 \le x \le 3$ by testing the endpoints and the turning point. E
4. Two functions are $f(x) = 2x + 1$ and $g(x) = 2(x + 1)$. Show with a table of values that they are not the same function, and explain the structural reason. R

3.3 Expanding and factorising

CRITERION A ▤ II • EQUIVALENCE • STRUCTURE

Expanding and factorising are the same relationship read in two directions. Expanding multiplies out; factorising finds the factors. Neither changes the value.

The distributive law is the engine of both operations: $a(b + c) = ab + ac$. Expanding a product of two binomials applies it twice, which is why four terms appear before like terms are collected. Factorising reverses the process, looking for a common factor or a pair of numbers with the right sum and product.

KEY IDEA

The pattern behind factorising

$(x + a)(x + b) = x^2 + (a + b)x + ab$. The coefficient of x is the sum of the two numbers; the constant term is their product. Factorising $x^2 + 7x + 12$ therefore means finding two numbers that add to 7 and multiply to 12.

WORKED EXAMPLE 3.3.1 — EXPANDING TWO BINOMIALS

PROBLEM

Expand and simplify $(2x - 5)(x + 3)$.

1. Multiply each term of the first bracket by each term of the second: $2x \times x = 2x^2$ and $2x \times 3 = 6x$.
2. Continue: $-5 \times x = -5x$ and $-5 \times 3 = -15$.
3. Collect like terms: $6x - 5x = x$.
4. Hence $(2x - 5)(x + 3) = 2x^2 + x - 15$.

ANSWER $2x^2 + x - 15$

Substitute $x = 1$: left side $(-3)(4) = -12$; right side $2 + 1 - 15 = -12$. Both agree.

WORKED EXAMPLE 3.3.2 FACTORISING A QUADRATIC

PROBLEM

Factorise $x^2 + 2x - 15$.

1. Look for two numbers whose product is -15 and whose sum is +2.
2. The product is negative, so the numbers have opposite signs.
3. Try the factor pairs of 15: (5, -3) has sum 2, which is correct.
4. Write the factors: $(x + 5)(x - 3)$.

ANSWER $(x + 5)(x - 3)$

Expand back: $x^2 - 3x + 5x - 15 = x^2 + 2x - 15$. Always expand to check a factorisation.

COMMON ERROR

Dropping the middle term

$(x + 3)^2 = x^2 + 9$ is wrong: it omits the middle term. $(x + 3)^2 = x^2 + 6x + 9$. Test with $x = 1$: the left side is 16, and $1 + 9 = 10$ is not 16.

COMMON ERROR

Using the common factor twice

Factorising $x^2 + 5x$ as $x(x + 5)x <$ the common factor is used once, not twice. The correct form is $x(x + 5)$.

PRACTICE 3.3 EXPANDING AND FACTORISING

SUPPORT 1–4 CORE 1–7 EXTENSION 8–10

1. Expand: (a) $3(2x - 5)$ (b) $-2(x + 4)$ (c) $x(x - 7)$. F
2. Expand and simplify: (a) $(x + 4)(x + 2)$ (b) $(x - 6)(x + 1)$ (c) $(2x + 1)(x - 3)$. F
3. Expand: (a) $(x + 5)^2$ (b) $(x - 4)^2$ (c) $(x + 7)(x - 7)$. F
4. Factorise fully: (a) $8x + 12$ (b) $x^2 + 6x$ (c) $3x^2 - 9x$. F
5. Factorise: (a) $x^2 + 8x + 15$ (b) $x^2 - 5x + 6$ (c) $x^2 - x - 12$. F
6. Explain, using areas of rectangles, why $(x + a)(x + b) = x^2 + (a + b)x + ab$. R
7. A rectangle has area $x^2 + 7x + 12$. Find expressions for its side lengths, and state what this tells you about the possible values of x. P
8. Show that $(x + 3)^2 - (x - 3)^2 = 12x$ by expanding both brackets, and explain what the result tells you about the difference of the two squares. E

PRACTICE 3.3 EXPANDING AND FACTORISING (CONTINUED)

SUPPORT 14 CORE 17 EXTENSION 80

1. Factorise $2x^2 + 7x + 3$ by writing it as $2x^2 + 6x + x + 3$ and grouping. Explain why this splitting works. E

3.4 Rearranging formulas

CRITERION A ≣ V • EQUIVALENCE • RELATIONSHIPS

A formula is an equation that is true for a whole family of cases. Rearranging it does not change the relationship; it changes which quantity is in plain view.

To make a variable the **subject** of a formula is to rewrite the formula so that the variable stands alone on one side. Every step must be applied to the whole of both sides, and must be reversible. The test is simple: the new formula must give the same values as the old one for every admissible input.

KEY IDEA

A reliable order of work

Undo the operations in reverse order. If a variable has been squared, take the square root last; if it has been multiplied and then had something added, subtract first. This is the same order of operations used to solve equations, applied to letters instead of numbers.

WORKED EXAMPLE 3.4.1 MAKING A VARIABLE THE SUBJECT

PROBLEM

Make r the subject of $A = \square r^2$.

1. Divide both sides by $\square$: $A/\square = r^2$.
2. Take the positive square root of both sides, because r is a length: $r = \square(A/\square)$.
3. State the domain: $A \square 0$.

ANSWER $r = \square(A/\square)$, for $A \square 0$

If $A = 12.57$, then $r = \square(12.57/3.1416) = 2.0$, and $\square(2)^2 = 12.57$.

WORKED EXAMPLE 3.4.2 A VARIABLE IN TWO PLACES

PROBLEM

Make x the subject of $y = (3x + 2)/(x - 1)$.

1. Multiply both sides by $(x - 1)$: $y(x - 1) = 3x + 2$.
2. Expand: $xy - y = 3x + 2$.
3. Collect the x terms on one side: $xy - 3x = y + 2$.
4. Factorise: $x(y - 3) = y + 2$.
5. Divide: $x = (y + 2)/(y - 3)$.

ANSWER $x = (y + 2)/(y - 3)$, for $y \neq 3$

The exclusion $y \neq 3$ arises because the original formula can never output 3; the denominator would be zero.



Figure 3.2 Matchstick triangles. Each new triangle shares one side with the previous one, so each figure needs two more matches than the last ▤ the structural reason the pattern is linear.

WORKED EXAMPLE 3.4.3 JUSTIFYING A GENERAL TERM

PROBLEM

The matchstick pattern has 3 matches in figure 1, 5 in figure 2 and 7 in figure 3. Find the number of matches in figure n and justify the formula.

1. The first difference is constant (2), so the rule is linear: $M = 2n + c$.
2. Substitute figure 1: $3 = 2(1) + c$, so $c = 1$.
3. The formula is $M = 2n + 1$.
4. Justify structurally: the first triangle needs 3 matches; each additional triangle shares one side and adds 2.
5. Hence figure n needs $3 + 2(n - 1) = 2n + 1$ matches.

ANSWER $M = 2n + 1$ matches

Test $n = 10$: formula gives 21; direct construction gives $3 + 2(9) = 21$.

COMMON ERROR

Dividing only one term

Dividing only one term of a side. To make x the subject of $2x + 3y = 12$, divide **every** term by 2: $x = (12 - 3y)/2$, not $x = 6 - 3y$.

PRACTICE 3.4 REARRANGING FORMULAS

SUPPORT 1–4 CORE 1–7 EXTENSION 8–9

1. Make x the subject: (a) $y = x + 7$ (b) $y = 4x$ (c) $y = 5x - 2$. F
2. Make t the subject: (a) $v = u + at$ (b) $s = 5t^2$ (c) $P = 2(l + t)$. F
3. Make a the subject of $A = \tfrac{1}{2}ab$. F
4. Make h the subject of $V = \tfrac{1}{3}\pi r^2 h$. F
5. Explain why making r the subject of $A = \pi r^2$ requires the positive root only. R
6. The total cost of a trip is $C = 45 + 0.6d$ for d kilometres. Make d the subject and find the distance possible on a budget of 120 units. P
7. Make x the subject of $y = (2x - 1)/(x + 3)$ and state the excluded value of y. E
8. A student writes: $\tfrac{3x + 2y}{} = 12$, so $x = 4 - 2y$. Identify the error, give the correct rearrangement, and test both with $y = 3$. R
9. Make n the subject of $S = \tfrac{n}{2}(2a + (n - 1)d)$ when $d = 0$, and interpret the result in words. E

3.5 Sequences, patterns and generalisation

CRITERION B, II • PATTERNS • JUSTIFICATION

A sequence is a list with a rule. Finding the rule is the easy half; showing that the rule must continue is the mathematics.

A sequence is **linear** when the difference between consecutive terms is constant. The constant difference is the coefficient of n in the general term. In the sequence 5, 9, 13, 17 the difference is 4, so the general term has the form $4n + c$; substituting $n = 1$ gives $c = 1$, so the nth term is $4n + 1$.

DEFINITION

The ***n*th term** (or general term) is an expression that gives the value of the sequence at position n. A sequence is **arithmetic** when consecutive terms differ by a constant.

KEY IDEA

Evidence is not proof

A pattern noticed in three or four cases is a **conjecture**, not a proof. It becomes a result only when it is derived from the structure of the pattern, or proved for a general n. Three cases can suggest the wrong rule entirely: the sequence 2, 4, 8 could be doubling, or it could be $n^2 - n + 2$.

WORKED EXAMPLE 3.5.1 FINDING AND TESTING A GENERAL

PROBLEM

Find the nth term of 7, 11, 15, 19 and use it to find the 20th term.

1. The first difference is constant: 11 - 7 = 4.
2. So the general term is $4n + c$.
3. Substitute $n = 1$: $7 = 4(1) + c$, so $c = 3$.
4. The nth term is $4n + 3$. Test $n = 3$: $4(3) + 3 = 15$, which matches.
5. For $n = 20$: $4(20) + 3 = 83$.

ANSWER $4n + 3$; the 20th term is 83

A linear rule grows by a constant amount, so the 20th term must be 7 + 19 × 4 = 83.

WORKED EXAMPLE 3.5.2 A PATTERN THAT IS NOT LINEAR

PROBLEM

The sequence 1, 4, 9, 16, … has first differences 3, 5, 7. Find its general term and justify it.

1. The first differences are not constant but they increase by 2, so the second difference is constant.
2. A constant second difference means the rule is quadratic: $an^2 + bn + c$.
3. Compare with the squares: $n = 1$ gives 1, $n = 2$ gives 4, $n = 3$ gives 9.
4. The terms are exactly the square numbers, so the nth term is n^2.
5. Justify structurally: the nth figure is an $n \times n$ square array, which contains n^2 dots.

ANSWER n^2

Test $n = 5$: 25, and the differences 3, 5, 7, 9 confirm the next term is 16 + 9 = 25.

COMMON ERROR

Confusing the differences

Concluding that a sequence is quadratic because the first differences are not constant. Check the *second* differences; only a constant second difference indicates a quadratic rule.

ATL

ATL ≠Thinking critically

Thinking skills. When you find a pattern, write down what would make it fail. Then look for that case. A conjecture you have tried to break is worth more than one you have merely confirmed.

PRACTICE 3.5 ▤SEQUENCES AND GENERALISATION

SUPPORT 1▤4 ▤CORE 1▤7 ▤EXTENSION 8▤0

1. Find the next two terms: (a) 6, 10, 14, < (b) 20, 17, 14, < (c) 2, 6, 12, 20, < F
2. Find the nth term: (a) 4, 7, 10, 13, < (b) 9, 14, 19, 24, < (c) 30, 26, 22, 18, < F
3. The nth term of a sequence is $3n - 2$. Write the first five terms and find which term equals 43. F
4. A sequence begins 1, 3, 6, 10, 15. Find the nth term and justify it using the differences. F
5. Explain why a sequence with constant first differences has a linear general term. R
6. A student says the sequence 2, 4, 8, 16 must be 2^n because the first four terms fit. Find a different rule that also fits the first four terms, and explain what this shows. R
7. A tiling pattern uses 5 tiles in figure 1 and adds 3 tiles per figure. Write the rule and find the figure with 62 tiles. P
8. The nth term is $n^2 + 2$. Show that the difference between consecutive terms is $2n + 1$, and explain why this is always odd. E
9. Discover and justify a formula for the sum of the first n odd numbers. Test it for $n = 8$. E

An investigation is not a longer exercise. It is a short piece of mathematics you do yourself: you gather evidence, look for a pattern, say what you think is true, and then test it until you can defend it.

INVESTIGATION

A row of equilateral triangles is built from matchsticks, each new triangle sharing one side with the previous one. Figure 1 uses 3 matches, figure 2 uses 5, figure 3 uses 7.

The matchstick triangles

Figure n	Matches used	First difference	Matches = 2n + 1
1	3	<	3
2	5	2	5
3	7	2	7
4	9	2	9
5	11	2	11

1. **Build.** Draw figures 1 to 4 and count the matches directly. Record your counts in a table.
2. **Describe.** Describe in words how each figure is made from the previous one. How many matches are added at each step?
3. **Generalise.** Write a formula for the number of matches in figure n. Explain how each part of your formula corresponds to part of the diagram.
4. **Test.** Use your formula to predict figure 12 and figure 50. Then draw a sketch to check figure 12 by counting.
5. **Extend.** The pattern is rearranged so that each new triangle shares **two** sides with the previous one. Find the new formula and explain the difference.
6. **Justify.** Your formula was found from a finite number of cases. Write a justification, not just an observation, that it holds for every n.

Present your table, your two formulas with their justifications, and one sentence explaining why the second pattern grows more slowly.

APPLICATION

Painting a community hall

A community hall is to be painted by a team paid 9 units per hour. A volunteer has already painted one wall of area A square metres. The team paints at 14 m< per hour and needs 3 hours for preparation.

Wall area painted by volunteer (m▒	Team hours	Total cost (units)
0	3.00	27.00
28	5.00	45.00
56	7.00	63.00
70	8.00	72.00

1. Write the total cost as a formula in terms of A. Identify the constant term and explain what it represents.
2. The budget is 108 units. Rearrange your formula to make A the subject and find the largest wall area the volunteer can leave unpainted.
3. Explain what happens to the model if the team's rate changes to 10 m< per hour, and state which term of the formula changes.
4. The volunteer works more slowly than the team. Explain why the model above assumes her time is free, and whether that is a reasonable assumption for a decision.
5. Recommend a plan to the committee, with a reason, if their real objective is to finish in one weekend rather than to minimise cost.

CRITERION C COMMUNICATING

The growing square frame

A square frame of width 1 unit is built around a central square of side n units, using unit tiles.

1. Draw the frames for $n = 1$, 2 and 3, and count the tiles in each.
2. Write down at least two different expressions for the number of tiles, and explain what each expression counts.
3. Show algebraically that your two expressions are equivalent.
4. Explain how your working demonstrates that the two ways of counting are the same, and why this is an argument rather than a coincidence.

1–2 Counts correctly for some cases; the expressions contain errors or are not explained.

3–4 Finds correct expressions and shows they agree numerically.

5–6 Gives two correct expressions, explains what each counts, and shows equivalence algebraically.

7–8 As above, and articulates why the algebraic identity is the justification, distinguishing it from a check on cases.

Chapter summary and review

Key ideas

- A term is a product of numbers and variables; like terms combine, unlike terms do not.
- A function assigns exactly one output to each input in its domain.
- The domain excludes inputs that make an operation undefined.
- $(x + a)(x + b) = x^2 + (a + b)x + ab$ governs both expanding and factorising.
- Rearranging a formula does not change it; it changes the subject.
- A linear sequence has a constant first difference, and its general term is $dn + c$.
- A conjecture supported by cases is not a proof; a structural justification is.

Vocabulary

term
A product of numbers and variables, separated by + or − signs.

coefficient
The numerical factor of a term.

constant term
A term with no variable part.

expression
A sum of terms; it has no equals sign.

function
A rule assigning exactly one output to each input.

domain
The set of input values for which a function is defined.

range
The set of output values a function can produce.

expand
Multiply out brackets to remove them.

factorise
Write an expression as a product of factors.

subject
The variable that stands alone on one side of a formula.

sequence
An ordered list of values generated by a rule.

conjecture
A statement believed true from evidence but not yet proved.

What students often think	What is actually true
$(x + 3)^2 = x^2 + 9$.	The middle term is doubled: $x^2 + 6x + 9$.
$3x$ and x^3 are like terms.	They have different variable parts; only identical parts combine.

Misconceptions worth confronting directly.

What students often think	What is actually true
A pattern shown by four cases must continue.	Cases suggest a rule; only a structural argument establishes it.
$-3^2 = 9$.	The exponent applies to 3, so the value is -9; $(-3)^2 = 9$.

Revision checklist

- ☐ I can name the terms, coefficients and constant term of an expression.
- ☐ I can substitute positive, negative and fractional values correctly.
- ☐ I can evaluate a function and solve $f(x) = k$.
- ☐ I can state the domain of a function with a restriction.
- ☐ I can expand and factorise quadratics of the form $x^2 + bx + c$.
- ☐ I can rearrange a formula, including with a repeated variable.
- ☐ I can find the general term of a linear sequence.
- ☐ I can distinguish a conjecture from a proof.

REFLECT

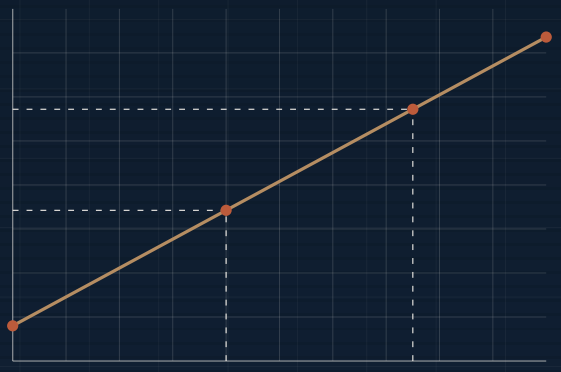
For each misconception in the table above, write the version you used to believe, and one sentence explaining why the correct version is true. If you never believed it, explain why another student might.

II

PART II

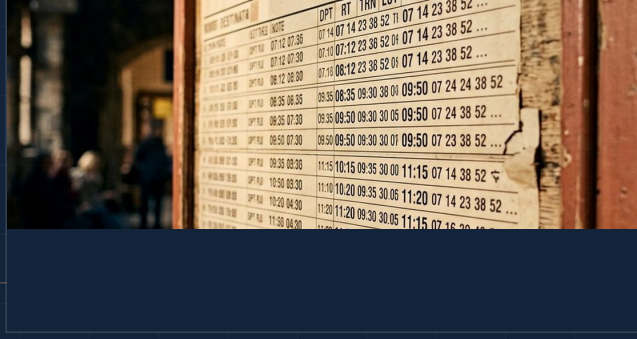
Relationships and equations

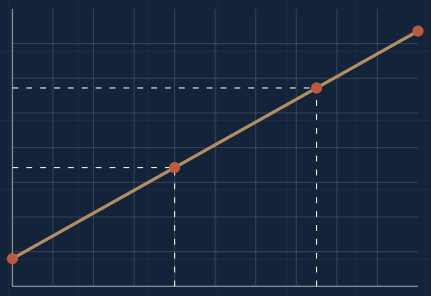
A relationship is a promise about how two quantities move together. Written as a table, a graph or an equation it is the same promise in three languages. This part teaches the translation, and then uses it to solve problems with more than one unknown.



04 Linear relationships

05 Equations, inequalities and systems





4

PATTERNS, CHANGE AND DECISION

Linear relationships

A straight line is the simplest way two quantities can be related. Because it is simple, it is everywhere: fares, wages, conversions, cooling, saving. Learning to read a line is learning to read a rate.

KEY CONCEPT ≡ RELATIONSHIPS | RELATED ≡ CHANGE | RELATED ≡ REPRESENTATION | CONTEXT ≡ FAIRNESS AND DEVELOPMENT

> INQUIRY
>
> When one quantity changes at a steady rate, what can a line tell us that a table cannot < and when does the line stop being a fair description of the situation?

ATL ≡ CRITICAL THINKING AND TRANSFER

Find two published tariffs that each charge a fixed fee plus a rate per unit. Plot both on the same axes and find the point where they are equal. Decide which is cheaper for a household like yours, and state the assumption your decision depends on.

MYP UNIT PLANNER ≣ THE WHOLE CHAPTER IN ONE VIEW

Framing the inquiry

Unit 4 ? approximately 16 hours of class time ? ATL focus: critical thinking

KEY AND RELATED CONCEPTS

Relationships. A linear relationship pairs each input with an output that changes by a constant amount. The constant amount is the gradient and it is the relationship's identity.

Change. Gradient is a rate: change in output per unit change in input. Reading a rate from a graph, a table or an equation should give the same number, in the same units.

Representation. A line can be given as a table, a graph, an equation, or a sentence. Moving between the four is the central skill, and each form hides something the others show.

GLOBAL CONTEXT AND EXPLORATION

Fairness and development. Tariffs, taxes, wages and subscriptions are linear models, and the choice between them is usually a question of fairness as much as arithmetic. A fixed charge that is small for a factory is large for a household; a rate that rewards volume penalises the small buyer. Modelling two options as two lines makes the argument about fairness precise enough to be settled with evidence.

STATEMENT OF INQUIRY

In order for a student to compare competing rates and decide between them on evidence, students must be able to move between tables, graphs and equations of a linear relationship and interpret gradient and intercept in the units of the context.

FACTUAL

- <What is the gradient of a line?
- <What does the *y*-intercept represent?
- <How do you find the equation of a line through two points?

CONCEPTUAL

- <Why does a constant rate of change always produce a straight line?

DEBATABLE

- <A model that fits the data perfectly for twelve months < should it be used to predict the next twelve?

Criterion	Strands	Evidence in this chapter
A ?Knowing and understanding	i, ii, iii	Find gradients and intercepts; write and rearrange the equation of a line; solve simultaneous linear problems graphically.
B ?Investigating patterns	i, ii, iii, iv	Investigate candle-burning data; test whether a linear model is adequate and justify the conclusion.
C ?Communicating	i, ii, iii	Interpret gradient and intercept in context; explain the choice of model and its limits.
D ?Applying mathematics	i, ii, iii, v	Model a real tariff, compare two options, and justify a recommendation.

Each teaching section closes with a graded practice set. The investigation addresses Criterion B; the application addresses Criterion D; the criterion task asks for extended written communication (Criterion C) about knowledge and understanding (Criterion A).

Learning intentions and success criteria

A linear model is a claim that something changes at a constant rate. Most of the work in this chapter is deciding whether that claim is true, and saying what it costs you when it is not.

YOU ARE LEARNING TO

- **Understand** that a constant rate of change produces a straight-line graph.
- **Know** how to calculate gradient from two points and from a graph.
- **Be able to** write the equation of a line in the form $y = mx + c$.
- **Be able to** find the equation of a line through two given points.
- **Be able to** interpret gradient and intercept in the units of a context.
- **Be able to** compare two linear models and justify a choice between them.

SUCCESS CRITERIA I CAN

- ☐ I can find the gradient of the line through (2, 5) and (6, 17).
- ☐ I can state the gradient and intercept of $3x + 2y = 12$ after rearranging it.
- ☐ I can write the equation of the line through (1, 4) and (3, 10).
- ☐ I can explain what the gradient means in a graph of cost against time.
- ☐ I can solve two linear equations graphically and check the solution algebraically.
- ☐ I can say when a linear model stops describing the data and why.

PREREQUISITE CHECK 10 MINUTES, GRAPH PAPER

1. Plot the points (1, 3), (2, 5), (3, 7) and describe the pattern.
2. Solve $4x + 3 = 19$.
3. Substitute $x = 2$ into $y = 3x - 1$ and state y.
4. Rearrange $2x + y = 7$ to make y the subject.
5. Calculate $(9 - 3)/(4 - 1)$.
6. Which is steeper, a line rising 2 for every 1 across, or one rising 3 for every 2 across?

Answers are in the chapter answer section. If more than two are incorrect, review plotting coordinates, substitution and rearranging before continuing.



A printed announcements board, read column by column. Every entry is dated, and the intervals between successive dates are what a reader uses to judge how quickly news is being posted ≣ the same idea as a constant difference in a sequence.
Original photograph made for this edition.

Explore: two ways to pay

Two phone plans are advertised. Plan A charges 8 units a month and 0.05 units per minute. Plan B charges 20 units a month and nothing per minute. Before reading further, work out when each is cheaper < and write down how you know.

YOUR TASK ≣ BEFORE YOU ARE TAUGHT

1. Make a table of monthly cost against minutes for 0, 50, 100, 200, 300 and 500 minutes for both plans.
2. Plot both sets of points on the same axes. Join each set with a straight line and explain why a straight line is appropriate.
3. Find from your graph the number of minutes at which the two plans cost the same. Read it as precisely as you can, then check it by solving an equation.
4. Write the monthly cost of each plan as a formula in terms of the minutes m. Which number in each formula is a rate, and which is a fixed charge?
5. Decide which plan is cheaper for a household using 180 minutes a month, and for one using 600 minutes a month. Justify each answer.
6. Write down one way the advertisement could mislead a reader even though every number in it is correct.

Section 4.3 gives this comparison its proper name: solving simultaneous equations. You have already met the idea as the intersection of two lines.

BEFORE YEAR 3	YEAR 3 THIS CHAPTER	LATER
▪ Plotting points in four quadrants. ▪ Substitution into simple expressions. ▪ Recognising that a proportional relationship doubles when the input doubles.	▪ Gradient and intercept of a line. ▪ Equations of lines and of parallel lines. ▪ Linear models of real situations. ▪ Solving two linear equations graphically.	▪ Simultaneous equations by elimination and substitution. ▪ Inequalities and linear programming. ▪ Rates of change in calculus.

Where this chapter sits in the sequence.

Inquiry pathway for this chapter

	Stage	What you do
1	Context	Two phone tariffs and a cooling cup of tea.
2	Question	How can a straight line describe a changing situation?
3	Concept	Gradient, intercept, rate of change, line of best fit.
4	Investigation	Burn a candle of known height and record its height over time.
5	Communication	Explain the gradient of a cost graph in units of the context.
6	Reflection	When does a constant rate stop being believable?

4.1 Gradient and rate of change

CRITERION A • RATE • GRADIENT

Gradient is a rate written as a single number. It answers the question <for each unit I go across, how far do I go up?<and its units are the units of the vertical axis divided by those of the horizontal.

For two points (x_1, y_1) and (x_2, y_2), the gradient is the change in y divided by the change in x. A positive gradient means the line rises left to right, a negative gradient means it falls, and a zero gradient means it is horizontal. A vertical line has no gradient, because the change in x is zero.

DEFINITION

Gradient $m = (y_2 - y_1)/(x_2 - x_1)$. It measures the change in y for each unit increase in x, in the units of the axes.

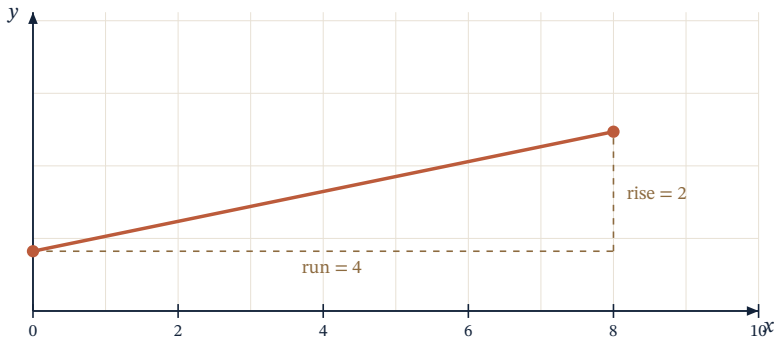


Figure 4.1 Gradient as a ratio of two changes. The triangles are similar on any part of the line, which is why the gradient does not depend on which two points you choose.

WORKED EXAMPLE 4.1.1 GRADIENT FROM TWO POINTS

PROBLEM

Find the gradient of the line through (2, 5) and (6, 17).

1. Identify the changes: y goes from 5 to 17, a change of 12.
2. x goes from 2 to 6, a change of 4.
3. Divide: $m = 12/4 = 3$.
4. The line rises 3 units for every 1 unit to the right.

ANSWER $m = 3$

Reversing the order gives (5 - 17)/(2 - 6) = -12/-4 = 3, the same value.

WORKED EXAMPLE 4.1.2 GRADIENT WITH A NEGATIVE

PROBLEM

A cup of tea cools from 88 °C to 64 °C in 8 minutes at a steady rate. Find the rate of cooling and interpret the sign.

1. The change in temperature is 64 - 88 = -24 °C.
2. The change in time is 8 minutes.
3. Gradient = -24/8 = -3 °C per minute.
4. The negative sign means the temperature is falling; the magnitude is the rate at which it falls.

ANSWER -3 °C per minute

After 8 minutes: 88 - 3 × 8 = 64 °C, as stated.

KEY IDEA

Units are part of the answer

A gradient always carries units, and the units are what make it interpretable. "The gradient is 3" means nothing until you can say three what per what. In a cost graph it might be 3 units per hour; in a distance graph, 3 metres per second.

COMMON ERROR

Upside-down gradient

Calculating $(x_2 - x_1)/(y_2 - y_1)$ and calling it the gradient. The vertical change goes on top: gradient is rise over run, not run over rise.

PRACTICE 4.1 GRADIENT AND RATE OF CHANGE

SUPPORT 1–4 CORE 1–6 EXTENSION 7–9

1. Find the gradient of the line through: (a) (1, 2) and (3, 8) (b) (0, 5) and (4, 1) (c) (-2, 3) and (2, 11). F
2. State whether each gradient is positive, negative or zero, and sketch the line: (a) $m = -2$ (b) $m = 0$ (c) $m = 0.5$. F
3. A cyclist travels 45 km in 3 hours at a steady speed. Find the gradient of the distance–time graph and give its units. F
4. A savings account holds 240 units and grows by 15 units a month. Find the rate of change and state whether the gradient is positive. F
5. Explain why a vertical line has no gradient, using the formula for m and the meaning of division by zero. R
6. Two lines have gradients 2 and -1/2. Draw both through the same point and describe how they are related. R
7. The temperature of a cooling object falls from 96 °C to 72 °C in 6 minutes. Assuming a constant rate, find when it reaches 40 °C. P
8. A graph shows distance against time for a walker who stops for ten minutes. Describe what the gradient does during the stop and explain what a gradient of zero means physically. P

4.2 The equation of a straight line

CRITERION A I • LINEAR EQUATION • INTERCEPT

Two numbers describe a straight line completely: how steep it is and where it crosses the vertical axis. Written as $y = mx + c$, they are both visible at once.

In $y = mx + c$, m is the gradient and c is the y-intercept, the value of y when $x = 0$. The intercept is where the line meets the vertical axis, and in an applied problem it usually represents a fixed cost, a starting value or an initial condition.

DEFINITION

The **gradient–intercept form** of a linear equation is $y = mx + c$, where m is the gradient and c the y-intercept. Two lines are **parallel** exactly when their gradients are equal.

WORKED EXAMPLE 4.2.1 READING GRADIENT AND INTERCEPT

PROBLEM

Find the gradient and y-intercept of $3x + 2y = 12$.

1. Rearrange to make y the subject: $2y = -3x + 12$.
2. Divide every term by 2: $y = -1.5x + 6$.
3. Compare with $y = mx + c$: $m = -1.5$ and $c = 6$.

ANSWER gradient -1.5; y-intercept 6

Set $x = 0$: $3(0) + 2y = 12$ gives $y = 6$. Set $y = 0$: $x = 4$, so the line crosses the axes at $(4, 0)$ and $(0, 6)$.

WORKED EXAMPLE 4.2.2 LINE THROUGH TWO POINTS

PROBLEM

Find the equation of the line through (1, 4) and (3, 10).

1. Find the gradient: $m = (10 - 4)/(3 - 1) = 6/2 = 3$.
2. Substitute one point into $y = 3x + c$: $4 = 3(1) + c$.
3. Solve for c: $c = 1$.
4. The equation is $y = 3x + 1$.

ANSWER $y = 3x + 1$

Test the other point: $3(3) + 1 = 10$, and the line passes through (3, 10).

KEY IDEA

One line, one equation

A line through two points is unique, so any method that uses both points correctly must give the same equation. If two methods give different answers, one of them has an arithmetic error < the disagreement is a check, not a matter of taste.

COMMON ERROR

Intercept taken before dividing

Reading c from an equation that is not in the form $y = mx + c$. In $2y = 6x + 8$ the intercept is **not** 8: divide first, giving $y = 3x + 4$.

PRACTICE 4.2 EQUATION OF A LINE

SUPPORT 1–4 CORE 1–7 EXTENSION 8–9

1. State the gradient and y-intercept: (a) $y = 2x + 5$ (b) $y = -x + 1$ (c) $y = 0.5x - 3$. F
2. Rearrange to the form $y = mx + c$: (a) $x + y = 6$ (b) $2x + y = 9$ (c) $4x - 2y = 10$. F
3. Write the equation of the line with gradient 4 passing through (0, -2). F
4. Write the equation of the line through: (a) (0, 1) and (2, 5) (b) (1, 7) and (4, 1). F
5. State whether each pair of lines is parallel: (a) $y = 3x + 1$ and $y = 3x - 4$ (b) $y = 2x$ and $y = -2x$ (c) $2y = 4x + 1$ and $y = 2x - 3$. R
6. Explain what the y-intercept represents in a graph of the cost of a taxi ride against distance. R

PRACTICE 4.2 EQUATION OF A LINE (CONTINUED)

SUPPORT 1–4 CORE 1–7 EXTENSION 8–9

1. A line passes through (2, -1) and is parallel to $y = 3x + 2$. Find its equation. P
2. A line passes through (1, 4) and (5, 4). Find its equation and explain what kind of line it is. P
3. Find the equation of the line through (-2, 5) and (4, -7), then find where it crosses both axes. E

4.3 Linear models and simultaneous lines

CRITERION A II • MODELLING • INTERSECTION

Two linear models can be compared by looking for where they cross. The crossing point is the value at which the two situations are equal, and it is usually the decision point of the problem.

When two lines are drawn on the same axes, the coordinates of their intersection satisfy both equations at once. That pair of values is the **simultaneous solution**. On a graph it can be read off; algebraically it can be found by substitution, which is exact where a reading from a graph is approximate.

KEY IDEA

Models have a domain of validity

A model is a simplification with a purpose. A linear model claims that a quantity changes at a constant rate < that the cost per unit never changes, that the cooling rate is steady, that the population grows by the same amount each year. The claim is usually reasonable over a limited range and false outside it. Stating the range is part of building the model, not an apology for it.

WORKED EXAMPLE 4.3.1 SOLVING GRAPHICALLY AND

PROBLEM

Solve the pair $y = 2x + 1$ and $y = -x + 7$.

1. At the intersection both expressions for y are equal, so $2x + 1 = -x + 7$.
2. Add x to both sides: $3x + 1 = 7$.
3. Subtract 1: $3x = 6$, so $x = 2$.
4. Substitute back: $y = 2(2) + 1 = 5$.

ANSWER $x = 2, y = 5$, that is the point $(2, 5)$

Both equations give $y = 5$ at $x = 2$: $-2 + 7 = 5$. On a graph the lines cross at $(2, 5)$.

WORKED EXAMPLE 4.3.2 A BREAK-EVEN COMPARISON

PROBLEM

Plan A costs $8 + 0.05m$ and Plan B costs 20, where m is minutes used. For how many minutes do they cost the same, and which is cheaper below that?

1. Set the costs equal: $8 + 0.05m = 20$.
2. Subtract 8: $0.05m = 12$.
3. Divide by 0.05: $m = 240$.
4. At 240 minutes both cost 20 units. Below 240 minutes Plan A is cheaper, because its fixed charge is lower.

ANSWER equal at 240 minutes; Plan A is cheaper below that

Try $m = 200$: Plan A $8 + 10 = 18$, Plan B 20. Try $m = 300$: Plan A 23, Plan B 20.

Minutes	Plan A	Plan B	Cheaper
100	13.00	20.00	A
200	18.00	20.00	A
240	20.00	20.00	equal
400	28.00	20.00	B

A table is a discrete model of a continuous relationship; the lines fill in the values the table leaves out.

COMMON ERROR

Treating a reading as exact

Solving simultaneous equations by reading a graph and reporting a value that is not on the grid lines. A graphical solution is approximate unless the crossing falls exactly on a grid intersection; substitute the answer into both equations to confirm it.

PRACTICE 4.3 LINEAR MODELS AND SIMULTANEOUS LINES

SUPPORT 1–3 CORE 1–6 EXTENSION 7–8

1. Solve by substitution: (a) $y = x + 3$ and $y = 2x + 1$ (b) $y = 3x - 2$ and $y = -x + 6$. F
2. Find where $y = 2x + 4$ crosses the x-axis and the y-axis. F
3. A hire company charges $15 + 3h$ and another charges $30 + 1.5h$. Find the number of hours at which they charge the same. F

PRACTICE 4.3 LINEAR MODELS AND SIMULTANEOUS LINES (CONTINUED)

SUPPORT 1 CORE 1 EXTENSION 7

1. Sketch $y = 2x - 1$ and $y = 6 - x$ on the same axes and read off the intersection; check your reading algebraically. R
2. Explain why two lines with the same gradient but different intercepts never meet, and what that means for two tariffs that have the same rate but different fixed charges. R
3. A gym charges 40 units joining fee plus 18 units per month; a second charges 100 units joining fee plus 8 units per month. Model both, find the break-even month, and recommend a choice for a member who will stay two years. P
4. A candle is 24 cm tall and burns at 1.5 cm per hour. Write a linear model for its height and find when it goes out. State the domain over which the model is valid. E
5. Two lines are $y = mx + 2$ and $y = 3x + c$. Find values of m and c for which the lines (a) are parallel, (b) meet at $(1, 5)$. E

4.4 Linear models from data

CRITERION B, II • MODELLING • FIT

Real data never lie exactly on a line. Fitting a line is the art of choosing the line that describes the trend without pretending the scatter is not there.

When data are plotted and the points lie roughly along a straight path, a line of best fit can summarise the trend. The line does not have to pass through any of the points; it should pass through the middle of them, with roughly equal numbers above and below. Its gradient estimates the rate, and its intercept estimates the starting value.

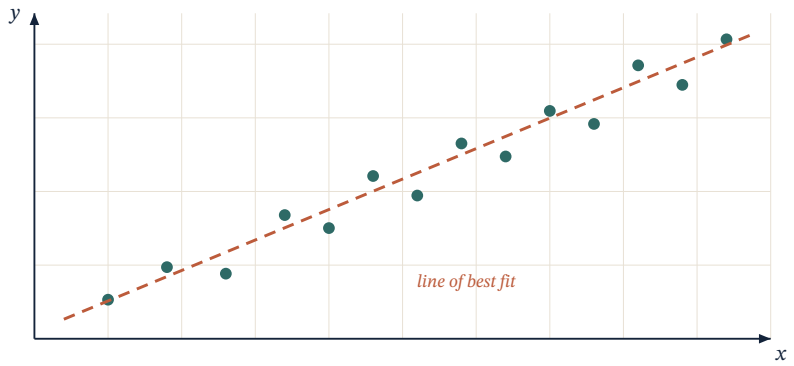


Figure 4.2 A line of best fit is chosen for the trend, not for the points. Moving the line to pass through one unusual point would misrepresent every other one.

KEY IDEA

Inside the data, and beyond it

Interpolation is estimating inside the range of the data; extrapolation is estimating outside it. Interpolation is usually defensible because the trend has been observed there. Extrapolation is a bet that the trend continues, and the further outside the range you go, the larger the bet.

WORKED EXAMPLE 4.4.1 FITTING A LINE AND USING IT

PROBLEM

A candle's height is measured every 20 minutes: 24.0, 21.2, 18.1, 15.3, 12.4, 9.6 cm. Fit a linear model and predict when the candle goes out.

1. The heights fall by about 2.8 cm every 20 minutes, so the gradient is about -2.8/20 = -0.14 cm per minute.
2. The intercept is the starting height, about 24.0 cm.
3. The model is $h = 24.0 - 0.14t$ for t in minutes.
4. The candle goes out when $h = 0$: $0.14t = 24.0$, so t ≈ 171 minutes.

ANSWER about 171 minutes, that is just under 3 hours

At 140 minutes the model gives 24.0 - 19.6 = 4.4 cm, and at 180 minutes it gives -1.2 cm < the negative value shows the model has been used past its domain.

COMMON ERROR

Precision borrowed from the calculator

Reporting a prediction from a fitted line to four decimal places. A line fitted by eye carries an uncertainty of at least a whole grid square; quoting 171.4286 minutes claims a precision the method cannot support.

NOTE

Correlation is not cause

A correlation between two quantities does not by itself show that one causes the other. Ice-cream sales and drowning incidents both rise in hot weather; neither causes the other.

PRACTICE 4.4 MODELS FROM DATA

SUPPORT 1 CORE 1 EXTENSION 7

1. Plot these points and sketch a line of best fit: (1, 2.1), (2, 3.9), (3, 6.2), (4, 7.8), (5, 10.1). Estimate the gradient. F
2. A machine's value falls from 12 000 units to 9 000 units over three years. Find the annual rate of depreciation. F
3. For the model $h = 24.0 - 0.14t$, find h at $t = 60$ and explain why $t = 200$ is outside its domain. F
4. Explain the difference between interpolation and extrapolation, using a graph you have drawn. R

PRACTICE 4.4 MODELS FROM DATA (CONTINUED)

SUPPORT 1 CORE 1 EXTENSION 7

1. A student fits a line through the first and last data points only. Explain why this may not be the best line of fit. R
2. A phone's battery falls from 100 per cent to 62 per cent in 90 minutes of video. Model the battery level and predict the remaining watch time. State one reason the prediction may be wrong. P
3. Data show that towns with more libraries have higher average incomes. Explain why this does not show that libraries raise incomes, and suggest a third variable. E
4. A model predicts a factory's output by $P = 1200 + 45t$ units per month. After 14 months the output is 1500 units. Evaluate how well the model has performed and suggest why. E

4.5 Inverse proportion and non-linear relations

CRITERION AV • PROPORTION • CHANGE

Not every relationship is linear. When doubling one quantity halves another, the graph is a curve and the rate of change is not constant < which is exactly what makes it worth recognising.

Two quantities are in **direct proportion** when their ratio is constant: doubling one doubles the other, and the graph is a straight line through the origin. They are in **inverse proportion** when their product is constant: doubling one halves the other, and the graph is a curve that never touches either axis.

DEFINITION

Direct proportion: $y = kx$. Inverse proportion: $y = k/x$, equivalently $xy = k$. In both cases k is the constant of proportion, found from one known pair of values.

Workers	Days needed	Workers × days
2	36	72
3	24	72
4	18	72
6	12	72
8	9	72

Inverse proportion: the product is constant, so the graph of days against workers is a curve, not a line.

WORKED EXAMPLE 4.5.1 USING INVERSE PROPORTION

PROBLEM

Six workers take 12 days to lay a floor. How long would eight workers take, at the same rate?

1. The work is constant, so *workers* × *days* = k.
2. From the known pair, $k = 6 \times 12 = 72$ worker-days.
3. For 8 workers: $8 \times d = 72$.
4. So $d = 9$ days.

ANSWER 9 days

The answer is less than 12 days, as it must be if more workers are employed.

WORKED EXAMPLE 4.5.2 WHY THE GRAPH CURVES

PROBLEM

Sketch $y = 12/x$ for x from 1 to 12 and explain why the gradient changes.

1. Make a table: (1, 12), (2, 6), (3, 4), (4, 3), (6, 2), (12, 1).
2. From 1 to 2 the value falls by 6; from 2 to 3 it falls by 2.
3. From 6 to 12 it falls by only 1 over six units.
4. The gradient is steep near the vertical axis and flattens as x increases; it is never constant.

ANSWER a falling curve, steep at first and flattening

The product $xy = 12$ at every point, which is what inverse proportion means.

COMMON ERROR

Reversing the direction of change

Assuming that if 4 workers take 10 days then 6 workers take 15 days. Adding workers cannot increase the time. Inverse proportion means more workers take fewer days.

PRACTICE 4.5 INVERSE PROPORTION

SUPPORT 1–3 CORE 1–5 EXTENSION 7–8

1. State whether each pair is in direct or inverse proportion: (a) cost and number of items at a fixed price (b) speed and time for a fixed journey (c) area and side of a square. F

PRACTICE 4.5 INVERSE PROPORTION (CONTINUED)

SUPPORT 1 CORE 1 EXTENSION 7

1. y is inversely proportional to x, and $y = 5$ when $x = 4$. Find y when $x = 10$. F
2. 4 taps fill a tank in 30 minutes. How long will 6 taps take? F
3. Sketch the graph of $y = 20/x$ for x from 2 to 10. F
4. Explain why the graph of an inverse proportion never touches either axis. R
5. A car travelling at 80 km/h takes 3 hours for a journey. How long would it take at 60 km/h? Explain why the answer is not 4 hours. P
6. The time to complete a job is inversely proportional to the number of workers. With 5 workers it takes 18 days. Find the constant and the time with 9 workers. P
7. Explain why ◁the more workers, the faster the job<stops being true in a real factory, and what that says about the model's domain. E

An investigation is not a longer exercise. It is a short piece of mathematics you do yourself: you gather evidence, look for a pattern, say what you think is true, and then test it until you can defend it.

INVESTIGATION

A candle 24.0 cm tall is lit and its height measured every 20 minutes. The question is not <what is the formula?<but <is a linear model adequate, and how would you know?<

The candle clock

Time (min)	Height (cm)	Change from previous (cm)
0	24.0	<
20	21.2	?2.8
40	18.1	?3.1
60	15.3	?2.8
80	12.4	?2.9
100	9.6	?2.8

1. **Plot.** Draw a scatter graph of height against time, with time on the horizontal axis. Choose a scale that lets you read to the nearest 0.5 cm.
2. **Fit.** Draw a line of best fit by eye. Record its gradient and intercept, and give both in units.
3. **Test.** Calculate what your model predicts at 60 minutes and compare with the measurement. Explain whether the difference is within reading error.
4. **Interpret.** Explain what the gradient means physically, and why it must be negative for a burning candle.
5. **Extend.** Use your model to predict when the candle will go out. State how far beyond the data you have extrapolated.
6. **Conclude.** Decide whether the linear model is adequate for this candle, and justify your decision with two pieces of evidence from your table or graph.

Present the graph with the fitted line, the model with units, the comparison at 60 minutes, and a conclusion that says clearly what the model supports and what it does not.

APPLICATION

Choosing a water tariff

A municipality offers two water tariffs. Tariff X: a fixed charge of 12 units per quarter plus 1.4 units per cubic metre. Tariff Y: no fixed charge but 2.6 units per cubic metre. A household uses about 18 m<a quarter.

Use (m	Tariff X	Tariff Y	Difference (X Y)
0	12.00	0.00	+12.00
5	19.00	13.00	+6.00
10	26.00	26.00	0.00
18	37.20	46.80	9.60
30	54.00	78.00	24.00

1. Write each tariff as a linear model in terms of u, the cubic metres used, and explain what the intercept of each represents.
2. Find algebraically the usage at which the tariffs cost the same, and check your answer with the table.
3. Recommend a tariff for the household described, for a small business using 30 m< and for a household that uses 6 m< Justify each.
4. A pensioner's household uses 4 m<a quarter. Discuss which tariff is fairer to that household, and whether <fairer<is the same as <cheaper<
5. The municipality raises Tariff X's fixed charge to 20 units. Recalculate the break-even point and explain how the change affects low users more than high users, using your model as evidence.

CRITERION C COMMUNICATING

Explaining a rate to someone who distrusts graphs

A community newsletter has published a graph of local bus fares rising over five years. A reader has written to say the graph is misleading. Write a one-page reply that explains what the graph does and does not show.

1. Explain what the gradient of the published graph represents, in the units of the axes, and what its intercept represents.
2. Identify two features of a linear graph that can make a change look larger or smaller than it is, and show how each would change the reader's impression.
3. Explain the difference between reading a value from the line inside the data and predicting beyond it, using the bus fare data as your example.
4. Write the reply in plain language. Your reader is intelligent but has no training in mathematics, and is suspicious of being persuaded by a picture.

1–2 Describes the graph; the interpretation of gradient or intercept is wrong or missing.

3–4 Interprets gradient and intercept correctly and identifies a misleading feature.

5–6 Explains both features clearly, distinguishes interpolation from extrapolation, and writes for the stated audience.

7–8 As above, and shows how the choice of axes changes the claim the graph makes, without overstating what can be concluded.

Chapter summary and review

Key ideas

- Gradient is the change in y per unit change in x, with units.
- A constant rate of change produces a straight line.
- $y = mx + c$ shows gradient and y-intercept directly.
- Two lines are parallel exactly when their gradients are equal.
- The intersection of two lines is the simultaneous solution.
- A line of best fit describes a trend and need not pass through any point.
- Interpolation is inside the data; extrapolation is beyond it and is a bet.
- Direct proportion is $y = kx$; inverse proportion is $y = k/x$.

Vocabulary

gradient
The change in the vertical quantity per unit change in the horizontal one.

intercept
Where a line crosses an axis; the y-intercept is the value when $x = 0$.

rate of change
How quickly one quantity changes relative to another.

simultaneous solution
A pair of values satisfying two equations at once.

line of best fit
A line describing the trend of scattered data.

interpolation
Estimating between known data values.

extrapolation
Estimating beyond the known data values.

direct proportion
$y = kx$: the ratio is constant.

inverse proportion
$y = k/x$: the product is constant.

domain of validity
The range over which a model can reasonably be trusted.

What students often think	What is actually true
Gradient is run over rise.	It is rise over run: the vertical change divided by the horizontal change.
The intercept of $2y = 6x + 8$ is 8.	Divide first; the intercept is 4.
A line of best fit must pass through all the points.	It passes through the middle of them, not through each one.

Misconceptions worth confronting directly.

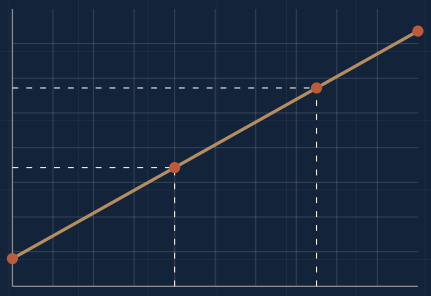
What students often think	What is actually true
If a model fits the data, it will predict the future.	A fit inside the data says nothing certain about behaviour outside it.
Doubling the workers always halves the time.	That is the inverse-proportion model; real work has a limit to how many people can work productively at once.

Revision checklist

- ☐ I can calculate a gradient from two points and from a graph.
- ☐ I can state the units of a gradient in context.
- ☐ I can rearrange a linear equation into $y = mx + c$.
- ☐ I can find the equation of a line through two points.
- ☐ I can find a simultaneous solution graphically and algebraically.
- ☐ I can fit a line to data and use it to interpolate.
- ☐ I can explain why an extrapolation is less reliable than an interpolation.
- ☐ I can recognise direct and inverse proportion and use each.

> **REFLECT**
>
> Choose one graph from a newspaper or a website. Identify its gradient in the units shown, state what the intercept represents, and write two sentences on what the graph does not tell you.





5

PATTERNS, CHANGE AND DECISION

Equations, inequalities and systems

An equation is a question with a hidden answer. An inequality is a question with a range of answers. Most decisions need the second: not ‹how much exactly?‹ but ‹how much is enough?‹

KEY CONCEPT ≣RELATIONSHIPS

RELATED ≣EQUIVALENCE

RELATED ≣CHANGE

CONTEXT ≣FAIRNESS AND DEVELOPMENT

INQUIRY

What does it mean to solve an inequality, and why does multiplying by a negative number change the answer set?

ATL ≣THINKING AND SELF-MANAGEMENT SKILLS

Set yourself six equations of increasing difficulty and time how long each takes. Identify the step that costs you most time across all six, and practise only that step. Record your time again after a week.

MYP UNIT PLANNER THE WHOLE CHAPTER IN ONE VIEW

Framing the inquiry

Unit 5 approximately 16 hours of class time ATL focus: thinking and self-management

KEY AND RELATED CONCEPTS

Relationships. An equation states that two expressions are equal; its solution is the value that makes the statement true. An inequality states an ordering and its solution is a set.

Equivalence. Each step in solving must produce an equivalent statement. Adding the same quantity preserves equivalence; multiplying by zero does not, which is why that step is forbidden.

Change. A system of two equations describes two constraints at once. Its solution is the point at which both hold, and its absence of solution is itself information.

GLOBAL CONTEXT AND EXPLORATION

Fairness and development. A budget is an inequality. A minimum wage, a maximum load, a fair allocation of a limited supply: each is a constraint that must be satisfied at the same time as others. Solving systems of equations and inequalities is how a committee turns several competing requirements into a feasible plan it can defend.

STATEMENT OF INQUIRY

In order for a student to express and satisfy several constraints at once, students must be able to solve linear equations and inequalities, use the balance principle to justify each step, and interpret the solution set in the language of the problem.

FACTUAL

<What is the balance principle?

<What happens to an inequality when both sides are multiplied by a negative number?

<How many solutions does a linear equation have?

CONCEPTUAL

<Why does adding the same quantity to both sides of an equation preserve the solution, while squaring both sides may not?

DEBATABLE

<Is a solution set with no values a failure of the mathematics or a result about the situation?

Criterion	Strands	Evidence in this chapter
A Knowing and understanding	i, ii, iii	Solve linear equations and inequalities; rearrange formulas; solve systems.
B Investigating patterns	i, ii, iii	Investigate when a system has no solution or infinitely many, and describe the conditions on the coefficients.
C Communicating	i, ii, iv	Justify each step of a solution; explain what a solution set means in context.
D Applying mathematics	i, ii, iii, v	Model a budget as an inequality, solve it, and justify the plan.

Each teaching section closes with a graded practice set. The investigation addresses Criterion B; the application addresses Criterion D; the criterion task asks for extended written communication (Criterion C) about knowledge and understanding (Criterion A).

Learning intentions and success criteria

The purpose of these techniques is not speed. It is to be able to say, at every step, why the new statement has exactly the same solutions as the old one.

YOU ARE LEARNING TO

- **Understand** the balance principle and why each step must be reversible.
- **Be able to** solve linear equations, including those with brackets and fractions.
- **Be able to** solve linear inequalities and represent the solution on a number line.
- **Know** and be able to explain the rule for multiplying an inequality by a negative number.
- **Be able to** solve a pair of linear equations by elimination and by substitution.
- **Understand** what it means for a system to have no solution or infinitely many.

SUCCESS CRITERIA I CAN

- ☐ I can solve $3(x - 2) = 4x + 5$ and check the answer in the original equation.
- ☐ I can solve $(x + 1)/3 - (x - 2)/2 = 1$.
- ☐ I can solve $4 - 2x \leq 10$ and represent the solution on a number line.
- ☐ I can explain with numbers why $-2x < 6$ gives $x > -3$.
- ☐ I can solve $2x + 3y = 12$ and $x - y = 1$ by elimination.
- ☐ I can give an example of a system with no solution and explain why.

PREREQUISITE CHECK 10 MINUTES, NO CALCULATOR

1. Solve $x + 7 = 12$.
2. Solve $3x = 21$.
3. Solve $2x - 5 = 9$.
4. Expand $4(x - 3)$.
5. Solve $-2x = 8$.
6. Simplify $(6x)/3$.

Answers are in the chapter answer section. If more than two are incorrect, review one- and two-step equations and expanding brackets before continuing.



Two measuring jugs and a set of balance weights. Every step in solving an equation is an assertion that the balance has not been disturbed. Original photograph made for this edition.

Explore: the heaviest box

A delivery van may carry at most 900 kg. A driver has already loaded four crates of 85 kg each. How many more 120 kg boxes can be loaded? Answer first without algebra, then with it.

YOUR TASK BEFORE YOU ARE TAUGHT

1. Work out the answer by arithmetic, showing each step. Record the number of boxes that fits and the number that does not.
2. Now write the condition as an inequality in b, the number of boxes, and solve it. Compare with your arithmetic answer.
3. Explain why the answer must be a whole number here, and where the rounding decision enters the calculation.
4. Change the crate weight to 95 kg. Re-solve and explain how the largest number of boxes changes and why.
5. Write a second condition: no single box may exceed 150 kg, and the total may not exceed 900 kg. Describe with a table which numbers of boxes satisfy both conditions.
6. Write down one question about loading that inequality 1 alone cannot answer.

Section 5.5 turns this into the algebra of inequalities, including the one rule that surprises everyone.

BEFORE YEAR 3	YEAR 3 THIS CHAPTER	LATER
▪ Solving one-step equations by inverse operations. ▪ Substituting values into expressions. ▪ Ordering integers on a number line.	▪ Equations with brackets and fractions. ▪ Linear inequalities and the number-line representation. ▪ Simultaneous equations by elimination and substitution. ▪ Conditions for no solution or infinitely many.	▪ Quadratic equations and factorisation. ▪ Linear programming with several constraints. ▪ Systems of equations in three variables.

Where this chapter sits in the sequence.

Inquiry pathway for this chapter

	Stage	What you do
1	Context	A loading limit and a community budget.
2	Question	How do you describe every value that satisfies a condition?
3	Concept	Equivalence, solution set, constraint, feasibility.
4	Investigation	When do two equations have no solution or infinitely many?
5	Communication	Justify every step with the balance principle.
6	Reflection	Why does multiplying by a negative reverse the ordering?

5.1 Solving linear equations

CRITERION A • EQUIVALENCE • BALANCE

Solving an equation is a sequence of equivalent statements. Each step must leave the solution set unchanged, and the reason is always the same: what is done to one side is done to the other.

The **balance principle** states that adding, subtracting, multiplying or dividing both sides of an equation by the same nonzero quantity produces an equivalent equation. Solving means applying it until the variable stands alone. The order of operations is undone in reverse: first the additions and subtractions, then the multiplications and divisions.

DEFINITION

Two equations are **equivalent** when they have exactly the same solutions. A linear equation in one variable has exactly one solution, unless the variable cancels, in which case it has none or infinitely many.

WORKED EXAMPLE 5.1.1 AN EQUATION WITH BRACKETS

PROBLEM

Solve $3(x - 2) = 4x + 5$.

1. Expand the bracket: $3x - 6 = 4x + 5$.
2. Subtract $3x$ from both sides: $-6 = x + 5$.
3. Subtract 5: $x = -11$.
4. Check in the original: left side $3(-13) = -39$; right side $4(-11) + 5 = -39$.

ANSWER $x = -11$

Both sides agree, so the solution is correct.

WORKED EXAMPLE 5.1.2 AN EQUATION WITH FRACTIONS

PROBLEM

Solve $(x + 1)/3 - (x - 2)/2 = 1$.

1. Multiply every term by 6, the lowest common denominator: $2(x + 1) - 3(x - 2) = 6$.
2. Expand: $2x + 2 - 3x + 6 = 6$.
3. Collect: $-x + 8 = 6$.
4. Solve: $x = 2$.
5. Check: $3/3 - 0/2 = 1$, correct.

ANSWER $x = 2$

Multiplying by the common denominator is a reversible step because the denominator is never zero.

COMMON ERROR

Half a multiplication

Multiplying only the first fraction by the common denominator. Every term must be multiplied, including the term on the right of the equals sign. In $(x+1)/3 - (x-2)/2 = 1$, the 1 becomes 6, not 1.

NOTE

When the variable disappears

When the variable cancels, look at what remains. $2x + 3 = 2x + 7$ reduces to $3 = 7$, which is false, so there is no solution. $2x + 3 = 2x + 3$ reduces to a true statement, so every value of x is a solution.

PRACTICE 5.1 LINEAR EQUATIONS

SUPPORT 1–4 CORE 1–7 EXTENSION 8–9

1. Solve: (a) $5x - 3 = 17$ (b) $4 - 2x = 10$ (c) $x/3 + 2 = 7$. F
2. Solve: (a) $2(x + 3) = 14$ (b) $5(x - 1) = 3x + 7$ (c) $4(2x + 1) = 3(x - 2)$. F
3. Solve: (a) $x/2 + x/3 = 5$ (b) $(x + 1)/4 = 3$ (c) $(2x - 1)/5 = (x + 2)/3$. F
4. Solve and check: $7 - 3(x - 2) = 2x + 1$. F
5. Explain why multiplying both sides by zero is not a permitted step, using a specific equation as an example. R
6. A student solves $3x + 6 = 12$ by dividing only the left side by 3. Show what goes wrong and give the correct solution. R

PRACTICE 5.1 LINEAR EQUATIONS (CONTINUED)

SUPPORT 1–4 CORE 1–7 EXTENSION 8–9

1. The perimeter of a rectangle is 46 cm and its length is 5 cm more than its width. Write an equation and solve it. P
2. Solve $2(3x - 1) - 3(2x + 1) = 4$ and describe what the result tells you about the equation. E
3. Find a value of k for which $kx + 5 = 3x + 5$ has no solution, and a value for which it has infinitely many. E

5.2 Forming equations from problems

CRITERION D, II • MODELLING • UNKNOWN

The difficult part of a word problem is not the algebra; it is deciding what the letter stands for. Choose the variable carefully and the equation usually writes itself.

A reliable sequence: name the unknown and state its units; write the relations the problem gives in words; translate each relation into an expression; write the equation; solve; check that the answer is physically sensible as well as algebraically correct.

KEY IDEA

Check against the situation

Always check the answer against the *situation*, not only against the equation. A negative length or a fractional number of people may satisfy the algebra while being impossible in the problem, which means the model has been applied outside its domain.

WORKED EXAMPLE 5.2.1 CONSECUTIVE NUMBERS

PROBLEM

The sum of three consecutive integers is 87. Find them.

1. Let the middle integer be n. The others are $n - 1$ and $n + 1$.
2. Write the sum: $(n - 1) + n + (n + 1) = 87$.
3. Simplify: $3n = 87$.
4. So $n = 29$, and the integers are 28, 29 and 30.

ANSWER 28, 29 and 30

$28 + 29 + 30 = 87$, and they are consecutive.

WORKED EXAMPLE 5.2.2 A PROBLEM WITH TWO CONDITIONS

PROBLEM

A theatre sells adult tickets at 12 units and child tickets at 7 units. On one night 240 tickets are sold for 2 430 units. How many of each were sold?

1. Let a be adult tickets and c child tickets.
2. Tickets: $a + c = 240$.
3. Money: $12a + 7c = 2430$.
4. From the first, $c = 240 - a$. Substitute: $12a + 7(240 - a) = 2430$.
5. Expand and collect: $5a + 1680 = 2430$, so $5a = 750$ and $a = 150$.
6. Then $c = 90$.

ANSWER 150 adult and 90 child tickets

$150 + 90 = 240$ and $12(150) + 7(90) = 1800 + 630 = 2430$. Both conditions hold.

COMMON ERROR

A variable with no content

Defining the variable as ‹the answer‹ ‹Let x be the number of adult tickets‹ is usable; ‹let x be the answer‹is not, because nothing can be written about it.

PRACTICE 5.2 FORMING EQUATIONS

SUPPORT 1–3 CORE 1–6 EXTENSION 7–8

1. The sum of two consecutive integers is 65. Find them. F
2. A number is multiplied by 4 and then 9 is subtracted; the result is 31. Find the number. F
3. A rectangle's length is three times its width and its perimeter is 64 cm. Find both dimensions. F
4. A father is four times as old as his son. In 12 years he will be twice as old. Find their present ages. R
5. Explain why a negative answer to a question about a number of people must be rejected even if it solves the equation. R
6. Tickets cost 15 units for adults and 9 for children. 320 tickets raise 3 660 units. Find the numbers of each sold. P
7. Two numbers differ by 14 and their sum is 96. Find them, then verify both conditions. P
8. A shop sells small and large boxes. 30 boxes weigh 138 kg; a small box weighs 3 kg and a large 6 kg. Find how many of each were sold, and explain what goes wrong if the total is 139 kg instead. E

5.3 Linear inequalities

CRITERION A • ORDER • SOLUTION SET

An inequality has a solution set, not a solution. Every step that preserves order is allowed < except multiplying or dividing by a negative number, which reverses it.

The symbols <, ?, > and ? describe order. Adding or subtracting the same quantity on both sides preserves the order, and so does multiplying or dividing by a positive number. Multiplying or dividing by a negative number **reverses** the order, because the number line is reflected about zero.

KEY IDEA

The reversal rule, tested

$-2x < 6$ does not give $x < -3$. Test $x = 4$: $-8 < 6$ is true, and 4 is greater than -3. Dividing by -2 reverses the inequality, giving $x > -3$. The test value settles the direction in one line.

WORKED EXAMPLE 5.3.1 A TWO-STEP INEQUALITY

PROBLEM

Solve $4 - 2x$? 10 and show the solution on a number line.

1. Subtract 4 from both sides: $-2x$? 6.
2. Divide by -2 and reverse the sign: x ? -3.
3. On a number line, mark a solid dot at -3 and shade to the left.
4. Test $x = -4$: $4 + 8 = 12$? 10, true.

ANSWER x ? -3

The dot is solid because -3 itself satisfies the inequality: $4 + 6 = 10$.

WORKED EXAMPLE 5.3.2 AN INEQUALITY FROM A CONTEXT

PROBLEM

A van may carry at most 900 kg. Four crates of 85 kg are already loaded. How many 120 kg boxes can be added?

1. Let b be the number of boxes.
2. The load is $4 \cdot 85 + 120b = 340 + 120b$.
3. The condition is $340 + 120b \le 900$.
4. Subtract 340: $120b \le 560$.
5. Divide by 120: $b \le 4.67$, so $b \le 4$.

ANSWER at most 4 boxes

Five boxes give $340 + 600 = 940$ kg, over the limit; four give 820 kg.

Step	Statement	Reason
1	$340 + 120b \le 900$	the constraint
2	$120b \le 560$	subtract 340, order kept
3	$b \le 4.67$	divide by 120 (positive), order kept
4	$b \le 4$	b is a whole number

Every step records why the statement remains true, which is what makes the answer defensible.

COMMON ERROR

The reversal that is easy to miss

Dividing by a negative and forgetting to reverse. $-3x > 12$ gives $x < -4$, not $x > -4$. Substitute $x = -5$ to check: $15 > 12$ is true.

COMMON ERROR

Forgetting that the variable counts objects

Reporting $b \le 4.67$ as the answer to <how many boxes<. The number of boxes is a whole number, so the solution set in the context is $\{0, 1, 2, 3, 4\}$.

PRACTICE 5.3 LINEAR INEQUALITIES

SUPPORT 1–4 CORE 1–7 EXTENSION 8–9

1. Solve and show on a number line: (a) $x + 5 > 2$ (b) $3x \le 15$ (c) $x/2 < 4$. F

PRACTICE 5.3 ▤ LINEAR INEQUALITIES (CONTINUED)

SUPPORT 1▤4 ▤ CORE 1▤7 ▤ EXTENSION 8▤9

1. Solve: (a) $-2x$? 8 (b) $5 - x > 2$ (c) $-4x < 12$. F
2. Solve: (a) $3x - 2 < x + 6$ (b) $2(x + 1)$? $5x - 4$. F
3. Write as an inequality: (a) speed must not exceed 50 km/h (b) at least 12 people are needed (c) the temperature stays below 4 ?C. F
4. Explain, with two numerical examples, why multiplying an inequality by a negative number reverses it. R
5. A lift can carry 480 kg. How many 68 kg people can it carry? State the solution set in context. P
6. A student has 60 units and needs x notebooks at 4.5 units each plus a bag costing 15 units. Write and solve an inequality for x. P
7. Solve $2 < 3x - 1$? 11 for integer x and describe the solution set in two ways. E
8. Explain why $x^2 > 9$ cannot be solved by taking square roots and writing $x > 3$. E

5.4 Simultaneous equations

CRITERION A II • SYSTEMS • EQUIVALENCE

Two equations, two unknowns: the solution is the pair that satisfies both at once. Elimination removes one unknown; substitution replaces it.

In **elimination**, the equations are combined so that one variable disappears: either the coefficients are already opposite, or they are made so by multiplying an equation through. In **substitution**, one equation is rearranged to express a variable in terms of the other, and that expression replaces the variable in the second equation. Both methods rely on the same principle: replacing a quantity by an equal quantity preserves the solution.

WORKED EXAMPLE 5.4.1 ELIMINATION

PROBLEM

Solve $2x + 3y = 12$ and $x - y = 1$.

1. Multiply the second equation by 2: $2x - 2y = 2$.
2. Subtract it from the first: $(2x + 3y) - (2x - 2y) = 12 - 2$.
3. Simplify: $5y = 10$, so $y = 2$.
4. Substitute into $x - y = 1$: $x = 3$.
5. Check both equations: $2(3) + 3(2) = 12$ and $3 - 2 = 1$.

ANSWER $x = 3, y = 2$

The pair satisfies both equations, which is the definition of a simultaneous solution.

WORKED EXAMPLE 5.4.2 SUBSTITUTION

PROBLEM

Solve $y = 2x - 3$ and $3x + 2y = 8$.

1. The first equation already gives y in terms of x.
2. Substitute into the second: $3x + 2(2x - 3) = 8$.
3. Expand and collect: $7x - 6 = 8$, so $7x = 14$ and $x = 2$.
4. Then $y = 2(2) - 3 = 1$.

ANSWER $x = 2, y = 1$

$3(2) + 2(1) = 8$ and $2(2) - 3 = 1$. Both hold.

KEY IDEA

Reading the number of solutions

If the equations are $y = mx + c$ and $y = nx + d$, then equal gradients with different intercepts give no solution (parallel lines), and equal gradients with equal intercepts give infinitely many. One solution requires different gradients.

COMMON ERROR

Substitution into the wrong equation

Substituting into the same equation that was rearranged. That step always produces a true but useless statement, because it says nothing about the second equation.

PRACTICE 5.4 SIMULTANEOUS EQUATIONS

SUPPORT 1–4 CORE 1–7 EXTENSION 8–9

1. Solve by elimination: (a) $x + y = 7, x - y = 3$ (b) $2x + y = 11, x + y = 7$. F
2. Solve by elimination: (a) $3x + 2y = 16, x + 2y = 8$ (b) $4x - y = 9, 2x + y = 9$. F
3. Solve by substitution: (a) $y = x + 4, 2x + y = 13$ (b) $x = 3y, x + 2y = 15$. F
4. Solve: $2x + 3y = 7$ and $3x - 2y = 4$. F
5. Explain why you must substitute into the equation you did not use, and what happens if you do not. R
6. Two coffees and three teas cost 21 units; one coffee and two teas cost 12 units. Find each price. P
7. Show that $y = 2x + 1$ and $2y = 4x + 2$ have infinitely many solutions, and explain why. E

PRACTICE 5.4 SIMULTANEOUS EQUATIONS (CONTINUED)

SUPPORT 1–4 CORE 1–7 EXTENSION 8–9

1. Show that $y = 3x + 1$ and $y = 3x - 4$ have no solution, and describe the graphs. E
2. Find k such that $kx + y = 5$ and $2x + y = 5$ have no solution for $x \neq 0$. E

5.5 Systems, constraints and feasibility

CRITERION D II, IV • CONSTRAINT • FEASIBLE

A real decision usually has several constraints that must hold at once. The set of points satisfying all of them is the feasible region, and the mathematics of choosing inside it is linear programming.

Each inequality describes a half-plane: all the points on one side of a line. Two inequalities together describe the overlap of two half-planes; several describe a polygon. A point inside the region satisfies every constraint and is called **feasible**; a point outside violates at least one.

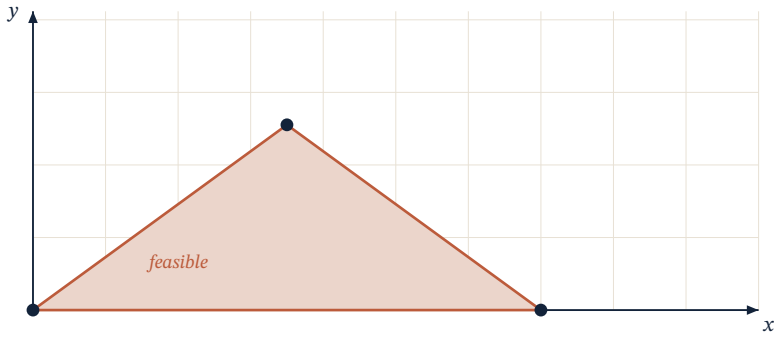


Figure 5.1 Two constraints and a non-negativity condition. The shaded region is the set of feasible plans; the vertex marked is a candidate for the best plan.

WORKED EXAMPLE 5.5.1 TESTING A PLAN

PROBLEM

A stall sells small and large bags. It can carry at most 60 kg, a small bag weighs 1 kg and a large 3 kg, and there is room for at most 40 bags. Test the plans (30, 5) and (20, 20).

1. Write the constraints: $s + 3l \leq 60$ and $s + l \leq 40$, with $s, l \geq 0$.
2. Test (30, 5): $30 + 15 = 45 \leq 60$ and $30 + 5 = 35 \leq 40$. Feasible.
3. Test (20, 20): $20 + 60 = 80 > 60$. Infeasible, though $20 + 20 = 40 \leq 40$ holds.
4. So (20, 20) breaks the weight limit even though it fits the space limit.

ANSWER (30, 5) is feasible; (20, 20) is not

A plan is feasible only when it satisfies every constraint, not merely one of them.

WORKED EXAMPLE 5.5.2 OPTIMISING AT A VERTEX

PROBLEM

With the constraints $s + 3l \leq 60$, $s + l \leq 40$ and $s, l \geq 0$, maximise the revenue $R = 4s + 9l$.

1. The vertices are (0, 0), (40, 0), (0, 20) and the intersection of the two lines.
2. Solve $s + 3l = 60$ and $s + l = 40$: subtract to get $2l = 20$, so $l = 10$ and $s = 30$. The vertex is (30, 10).
3. Evaluate R at each vertex: 0, 160, 180 and $120 + 90 = 210$.
4. The maximum is 210 at (30, 10).

ANSWER $s = 30$, $l = 10$, giving $R = 210$

For a linear objective, the maximum on a polygon occurs at a vertex; checking all four is therefore sufficient.

COMMON ERROR

Checking only some constraints

Assuming a plan that satisfies two constraints satisfies all of them. Feasibility is a condition on the whole set, and one violated constraint is enough to make a plan impossible.

PRACTICE 5.5 ▪ CONSTRAINTS AND FEASIBILITY

SUPPORT 1 ▪ CORE 1 ▪ EXTENSION 7

1. Test whether each point satisfies $x + 2y \le 10$ and $x \ge 0$: (a) (2, 3) (b) (6, 3) (c) (0, 5). F
2. Sketch the region defined by $x + y \le 6$, $x \ge 0$, $y \ge 0$. F
3. Sketch $y \le 2x - 1$ and shade the region containing (0, 2). F
4. Explain what a vertex of the feasible region represents in a manufacturing problem. R
5. A bakery can bake at most 200 items; loaves take 2 units of oven time and rolls 1. It has 320 units of oven time. Write the constraints. P
6. With the constraints $x + y \le 8$, $x \ge 5$, $y \ge 5$ and $x, y \ge 0$, maximise $P = 3x + 2y$. P
7. Explain why the maximum of $P = 3x + 2y$ on a polygon must occur at a vertex, by describing what happens to P as you move along an edge. E
8. A committee must choose between two projects. Each has a cost and a benefit, and the budget is limited. Describe how you would set the problem up as constraints and an objective, and what the solution would tell the committee. E

An investigation is not a longer exercise. It is a short piece of mathematics you do yourself: you gather evidence, look for a pattern, say what you think is true, and then test it until you can defend it.

INVESTIGATION

When there is no answer at all

Some pairs of equations have one solution, some have none and some have infinitely many. The mathematics is to find out exactly which feature of the equations decides which case.

System	Gradients	Intercepts	Solutions
$y = 2x + 1, y = 2x - 4$	equal	different	none
$y = 2x + 1, 2y = 4x + 2$	equal	equal	infinitely many
$y = 2x + 1, y = -x + 4$	different	different	one

1. **Gather.** Solve four systems of your own, two with equal gradients and two with different ones. Record what happens to the variable when you eliminate.
2. **Tabulate.** For each system record the gradients, the intercepts and the number of solutions. Look for the pattern.
3. **Conjecture.** Write a rule that predicts the number of solutions from the gradients and intercepts alone.
4. **Test.** Use your rule to predict the number of solutions of $3x - y = 4$ and $6x - 2y = 9$, then solve algebraically to check.
5. **Explain.** Describe the graphs of the three cases in words a classmate could draw from your description alone.
6. **Justify.** Explain why the algebraic step that makes the variable vanish leaves $5 = 5$ in one case and $1 = 4$ in another.

Present the table, the rule, the prediction and test, and one paragraph explaining why the algebra and the graph agree.

APPLICATION

Planning a community budget

A youth club has 1 200 units to spend. It wants to run workshops at 90 units each and buy equipment kits at 150 units each. At least 4 workshops are required by the funder, and no more than 8 kits can be stored.

Workshops w	Kits k	Cost	Feasible?
4	5	1 110	yes
6	4	1 140	yes
8	3	1 170	yes
10	2	1 200	yes
12	1	1 230	no: over budget

1. Write the constraints as inequalities in w and k, including the funder's requirement and the storage limit.
2. Sketch the feasible region and mark the vertices.
3. Identify three feasible plans from the region, and one plan that is almost feasible, explaining which constraint it breaks.
4. Suppose each workshop reaches 25 young people and each kit serves 60. Write the total reach as an expression and find the feasible plan that maximises it.
5. Recommend a plan and justify it. State one value judgement your recommendation contains that the mathematics alone cannot settle.

CRITERION C COMMUNICATING

Justifying every step

A classmate has solved four equations correctly but cannot say why the steps are allowed. Write a one-page guide that supplies the reason for each kind of step, using their four solutions as examples.

1. Explain the balance principle and why it preserves the solution set.
2. Give the reason for each of: adding a quantity to both sides, expanding a bracket, multiplying by a common denominator, dividing by a coefficient.
3. Explain the one step that is allowed for equations but changes the answer set for inequalities, and say exactly what changes.
4. Explain why substituting a candidate answer back into the original equation is a proof that it is a solution.

1–2 Gives correct solutions but the justifications are missing or circular.
3–4 States the balance principle and applies it to most steps.
5–6 Justifies each kind of step correctly and explains the inequality reversal.
7–8 As above, and explains why a check by substitution is a proof rather than a convenience.

Chapter summary and review

Key ideas

- The balance principle generates equivalent equations.
- Solving undoes operations in reverse order.
- Multiplying or dividing an inequality by a negative number reverses it.
- An inequality has a solution set; represent it on a number line.
- Elimination and substitution both preserve the solution set.
- Equal gradients with different intercepts: no solution.
- All constraints must hold for a plan to be feasible.
- For a linear objective, the optimum on a polygon is at a vertex.

Vocabulary

equation
A statement that two expressions are equal.

equivalent
Having exactly the same solutions.

balance principle
Whatever is done to one side must be done to the other.

inequality
A statement of order between two expressions.

solution set
All the values that satisfy a statement.

elimination
Removing a variable by combining equations.

substitution
Replacing a variable by an equal expression.

constraint
A condition a solution must satisfy.

feasible region
The set of points satisfying all constraints.

objective
The quantity to be maximised or minimised.

What students often think	What is actually true
$-2x < 6$ gives $x < -3$.	Dividing by a negative reverses it: $x > -3$.
Multiplying both sides by the common denominator of the first fraction only.	Every term must be multiplied, including constants on the other side.
A system always has a solution.	Parallel lines give none; identical lines give infinitely many.

Misconceptions worth confronting directly.

What students often think	What is actually true
A feasible plan is the best plan.	Feasible means allowed; optimal needs an objective.

Revision checklist

- ☐ I can solve equations with brackets and fractions.
- ☐ I can say why each step is permitted.
- ☐ I can solve a linear inequality and represent the solution set.
- ☐ I can explain the reversal rule with a numerical test.
- ☐ I can solve simultaneous equations by both methods.
- ☐ I can predict the number of solutions from gradients and intercepts.
- ☐ I can write constraints from a description and test a plan.
- ☐ I can optimise a linear objective on a feasible region.

REFLECT

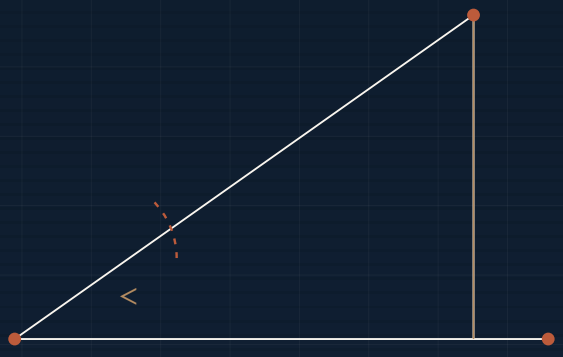
Write three equations of your own: one with a solution, one with no solution and one with infinitely many. Exchange them with a partner and identify which is which without solving, using only the structure.

III

PART III

Geometry, proof and space

Geometry is the part of mathematics where a diagram can be an argument < but only if the diagram is read as a claim to be justified. This part moves from moving shapes without changing them, through similarity and trigonometry, to solids whose measurements carry an honest error.



06 Transformations and congruence

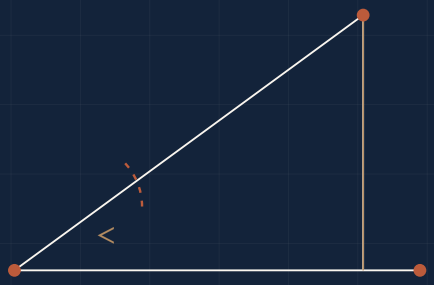
07 Similarity and trigonometry

08 Solids, measurement and accuracy



6

SHAPE, SPACE AND REASONING



Transformations and congruence

A transformation moves a shape without changing what it is. That sentence contains the whole chapter: what changes is position, orientation or size; what does not change is the information that makes the shape the shape.

KEY CONCEPT ▪ FORM | RELATED ▪ RELATIONSHIPS | RELATED ▪ MOVEMENT

CONTEXT ▪ ORIENTATION IN SPACE AND TIME

> **INQUIRY**
>
> When a shape is moved, what exactly is preserved < and how can we prove that two shapes are the same without moving either of them?

ATL ▪ THINKING AND COMMUNICATION SKILLS

Take a wallpaper or tiling pattern you can see around you. Identify the symmetries it contains and write down the smallest transformation that maps the pattern onto itself. Explain your answer so that someone could check it without seeing the wall.

MYP UNIT PLANNER ≣THE WHOLE CHAPTER IN ONE VIEW

Framing the inquiry

Unit 6 ?approximately 15 hours of class time ?ATL focus: thinking and communication

KEY AND RELATED CONCEPTS

Form. Congruence is a statement about form: two shapes are congruent when one can be transformed into the other by movements that preserve distance and angle.

Relationships. Every transformation has an inverse, and the composition of two transformations is another transformation. These relationships are what make the set of transformations a structure rather than a list.

Movement. Translation, rotation and reflection change position, orientation or handedness; only enlargement changes size. Which quantities each one preserves is the key information.

GLOBAL CONTEXT AND EXPLORATION

Orientation in space and time. Every map, every architectural drawing and every pattern on a wall depends on transformations. A tile pattern is a design generated by repeating a motif under translations, rotations and reflections; a floor plan is a projection of a three-dimensional arrangement. Reading orientation < which way is north, which way is the reflection < is a practical skill that rests on this mathematics.

STATEMENT OF INQUIRY

In order for a student to determine whether two figures are the same shape, students must be able to describe a transformation precisely, identify the properties it preserves, and use the criteria for congruent triangles to justify a conclusion.

FACTUAL

<What is a translation?

<What is the difference between congruence and similarity?

<What are the four congruence criteria for triangles?

CONCEPTUAL

<Why is a reflection not achievable by any sequence of translations and rotations in the plane?

DEBATABLE

<If two shapes are congruent only when one can be moved onto the other, does the mathematics depend on being able to move them?

Criterion	Strands	Evidence in this chapter
A □ Knowing and understanding	i, ii, iii	Describe and perform translations, rotations, reflections and enlargements; use congruence criteria.
B □ Investigating patterns	i, ii, iii, iv	Investigate the symmetries of a tiling; find the smallest repeating transformation and justify it.
C □ Communicating	i, ii, iii	Describe a transformation precisely enough to be reproduced; write a congruence proof.
D □ Applying mathematics	i, ii, iii, v	Use transformations to design a repeating pattern and to solve an orientation problem.

Each teaching section closes with a graded practice set. The investigation addresses Criterion B; the application addresses Criterion D; the criterion task asks for extended written communication (Criterion C) about knowledge and understanding (Criterion A).

Learning intentions and success criteria

The vocabulary here is unusually important. ⊲Move it left⊲is not a description; ⊲translate 3 units left and 2 units down⊲is.

YOU ARE LEARNING TO

- **Understand** that translations, rotations and reflections preserve length and angle, and that enlargements do not.
- **Be able to** describe a transformation precisely, including a centre and angle or a mirror line.
- **Be able to** perform and combine transformations on a coordinate grid.
- **Know** the four congruence criteria for triangles and when each applies.
- **Be able to** write a congruence proof, justifying each statement.
- **Understand** the symmetries of a shape and how to find them.

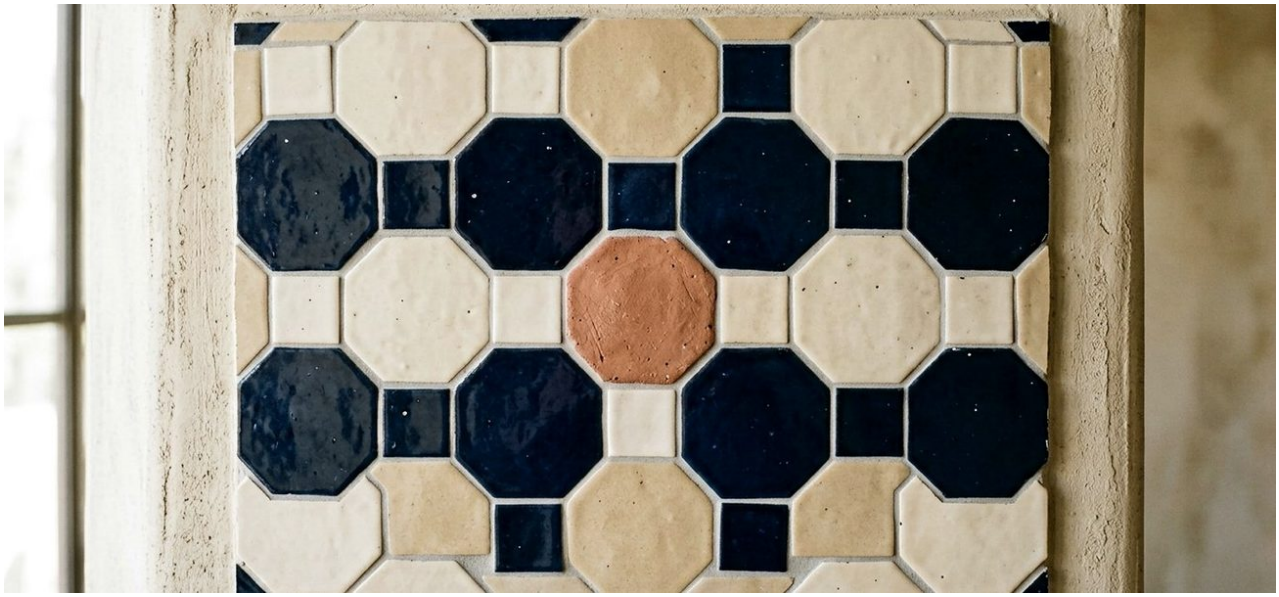
SUCCESS CRITERIA I CAN

- ☐ I can describe a rotation fully: centre, angle and direction.
- ☐ I can reflect a shape in a given line and state the coordinates of the image.
- ☐ I can enlarge a shape by a given scale factor from a given centre.
- ☐ I can decide whether two triangles are congruent and say which criterion I used.
- ☐ I can write a proof that two triangles are congruent, with a reason for each statement.
- ☐ I can find all the lines of symmetry of a shape and state its order of rotational symmetry.

PREREQUISITE CHECK 10 MINUTES, GRID PAPER

1. Plot (3, 4) and state its distance from the *y*-axis.
2. State the coordinates of the point 3 units left of (5, -2).
3. Name a triangle with all sides equal.
4. State the size of each angle in an equilateral triangle.
5. Calculate < □ 6 □ 4.
6. State the sum of the angles in a quadrilateral.

Answers are in the chapter answer section. If more than two are incorrect, review coordinates in four quadrants, angle facts and the names of triangles before continuing.



A tiled surface. The pattern is generated by repeating one motif under translations and reflections; the grout lines make the symmetries visible. Original photograph made for this edition.

Explore: what survives the move

Draw a scalene triangle on squared paper and cut it out. You will move it four ways and record what changes and what does not.

YOUR TASK BEFORE YOU ARE TAUGHT

1. Record the side lengths and angles of your triangle in a table before you start.
2. Slide the triangle 5 squares right and 3 up. Measure the sides and angles of the new position and add them to the table.
3. Turn the triangle a quarter turn about one of its vertices. Record the measurements again.
4. Flip the triangle over. Record the measurements, and note anything that has changed that did not change in the first two moves.
5. Compare the four rows. Write a statement beginning <A translation, rotation or reflection preserves < <and one beginning <but changes < <.
6. Predict which measurements would change if the triangle were enlarged instead, and test your prediction.

Section 6.1 gives these observations their names, and section 6.4 turns them into a proof technique.

BEFORE YEAR 3	YEAR 3 THIS CHAPTER	LATER
▪ Plotting and reading coordinates. ▪ Naming and classifying triangles and quadrilaterals. ▪ Angle facts: straight line, around a point, in a triangle.	▪ Translations, rotations, reflections and enlargements. ▪ Congruence and the four criteria. ▪ Symmetry, tessellation and congruence proofs.	▪ Similarity and scale factors in two and three dimensions. ▪ Vectors and matrix transformations. ▪ Group structure of transformations.

Where this chapter sits in the sequence.

Inquiry pathway for this chapter

	Stage	What you do
1	Context	A tiled wall and a floor plan drawn to scale.
2	Question	What stays the same when a shape is moved?
3	Concept	Congruence, isometry, orientation, symmetry.
4	Investigation	Find the smallest transformation that maps a tiling onto itself.
5	Communication	Write instructions that reproduce a transformation exactly.
6	Reflection	Why can a reflection not be achieved by sliding and turning?

6.1 Translations, rotations and reflections

CRITERION A • ISOMETRY • ORIENTATION

Three movements preserve every length and every angle. They are called isometries, and they differ only in what they do to orientation.

A **translation** slides every point the same distance in the same direction, given as a column vector. A **rotation** turns every point about a fixed centre through a given angle, clockwise or anticlockwise. A **reflection** flips every point across a fixed mirror line. All three preserve distance and angle, so the image is congruent to the object.

DEFINITION

An **isometry** is a transformation that preserves distance. Translations, rotations and reflections are isometries; enlargements are not.

Transformation	What it preserves	What it changes	Orientation
Translation	length, angle, orientation	position	unchanged
Rotation	length, angle, orientation	position, heading	unchanged
Reflection	length, angle	position, handedness	reversed
Enlargement	angle only	length, area	unchanged

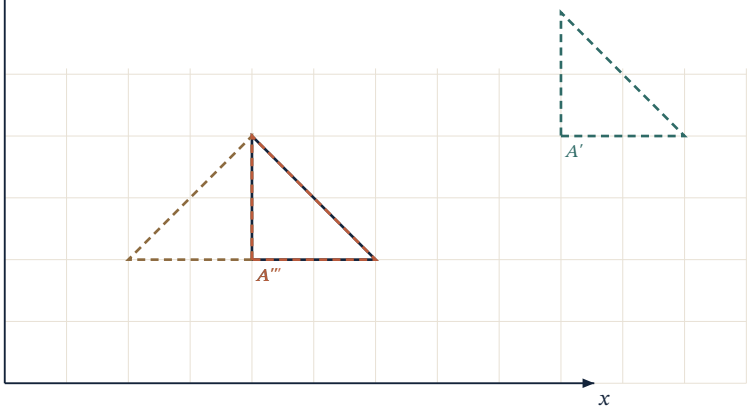


Figure 6.1 One triangle under four transformations. The reflected image cannot be slid or turned onto the original, because its handedness is reversed.

WORKED EXAMPLE 6.1.1 DESCRIBING A ROTATION

PROBLEM

Triangle A has vertices (1, 1), (3, 1), (1, 2). Describe the rotation that takes A to the triangle with vertices (-1, 1), (-1, 3), (-2, 1).

1. Match one vertex to its image: (3, 1) goes to (-1, 3).
2. Test the origin as centre: a quarter turn anticlockwise sends (x, y) to $(-y, x)$.
3. Apply it: (3, 1) gives (-1, 3), (1, 1) gives (-1, 1), (1, 2) gives (-2, 1).
4. All three vertices match, so the centre is the origin.

ANSWER rotation of 90° anticlockwise about the origin

Check a second vertex independently: (1, 2) must go to (-2, 1), and it does.

WORKED EXAMPLE 6.1.2 REFLECTING IN A LINE

PROBLEM

Reflect the point (4, 1) in the line $y = x$ and in the line $x = 2$.

1. In the line $y = x$, coordinates swap: (4, 1) goes to (1, 4).
2. In the vertical line $x = 2$, the y-coordinate is unchanged and the x-coordinate is reflected about 2.
3. The distance from $x = 4$ to the line is 2, so the image is 2 on the other side: $x = 0$.
4. The image is (0, 1).

ANSWER (1, 4) in $y = x$; (0, 1) in $x = 2$

The midpoint of (4, 1) and (0, 1) is (2, 1), which lies on the mirror line, and the segment joining them is perpendicular to it.

COMMON ERROR

An incomplete description

Describing a rotation as ◁a turn to the right< without giving the centre. The same angle about a different centre produces a different image, so the description is incomplete and cannot be reproduced.

KEY IDEA

The one that cannot be imitated

A reflection is the only isometry that reverses handedness. This is why a left hand cannot be made to match a right hand by sliding and turning: no combination of translations and rotations reverses orientation.

PRACTICE 6.1 ▪ TRANSLATIONS, ROTATIONS AND REFLECTIONS

SUPPORT 1▪4 ▪ CORE 1▪6 ▪ EXTENSION 7▪9

1. Translate (2, 5) by the vector (3, -4). State the image. F
2. Reflect (3, 2) in (a) the x-axis (b) the y-axis (c) the line $y = x$. F
3. Rotate (2, 0) by 90□ anticlockwise about the origin, then by 180□. F
4. Triangle T has vertices (0, 0), (4, 0), (0, 3). Find the image after a translation of (-2, 1). F
5. Explain why a reflection changes the order of the vertices when they are read clockwise. R
6. Describe fully the single transformation that maps (1, 2) to (2, 1). R
7. A shape is rotated 90□ clockwise about the origin and then reflected in the x-axis. Describe the single transformation that would have the same effect. P
8. Explain why the composition of two reflections in parallel lines is a translation, and state the distance in terms of the lines. E
9. Find the image of (5, 1) after a rotation of 180□ about (2, 3), and explain your method. E

6.2 Enlargement and scale

CRITERION A I • SCALE • PROPORTION

Enlargement changes size and keeps shape. Angles survive; lengths are multiplied by the scale factor; areas by its square.

An enlargement with centre O and scale factor k sends each point P to a point P' on the ray OP with $OP' = k \times OP$. When $k > 1$ the image is larger; when $0 < k < 1$ it is smaller; when k is negative the image lies on the opposite side of the centre and is rotated through 180°.

KEY IDEA

Lengths scale, areas scale squared

A scale factor of k multiplies every length by k and every area by k^2. This is why doubling a photograph's dimensions quadruples the paper needed, and why a model at $1 : 50$ uses $50^2 = 2500$ times less paint than the real object.

WORKED EXAMPLE 6.2.1 FINDING A SCALE FACTOR

PROBLEM

A triangle with vertices $(1, 1)$, $(3, 1)$, $(1, 2)$ is enlarged about the origin to an image with vertices $(2, 2)$, $(6, 2)$, $(2, 4)$. Find the scale factor.

1. Compare corresponding vertices: $(1, 1)$ goes to $(2, 2)$.
2. Divide the image coordinates by the object coordinates: $2 \div 1 = 2$.
3. Check with another vertex: $(3, 1)$ goes to $(6, 2)$, and $6 \div 3 = 2$.
4. The scale factor is 2.

ANSWER scale factor 2, centre the origin

The area of the image is $2^2 = 4$ times the area of the object, as the area rule requires.

WORKED EXAMPLE 6.2.2 AREA UNDER ENLARGEMENT

PROBLEM

A rectangle 6 cm by 4 cm is enlarged by scale factor 2.5. Find the dimensions and area of the image.

1. Multiply each length by 2.5: 15 cm by 10 cm.
2. The area of the image is 15 × 10 = 150 cm<
3. Check with the area rule: the original area is 24 cm<and $2.5^2 = 6.25$, giving 24 × 6.25 = 150 cm<

ANSWER 15 cm by 10 cm; area 150 cm<

Both methods agree, which confirms that area scales as the square of the scale factor.

COMMON ERROR

Area doubled instead of squared

Enlarging by scale factor 2 and expecting the area to double. Every length doubles, so the area is multiplied by 2 × 2 = 4.

PRACTICE 6.2 ENLARGEMENT AND SCALE

SUPPORT 1–3 CORE 1–6 EXTENSION 7–8

1. Enlarge (2, 3) by scale factor 3 about the origin. F
2. A 5 cm by 8 cm rectangle is enlarged by scale factor 1.5. Find its new dimensions and area. F
3. Find the scale factor when a 4 cm segment becomes 10 cm. F
4. A map uses a scale of 1 : 25 000. A road is 6 cm on the map. Find its real length in kilometres. F
5. Explain why the angles of a shape do not change under enlargement. R
6. A photograph is enlarged to three times its width and height. Explain how much more paper is needed. R
7. A model car is built at 1 : 24. The model's paint area is 250 cm< Estimate the paint area of the real car in m< P
8. A triangle is enlarged by scale factor -1 about the origin. Describe the image and explain why it is congruent to the object. E

6.3 Congruence and the triangle criteria

CRITERION A≣II • CONGRUENCE • PROOF

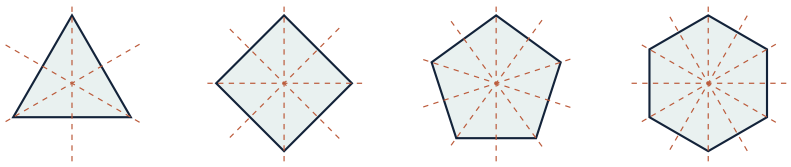
Two figures are congruent when one can be moved onto the other by an isometry. For triangles, four short criteria are enough to decide this without moving anything.

Triangles are congruent if any one of the following holds: three sides match (SSS); two sides and the included angle match (SAS); two angles and a corresponding side match (AAS); or the hypotenuse and one leg of a right angle match (RHS). Angle<angle<angle and side<side<angle are **not** criteria, and knowing why is as important as knowing the four that are.

DEFINITION

Congruent figures have exactly the same shape and size; the symbol is □. Congruent triangles have all three sides and all three angles equal.

Criterion	What is given	Why it fixes the triangle
SSS	three sides	the angles are forced by the side lengths
SAS	two sides and the angle between them	the third side is fixed
AAS	two angles and a matching side	the third angle is forced, then the sides scale
RHS	right angle, hypotenuse, one leg	Pythagoras fixes the third side



equilateral
3 lines

square
4 lines

regular pentagon
5 lines

regular hexagon
6 lines

Figure 6.2 Symmetry of plane figures. A line of symmetry is a reflection that maps the figure onto itself; the order of rotational symmetry counts the rotations that do the same.

WORKED EXAMPLE 6.3.1 CHOOSING THE CRITERION

PROBLEM

In triangles ABC and DEF, $AB = DE$, $AC = DF$ and angle A = angle D. Are the triangles congruent?

1. The two given sides meet at A in the first triangle and at D in the second.
2. So the equal angles are the angles **included** between the equal sides.
3. The criterion SAS applies.
4. The triangles are congruent.

ANSWER yes, by SAS

If the equal angles were not the included ones, the criterion would be SSA, which does not guarantee congruence.

WORKED EXAMPLE 6.3.2 A CONGRUENCE PROOF

PROBLEM

$ABCD$ is a kite with $AB = AD$ and $CB = CD$. Prove that triangle ABC is congruent to triangle ADC, and hence that AC bisects angle BAD.

1. $AB = AD$ (given).
2. $CB = CD$ (given).
3. AC is common to both triangles.
4. Therefore triangle ABC ≡ triangle ADC (SSS).
5. Corresponding angles are equal, so angle BAC = angle DAC; hence AC bisects angle BAD.

ANSWER congruent by SSS; the bisector follows from corresponding parts

Every statement has a reason, and the conclusion uses the fact that corresponding parts of congruent triangles are equal.

COMMON ERROR

Statements without reasons

Writing a proof as a list of statements without reasons. A statement without a reason is an assertion; the reason is what makes it mathematics. Each line should say either ‹given›, ‹common›, or the name of the fact used.

COMMON ERROR

The criterion that does not exist

Using SSA as a congruence criterion. Two sides and a non-included angle can produce two different triangles, so it does not determine the shape.

PRACTICE 6.3 CONGRUENCE AND PROOF

SUPPORT 1–3 CORE 1–5 EXTENSION 7–9

1. State the criterion that shows each pair of triangles congruent: (a) sides 5, 7, 9 in both (b) sides 4 and 6 with a 50° angle between them in both. F
2. Triangles have angles 40°, 60°, 80° in both, with the side between the 40° and 60° angles 8 cm in both. Are they congruent? State the criterion. F
3. Explain why AAA does not guarantee congruence. R
4. $ABCD$ is a parallelogram. Prove that triangle ABC is congruent to triangle CDA. P

PRACTICE 6.3 CONGRUENCE AND PROOF (CONTINUED)

SUPPORT 1–3 CORE 1–6 EXTENSION 7–9

1. In the diagram, $AB = AC$ and D is the midpoint of BC. Prove that AD is perpendicular to BC. P
2. Explain the difference between ‹congruent› and ‹similar›, using an example of each. R
3. Prove that the diagonals of a rectangle are equal in length, using congruent triangles. E
4. A student writes ‹the triangles are congruent because they look the same›. Explain why this is not a proof and what would be needed. E
5. Find all the lines of symmetry and the order of rotational symmetry of a regular hexagon, and justify the count. E

6.4 Symmetry and tessellation

CRITERION B ▤, II • SYMMETRY • PATTERN

A shape has symmetry when a transformation maps it onto itself. A plane can be tiled when copies of a shape fit together without gaps or overlaps, and the two ideas are connected through angle.

A **line of symmetry** is a mirror line that maps a figure onto itself. The **order of rotational symmetry** is the number of distinct rotations less than a full turn that also map the figure onto itself. A square has four lines of symmetry and rotational symmetry of order four; a parallelogram has none and order two.

A **tessellation** is a covering of the plane with no gaps and no overlaps. Any triangle and any quadrilateral tessellates, because copies can be arranged so that their angles meet at a point and sum to 360□ Regular pentagons do not tessellate, because 108□does not divide 360□

Regular polygon	Interior angle	Tessellates?
Triangle	60□	yes (six meet)
Square	90□	yes (four meet)
Hexagon	120□	yes (three meet)
Pentagon	108□	no (360 □ 108 is not a whole number)

The tessellation test is arithmetic: the interior angle must divide 360□exactly.

WORKED EXAMPLE 6.4.1 COUNTING THE SYMMETRIES

PROBLEM

Find the lines of symmetry and the order of rotational symmetry of a regular hexagon.

1. A regular n-gon has n lines of symmetry, through opposite vertices or opposite edge midpoints.
2. For a hexagon, $n = 6$, so there are six lines.
3. Rotation by 360□□ 6 = 60□maps it onto itself, so the order is 6.
4. Every multiple of 60□less than 360□works, which gives six rotations.

ANSWER six lines; rotational symmetry of order 6

The two counts agree, as they must for any regular polygon.

COMMON ERROR

Diagonals are not mirror lines

Counting a shape's diagonals as lines of symmetry. A rectangle's diagonals are not mirror lines: reflecting in a diagonal does not map the rectangle onto itself unless it is a square.

PRACTICE 6.4 SYMMETRY AND TESSELLATION

SUPPORT 1–3 CORE 1–6 EXTENSION 7–8

1. State the number of lines of symmetry and the order of rotational symmetry: (a) equilateral triangle (b) square (c) regular octagon (d) isosceles trapezium. F
2. Explain why a scalene triangle has no line of symmetry. F
3. Decide whether each tessellates, with a reason: (a) regular hexagon (b) regular pentagon (c) any parallelogram. F
4. Show that the interior angle of a regular pentagon is 108° and use it to explain why pentagons do not tessellate. R
5. A shape has rotational symmetry of order 4 and no lines of symmetry. Describe a shape that fits. P
6. Design a tessellation from an arbitrary quadrilateral and explain why it works using the angle sum. P
7. Explain why every triangle tessellates, whatever its shape. E
8. Two shapes have the same area but different symmetry. Explain why area and symmetry are independent properties. E

6.5 Congruence, coordinates and proof

CRITERION A V • COORDINATES • JUSTIFICATION

Coordinates turn a geometric claim into arithmetic. That is powerful and risky: arithmetic can establish a specific case, but a general claim still needs a general argument.

On a coordinate grid, a claim about a shape can often be tested by computing. Side lengths come from the distance between coordinates; midpoints from averaging them; perpendicularity from gradients whose product is -1. A computation establishes a claim about that particular figure, and can suggest a general result < but it does not prove one.

KEY IDEA

Example, counterexample, proof

A single example can disprove a universal claim but cannot prove it. Showing that one particular quadrilateral's diagonals bisect each other does not show that every parallelogram's do; the proof needs letters for coordinates, so that the argument covers every case at once.

WORKED EXAMPLE 6.5.1 TESTING A CLAIM BY COORDINATES

PROBLEM

Show that the quadrilateral with vertices $A(1, 1)$, $B(5, 1)$, $C(5, 4)$, $D(1, 4)$ is a rectangle, and find the length of its diagonal.

1. AB is horizontal with length 4; DC is horizontal with length 4.
2. BC is vertical with length 3; AD is vertical with length 3.
3. Adjacent sides are perpendicular because one is horizontal and the other vertical, so the figure is a rectangle.
4. The diagonal is $\sqrt{(4^2 + 3^2)} = \sqrt{25} = 5$.

ANSWER a rectangle with diagonal 5

Both diagonals are 5, as they must be in a rectangle; the second diagonal confirms the first.

WORKED EXAMPLE 6.5.2 A GENERAL ARGUMENT

PROBLEM

Prove that the diagonals of any parallelogram bisect each other.

1. Let the vertices be $A(0, 0)$, $B(a, 0)$, $D(b, c)$ and $C(a + b, c)$, so that AB is parallel to DC.
2. The midpoint of AC is $((a + b)/2, c/2)$.
3. The midpoint of BD is $((a + b)/2, c/2)$.
4. The midpoints coincide, so the diagonals bisect each other.

ANSWER proved for a general parallelogram

Because a, b, c are arbitrary, the argument covers every parallelogram, not one example.

COMMON ERROR

One case is not a proof

Proving a general statement with one example. Calculating for a single parallelogram shows only that the statement is true in that case; a proof must use general coordinates or a general argument.

PRACTICE 6.5 CONGRUENCE, COORDINATES AND PROOF

SUPPORT 1–3 CORE 1–6 EXTENSION 7–8

1. Find the length of the segment joining (1, 2) and (4, 6). F
2. Find the midpoint of (-2, 5) and (4, -1). F
3. Show that (0, 0), (3, 4), (7, 1) form an isosceles triangle. F
4. Show that the quadrilateral (0, 0), (4, 0), (4, 2), (0, 2) is a rectangle and state its perimeter. F
5. Explain why showing one diagonal bisects the other in one quadrilateral does not prove it for all parallelograms. R
6. Prove that the midpoint of the hypotenuse of a right triangle is equidistant from all three vertices, using coordinates. P
7. Prove that a quadrilateral whose diagonals bisect each other is a parallelogram. E
8. A student claims all rhombuses are squares. Find a counterexample with coordinates and explain why the claim fails. E

An investigation is not a longer exercise. It is a short piece of mathematics you do yourself: you gather evidence, look for a pattern, say what you think is true, and then test it until you can defend it.

INVESTIGATION

Every repeating pattern has a smallest transformation that maps it onto itself. Finding it is the first step in describing the pattern precisely.

The smallest repeat

Pattern	Smallest translation	Rotational symmetry	Mirror lines
Square grid	one square right	order 4	4
Staggered brick	two bricks right, one up	order 2	0
Regular hexagon tiling	one hexagon across	order 6	6
Herringbone	two units along the herringbone	order 2	0

1. **Record.** Choose a repeating pattern you can see, photograph it or copy it onto grid paper, and mark the boundary of one repeating unit.
2. **Describe.** Write the translation vector that maps the unit onto the next unit, in units of the grid.
3. **Test.** Apply your transformation to a point on the boundary of the unit and check that the image lands on a matching point of the next unit.
4. **Extend.** Decide whether the pattern also has rotational or reflective symmetry, and if so describe the centre or the mirror line.
5. **Compare.** Exchange patterns with another group and decide whether their smallest translation is genuinely the smallest, or whether a smaller one exists.
6. **Justify.** Explain why no smaller translation can work for your pattern.

Present a marked diagram of one repeating unit, the translation vector, the full symmetry description, and a justification of minimality.

APPLICATION

EXTENSION

A floor plan for a small room

A room measures 4.2 m by 3.6 m. Tiles are 30 cm square. The design must use a rotational pattern and a border of one tile width.

Item	Dimension	Number of tiles
Room	4.2 m × 3.6 m	<
One tile	0.30 m × 0.30 m	<
Along the length	4.2 m	14
Along the width	3.6 m	12
Whole room	<	168

1. Calculate how many tiles fit along each wall, and explain why the answer must be a whole number for the design you are given.
2. Describe a rotational design that maps the floor onto itself under a half turn about the centre of the room.
3. Draw a quarter of the floor and use the rotational symmetry to complete the design, explaining which transformation produces each quadrant.
4. The tiles come in boxes of 12 with 5 per cent wastage allowed. Calculate the number of boxes to order and justify the rounding.
5. A border of one tile width is added around the edge. Recalculate the number of tiles needed and explain how the border changes the symmetry of the design.

CRITERION C COMMUNICATING

Instructions that cannot be misread

A classmate will be given your written instructions and must reproduce a transformation exactly, without seeing your diagram. Write the instructions.

1. Choose a shape and a sequence of three transformations. Write instructions for each, giving every piece of information the reader needs.
2. Test your instructions on a partner and record where they were ambiguous.
3. Revise the instructions so that each ambiguity is closed, and explain what you changed and why.
4. Add a short paragraph explaining why <move it over there< fails as an instruction, in terms of the properties the transformation preserves.

1–2 Instructions are incomplete; the reader cannot reproduce the result.

3–4 Instructions are mostly complete but omit a centre, direction or mirror line.

5–6 Instructions are complete and reproducible, and ambiguities are identified and fixed.

7–8 As above, and the explanation of precision connects to what each transformation preserves.

Chapter summary and review

Key ideas

- Translations, rotations and reflections preserve length and angle.
- Only a reflection reverses orientation.
- A rotation must be described by centre, angle and direction.
- An enlargement by scale factor k multiplies lengths by k and areas by k^2.
- Triangles are congruent by SSS, SAS, AAS or RHS.
- AAA and SSA are not congruence criteria.
- Every congruence proof needs a reason for each statement.
- A regular n-gon has n lines of symmetry and rotational symmetry of order n.

Vocabulary

transformation
A rule that moves every point of a figure.

isometry
A transformation that preserves distance.

translation
A slide by a fixed vector.

rotation
A turn about a centre through a given angle.

reflection
A flip in a mirror line.

enlargement
A scaling about a centre by a scale factor.

congruent
Exactly the same shape and size.

similar
The same shape, possibly a different size.

tessellation
A tiling of the plane with no gaps or overlaps.

line of symmetry
A mirror line mapping a figure onto itself.

What students often think	What is actually true
Reflection and rotation are the same kind of move.	Only reflection reverses handedness; no rotation can do it.
Enlarging by 2 doubles the area.	Area is multiplied by 4.
AAA proves congruence.	It proves similarity only; the size is not fixed.
A diagonal of a rectangle is a line of symmetry.	Reflecting in a diagonal does not map a rectangle onto itself.

Misconceptions worth confronting directly.

Revision checklist

- ☐ I can describe a translation, rotation or reflection completely.
- ☐ I can find the image of a point or shape under any of them.
- ☐ I can find a scale factor from an object and its image.
- ☐ I can use the area rule for enlargement.
- ☐ I can choose and name the correct congruence criterion.
- ☐ I can write a congruence proof with reasons.
- ☐ I can find the symmetries of a shape.
- ☐ I can decide whether a regular polygon tessellates, with a reason.

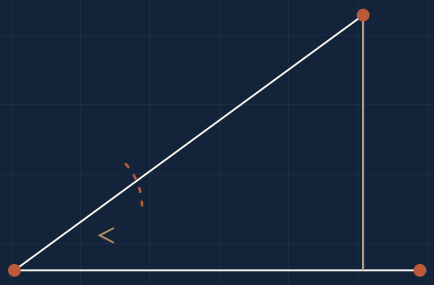
REFLECT

Draw a shape with exactly one line of symmetry and rotational symmetry of order 1, then one with two lines and order 2, then one with no lines and order 2. Explain how you know each count is correct.



7

SHAPE, SPACE AND REASONING



Similarity and trigonometry

Similarity says that shape survives scaling. Trigonometry turns that statement into measurement: once a triangle's angles are fixed, its side ratios are fixed too, and those ratios let us measure what we cannot reach.

KEY CONCEPT RELATIONSHIPS | RELATED FORM | RELATED SPACE | CONTEXT IDENTITIES AND RELATIONSHIPS

INQUIRY

Why do the ratios of a right triangle's sides depend only on its angles < and how does that let us measure a height we could never reach with a ruler?

ATL RESEARCH AND THINKING SKILLS

Find how an ancient or modern culture estimated a height it could not measure directly < a pyramid, a tree, a mountain. Describe the method and check whether the geometry in it is sound, giving reasons.

MYP UNIT PLANNER ≣THE WHOLE CHAPTER IN ONE VIEW

Framing the inquiry

Unit 7 ?approximately 18 hours of class time ?ATL focus: research and thinking

KEY AND RELATED CONCEPTS

Relationships. In similar triangles, corresponding side ratios are equal. This single relationship links lengths we can measure to lengths we cannot.

Form. The trigonometric ratios are properties of an angle, not of a particular triangle. That is why a table of ratios works for every right triangle with that angle.

Space. Similarity extends to three dimensions, where it links lengths, areas and volumes through the square and cube of the scale factor.

GLOBAL CONTEXT AND EXPLORATION

Identities and relationships. Similarity is how we compare things of different size: a model and the object it represents, a map and the terrain, a photograph and the scene. Trigonometry is how we compare what we can measure with what we cannot. Both depend on recognising that the same relationship can hold between quantities whose absolute sizes are completely different.

STATEMENT OF INQUIRY

In order for a student to determine a length or angle they cannot measure directly, students must be able to recognise similar triangles, use the constant ratio of corresponding sides, and select the trigonometric ratio appropriate to the information they have.

FACTUAL

<When are two triangles similar?

<What are the three trigonometric ratios?

<What is an angle of elevation?

CONCEPTUAL

<Why does sin 35?have a single value although infinitely many right triangles contain a 35?angle?

DEBATABLE

<If a scale model preserves shape but not size, what can it honestly be used to demonstrate?

Criterion	Strands	Evidence in this chapter
A ?Knowing and understanding	i, ii, iii	Use similarity ratios; apply sine, cosine and tangent; use Pythagoras' theorem; solve right triangles.
B ?Investigating patterns	i, ii, iii, iv	Investigate the ratios of right triangles at a fixed angle and establish their constancy.
C ?Communicating	i, ii, iv	Explain a measurement method; justify the choice of ratio; present working with appropriate accuracy.
D ?Applying mathematics	i, ii, iii, v	Measure an inaccessible height or distance and evaluate the reliability of the result.

Each teaching section closes with a graded practice set. The investigation addresses Criterion B; the application addresses Criterion D; the criterion task asks for extended written communication (Criterion C) about knowledge and understanding (Criterion A).

Learning intentions and success criteria

This chapter gives you a method for measuring things you cannot reach. The method depends entirely on one fact about similar triangles, so the reasoning matters more than the buttons on a calculator.

YOU ARE LEARNING TO

- **Understand** the conditions for two triangles to be similar.
- **Be able to** use the constant ratio of corresponding sides to find unknown lengths.
- **Be able to** apply Pythagoras' theorem to find a side of a right triangle.
- **Know** the definitions of sine, cosine and tangent, and choose the right one.
- **Be able to** solve a right triangle given a side and an angle.
- **Be able to** solve problems involving angles of elevation and depression.

SUCCESS CRITERIA I CAN

- ☐ I can decide whether two triangles are similar and state the scale factor.
- ☐ I can find an unknown length using a similarity ratio.
- ☐ I can use Pythagoras to find the hypotenuse and a shorter side.
- ☐ I can state which ratio to use and why, given the information I have.
- ☐ I can find an unknown angle using the inverse trigonometric functions.
- ☐ I can solve a height problem involving an angle of elevation.

PREREQUISITE CHECK 10 MINUTES, CALCULATOR

1. State the sum of the angles in a triangle.
2. Find $3^2 + 4^2$ and its square root.
3. Solve $x/4 = 9/6$.
4. Name the longest side of a right triangle.
5. Calculate $24 \div 1.6$.
6. Convert $45°$ to a decimal of a right angle.

Answers are in the chapter answer section. If more than two are incorrect, review ratio, Pythagoras' theorem and solving simple equations before continuing.



A vertical post and the shadow it casts. The ratio of height to shadow length is the same for every upright object at the same time of day, which is how a height can be measured with a tape.
Original photograph made for this edition.

Explore: the ratio that does not change

Draw three right triangles with a 35° angle but different sizes. Measure their sides and compare the ratios. If the ratios are the same, you have found the fact that makes trigonometry possible.

YOUR TASK BEFORE YOU ARE TAUGHT

1. Draw a right triangle with a 35° angle and a horizontal side of 4 cm. Measure the opposite side to the nearest millimetre.
2. Repeat with horizontal sides of 6 cm and 10 cm, keeping the 35° angle.
3. For each triangle, record the opposite side, the adjacent side, the hypotenuse, and the ratios opposite ÷ hypotenuse and opposite ÷ adjacent.
4. Compare the ratios across your three triangles. Record the largest difference between them.
5. Write a conjecture about the ratios of right triangles with equal angles, and state what would falsify it.
6. Explain why measurement error makes it impossible to confirm the conjecture by drawing alone.

Section 7.2 shows why the ratios must be equal, from similarity, and gives them names.

Where this chapter sits in the sequence.

BEFORE YEAR 3

- Ratio and proportion.
- Pythagoras' theorem for right triangles.
- Angle facts in triangles and parallel lines.

YEAR 3 THIS CHAPTER

- Similar triangles and scale factors.
- The three trigonometric ratios.
- Solving right triangles; elevation and depression.
- Similarity in three dimensions.

LATER

- The sine and cosine rules for non-right triangles.
- Trigonometric graphs and identities.
- Vectors and coordinate geometry of circles.

Inquiry pathway for this chapter

	Stage	What you do
1	Context	Measuring the height of a building from its shadow.
2	Question	How can a length be found without being measured?
3	Concept	Similarity, ratio, sine, cosine, tangent.
4	Investigation	Compare side ratios of right triangles at a fixed angle.
5	Communication	Present a measurement method another group can repeat.
6	Reflection	How accurate is a height measured this way, and why?

7.1 Similar triangles

CRITERION A≣ • SIMILARITY • SCALE

Similar triangles have the same angles and proportional sides. The second follows from the first, which is what makes similarity useful: you can establish it from angles alone.

Two triangles are **similar** when their corresponding angles are equal, or equivalently when their corresponding sides are in the same ratio. The constant ratio is the **scale factor** between them. Because the angle sum is 180□ two equal angles are enough: the third is forced.

DEFINITION

Triangles ABC and DEF are similar, written $ABC \square DEF$, when $A = D$, $B = E$, $C = F$ and $AB/DE = BC/EF = CA/FD = k$.

KEY IDEA

One fact, many consequences

Similarity is congruence without the size condition. Everything that follows < the trigonometric ratios, the area rule, the volume rule for models < is a consequence of the single fact that corresponding ratios in similar figures are equal.

WORKED EXAMPLE 7.1.1 FINDING AN UNKNOWN LENGTH

PROBLEM

Triangles ABC and DEF are similar with $AB = 6$ cm, $BC = 9$ cm and $DE = 8$ cm. Find EF.

1. Match corresponding sides: AB with DE, BC with EF.
2. Write the ratio: $AB/DE = BC/EF$, that is $6/8 = 9/EF$.
3. Cross-multiply: $6 \square EF = 72$.
4. So $EF = 12$ cm.

ANSWER $EF = 12$ cm

The scale factor is $8/6 = 4/3$; $9 \square 4/3 = 12$, which agrees.

WORKED EXAMPLE 7.1.2 ESTABLISHING SIMILARITY

PROBLEM

In triangle ABC, DE is parallel to BC with D on AB and E on AC. Prove that triangle ADE is similar to triangle ABC.

1. Angle ADE = angle ABC (corresponding angles, DE parallel to BC).
2. Angle AED = angle ACB (corresponding angles, same reason).
3. Angle A is common to both triangles.
4. All three angles are equal, so ADE ~ ABC.

ANSWER similar because all corresponding angles are equal

Only two equal angles were needed: the third follows from the angle sum.

COMMON ERROR

Mismatched corresponding sides

Matching sides by their order of appearance rather than by their position in the triangle. Corresponding sides are those **between the same pair of equal angles**, not simply the first, second and third listed.

PRACTICE 7.1 SIMILAR TRIANGLES

SUPPORT 1–4 CORE 1–6 EXTENSION 7–9

1. State whether each pair is similar, with a reason: (a) angles 40° 60° in both (b) sides 3, 4, 5 and 6, 8, 10 (c) sides 3, 4, 5 and 4, 5, 6. F
2. Two similar triangles have sides 5, 7, 9 and 10, 14, 18. Find the scale factor. F
3. In similar triangles, $AB = 4$, $BC = 6$, $DE = 10$. Find EF. F
4. A tree casts a 12 m shadow while a 1.8 m person casts a 2.4 m shadow. Find the tree's height. F
5. Explain why knowing two angles of a triangle is enough to establish similarity. R
6. Explain why AAA gives similarity but not congruence. R
7. A photograph 10 cm by 15 cm is enlarged so its width becomes 25 cm. Find the new height and the ratio of the areas. P
8. In triangle ABC, D lies on AB and E on AC with DE parallel to BC. If $AD = 3$, $DB = 5$ and $AE = 4$, find EC. P
9. Two similar triangles have areas 36 cm<and 81 cm< Find the ratio of their corresponding sides. E

7.2 Pythagoras' theorem

CRITERION A • RIGHT TRIANGLE • PROOF

In a right triangle the squares on the two shorter sides add to the square on the hypotenuse. It is the oldest and most used relationship in geometry.

For a right triangle with legs a and b and hypotenuse c, $a^2 + b^2 = c^2$. The converse is also true: if the three sides satisfy the relationship, the triangle is right-angled. The theorem lets you find any side given the other two, and it is the tool that converts trigonometry's ratios into lengths.

THEOREM

Theorem

Pythagoras' theorem. In a right triangle, the square of the hypotenuse equals the sum of the squares of the other two sides: $a^2 + b^2 = c^2$.

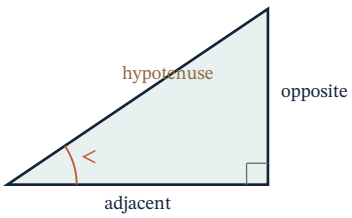


Figure 7.1 The right triangle that defines the three ratios. The labels are relative to the marked angle: opposite, adjacent, hypotenuse which is why the same side can be opposite or adjacent depending on the angle you choose.

WORKED EXAMPLE 7.2.1 FINDING THE HYPOTENUSE

PROBLEM

A right triangle has legs 7 cm and 24 cm. Find the hypotenuse.

1. Apply the theorem: $c^2 = 7^2 + 24^2$.
2. Compute: $49 + 576 = 625$.
3. Take the positive root, since a length is positive: $c = 25$ cm.

ANSWER 25 cm

7, 24, 25 is a Pythagorean triple: $49 + 576 = 625$.

WORKED EXAMPLE 7.2.2 FINDING A SHORTER SIDE

PROBLEM

A ladder 5.2 m long leans against a wall with its foot 1.5 m from the wall. How high does it reach?

1. The ladder is the hypotenuse: $5.2^2 = 1.5^2 + h^2$.
2. Compute: $27.04 = 2.25 + h^2$.
3. Subtract: $h^2 = 24.79$.
4. Take the root: $h \approx 4.98$ m.

ANSWER about 5.0 m

The height must be less than the ladder's length and close to it, because the foot is near the wall.

COMMON ERROR

Adding when you should subtract

Adding the squares when a shorter side is wanted. If the hypotenuse is known, the required square is the difference of the other two: $h^2 = c^2 - a^2$.

COMMON ERROR

A negative length

Taking the negative square root and reporting a negative length. Lengths are positive; the equation has two roots but only one is a possible answer.

PRACTICE 7.2 PYTHAGORAS' THEOREM

SUPPORT 1–4 CORE 1–7 EXTENSION 8–9

1. Find the hypotenuse: (a) legs 3, 4 (b) legs 5, 12 (c) legs 8, 15. F
2. Find the missing shorter side: (a) hypotenuse 13, leg 5 (b) hypotenuse 10, leg 6. F
3. Decide whether each triangle is right-angled: (a) 6, 8, 10 (b) 5, 6, 8 (c) 9, 12, 15. F
4. A rectangle is 12 cm by 5 cm. Find the length of its diagonal. F
5. Explain why the theorem fails for a triangle that is not right-angled, and what relationship would hold instead. R
6. A 6 m ladder leans against a wall and reaches 5.5 m up. How far is its foot from the wall? P
7. Find the length of the diagonal of a cube with edge 4 cm, giving your answer in the form $k\sqrt{3}$. P
8. A triangle has sides $n^2 - 1$, $2n$ and $n^2 + 1$. Show that it is right-angled for every $n > 1$, and generate two examples. E
9. Explain why a triangle with sides 5, 5 and 11 cannot exist, using the triangle inequality as well as Pythagoras. E

7.3 The trigonometric ratios

CRITERION A II • RATIO • ANGLE

Because all right triangles with a given acute angle are similar, the ratios of their sides depend only on the angle. Those ratios are the sine, cosine and tangent.

For an acute angle θ in a right triangle, the **sine** is the opposite side over the hypotenuse, the **cosine** is the adjacent over the hypotenuse, and the **tangent** is the opposite over the adjacent. The ratios are properties of the angle, not of the triangle, which is why a calculator can return a single value for sin 35°.

DEFINITION

$\sin\theta = \mathit{opp}/\mathit{hyp}$, $\cos\theta = \mathit{adj}/\mathit{hyp}$, $\tan\theta = \mathit{opp}/\mathit{adj}$. The mnemonic SOH-CAH-TOA names the three definitions in order.

KEY IDEA

Why one value is enough

The ratios' constancy is a consequence of similarity, not an independent fact. Any two right triangles with a 35° angle are similar (they have two equal angles), so their corresponding ratios are equal. If the triangles were not similar, no table of values would be possible.

WORKED EXAMPLE 7.3.1 CHOOSING AND USING A RATIO

PROBLEM

In a right triangle, the angle □ = 32□and the hypotenuse is 18 cm. Find the side opposite □.

1. The known side is the hypotenuse and the required side is the opposite, so use sine.
2. sin 32□= *opp*/18.
3. Rearrange: *opp* = 18 □ sin 32□
4. Compute: 18 □ 0.5299 □ 9.54 cm.

ANSWER about 9.5 cm

The opposite side must be shorter than the hypotenuse, and it is.

WORKED EXAMPLE 7.3.2 FINDING AN ANGLE

PROBLEM

In a right triangle the side opposite an angle is 7 cm and the adjacent side is 12 cm. Find the angle.

1. Opposite and adjacent are known, so use tangent.
2. tan □= 7/12 □ 0.5833.
3. Use the inverse function: □= $\tan^{-1}(0.5833)$.
4. □□ 30.3□

ANSWER about 30□

The tangent of 30□is 1/□3 □ 0.577, and 0.583 is close to it.

COMMON ERROR

Opposite and adjacent swapped

Using the wrong ratio because the sides were chosen by their position on the page rather than by their relationship to the angle. <Opposite<means opposite the angle you are using; if you change the angle, the labels change too.

PRACTICE 7.3 ≣TRIGONOMETRIC RATIOS

SUPPORT 1≣4 ≣CORE 1≣7 ≣EXTENSION 8≣9

1. Write the three ratios for the angle marked in a 3<4<5 right triangle, where the side opposite is 3. F

PRACTICE 7.3 TRIGONOMETRIC RATIOS (CONTINUED)

SUPPORT 1–4 CORE 1–7 EXTENSION 8–9

1. Find, to 3 significant figures: (a) $\sin 48°$ (b) $\cos 71°$ (c) $\tan 15°$ F
2. Find the angle, to the nearest degree: (a) $\sin^{-1}(0.5)$ (b) $\cos^{-1}(0.5)$ (c) $\tan^{-1}(1)$. F
3. A right triangle has hypotenuse 20 cm and one angle $40°$. Find both shorter sides. F
4. Explain why $\sin 30° = 0.5$ exactly, using an equilateral triangle. R
5. Explain why the tangent of an angle can be greater than 1 while the sine cannot. R
6. A ramp rises 1.2 m over a horizontal run of 8.5 m. Find the angle of the ramp. P
7. Show that $\sin^2\theta + \cos^2\theta = 1$ using the definitions and Pythagoras' theorem. E
8. Find the exact values of $\sin 45°$, $\cos 45°$ and $\tan 45°$ from an isosceles right triangle. E

7.4 Solving right triangles

CRITERION A V • MODELLING • MEASUREMENT

To solve a right triangle is to find every unknown side and angle. It takes one side and one acute angle, or two sides, and a decision about which ratio uses only what you know.

The procedure is always the same: draw and label the triangle, mark the angle you are using, identify which sides are opposite, adjacent and hypotenuse, choose the ratio that connects the known and required quantities, then solve and check that the answer is sensible.

You know	You want	Use
angle and hypotenuse	opposite	sine
angle and hypotenuse	adjacent	cosine
angle and adjacent	opposite	tangent
two sides	the angle between them	inverse tangent

A decision table for choosing the ratio. Reading it before reaching for the calculator prevents most errors.

WORKED EXAMPLE 7.4.1 A HEIGHT FROM A DISTANCE

PROBLEM

From a point 30 m from the base of a tower, the angle of elevation of the top is 62°. Find the height of the tower.

1. The horizontal distance is adjacent to the 62° angle; the height is opposite it.
2. So use tangent: $\tan 62° = h/30$.
3. Rearrange: $h = 30 \times \tan 62°$.
4. Compute: $30 \times 1.8807 \approx 56.4$ m.

ANSWER about 56 m

The angle is steep, so the height should be much greater than the distance; 56 m against 30 m is consistent.

WORKED EXAMPLE 7.4.2 AN ANGLE OF DEPRESSION

PROBLEM

From the top of a 40 m cliff, a boat is seen at an angle of depression of 25°. How far is the boat from the foot of the cliff?

1. The angle of depression equals the angle of elevation from the boat, by alternate angles: 25°.
2. The cliff height is opposite that angle; the distance is adjacent.
3. $\tan 25° = 40/d$.
4. So $d = 40 / \tan 25° \approx 40 / 0.4663 \approx 85.8$ m.

ANSWER about 86 m

A shallow angle of 25° should give a distance much greater than the height, and it does.

COMMON ERROR

Measuring the angle from the wrong line

Using the angle of depression as if it were measured from the vertical. An angle of depression is measured from the horizontal, and it equals the angle of elevation from the object by alternate angles.

NOTE

Reporting the answer honestly

Angles measured with a clinometer are rarely better than to the nearest degree, and distances to the nearest metre. A height calculated from them should be reported to two significant figures, not six.

PRACTICE 7.4 SOLVING RIGHT TRIANGLES

SUPPORT 1–4 CORE 1–6 EXTENSION 7–9

1. Find the unknown side: a right triangle with hypotenuse 15 cm and an angle of 35° find the side opposite. F
2. Find the unknown angle: a right triangle with opposite 4.5 cm and adjacent 7.2 cm. F
3. A flagpole casts a 9 m shadow when the sun's elevation is 48°. Find the pole's height. F
4. From 50 m away, the angle of elevation of a building's top is 31°. Find its height. F

PRACTICE 7.4 SOLVING RIGHT TRIANGLES (CONTINUED)

SUPPORT 1–4 CORE 1–6 EXTENSION 7–9

1. Explain why the angle of depression from A to B equals the angle of elevation from B to A. R
2. A ladder 4.5 m long makes an angle of 68° with the ground. How far up the wall does it reach, and how far is its foot from the wall? P
3. An aircraft is 1 200 m above the ground and 3 500 m horizontally from the start of the runway. Find its angle of descent. P
4. Two observers 80 m apart on level ground measure the elevation of a balloon as 34° and 52°. Explain why the balloon is not necessarily above the midpoint, and describe how to find its height. E
5. A clinometer reading has an uncertainty of ±1°. For an angle of 45° at 30 m, estimate how much the calculated height could change. E

7.5 Similarity in three dimensions

CRITERION A • SCALE • VOLUME

In three dimensions, a scale factor k multiplies lengths by k, surface areas by k^2 and volumes by k^3. This one fact explains why models are misleading and why large animals need thick legs.

Two solids are similar when one is an enlargement of the other. Their corresponding lengths are in the ratio k, their surface areas in the ratio k^2, and their volumes in the ratio k^3. The consequences are physical: doubling a solid's dimensions multiplies its weight by 8 while its cross-section only grows by 4, so the stress on its structure increases.

KEY IDEA

The cube rule and what it forbids

This is why a scale model cannot be used to test strength, and why a 1 : 24 model car does not weigh 1/24 of the real car < it weighs $1/24^3$, that is 1/13 824. A model can test shape; it cannot test load without a correction.

WORKED EXAMPLE 7.5.1 VOLUMES OF SIMILAR SOLIDS

PROBLEM

Two similar cans have heights 10 cm and 15 cm. The smaller holds 330 mL. Find the capacity of the larger.

1. The scale factor is $15/10 = 1.5$.
2. Volumes scale as the cube: $1.5^3 = 3.375$.
3. Capacity: $330 \times 3.375 = 1113.75$ mL.

ANSWER about 1 110 mL, that is 1.11 litres

The larger can is half as tall again but holds more than three times as much < which is the point of the cube rule.

WORKED EXAMPLE 7.5.2 SURFACE AREA OF A MODEL

PROBLEM

A 1 : 50 model of a building has a surface area of 2.4 m< Find the surface area of the real building.

1. Lengths scale by 50, so areas scale by $50^2 = 2500$.
2. Multiply: 2.4 ? 2500 = 6000 m<

ANSWER 6 000 m<

The answer is 2 500 times larger, and for a 50-fold enlargement that is the expected order of magnitude.

COMMON ERROR

Scaling volume by k instead of k^3

Scaling a volume by the scale factor rather than its cube. A model at 1 : 10 needs $10^3 = 1000$ times less material, not 10 times less.

PRACTICE 7.5 SIMILARITY IN THREE DIMENSIONS

SUPPORT 1 CORE 1 EXTENSION 7

1. Two similar cubes have edges 3 cm and 6 cm. Find the ratios of their surface areas and volumes. F
2. A model is built at 1 : 20. By what factor are lengths, areas and volumes scaled? F
3. Two similar bottles hold 250 mL and 2 000 mL. Find the ratio of their heights. F
4. A photograph on a poster is 4 times the width and height of the original. Explain how much more ink the larger print needs. F
5. Explain, using the cube rule, why an elephant's legs are proportionally much thicker than a mouse's. R
6. A statue 2 m tall is modelled at 1 : 8. The model needs 300 g of clay. Estimate the clay needed for the statue at the same density. P
7. Two similar cones have volumes in the ratio 27 : 64. Find the ratio of their radii and of their surface areas. P
8. Explain why a scale model cannot be used to test whether a bridge design will hold, even if the shape is exact. E

An investigation is not a longer exercise. It is a short piece of mathematics you do yourself: you gather evidence, look for a pattern, say what you think is true, and then test it until you can defend it.

INVESTIGATION

Measuring the unmeasurable

Choose a height you cannot measure directly — a building, a tree, a lamp post. You will measure it two ways and compare the results.

Method	What you measure	What you calculate	Assumption
Shadow	shadow lengths of object and metre rule	similar triangles	the sun's rays are parallel
Clinometer	distance and angle of elevation	tan ratio	the ground is level
Photograph	pixel heights of object and a person	ratio	both are the same distance away

1. **Plan.** Choose your object and two methods from the table. Describe exactly what you will measure and with what instrument.
2. **Measure.** Record every measurement with its precision, and repeat each one three times.
3. **Calculate.** Compute the height by both methods, showing the geometry and the ratio you used.
4. **Compare.** State the two results and their difference as a percentage of the mean.
5. **Explain.** Identify the assumption in each method and say how a violation of it would bias the result.
6. **Conclude.** Decide which method you trust more, and justify your decision using your own data.

Present a labelled diagram for each method, a table of readings, both calculations, and a conclusion that compares the methods on evidence.

APPLICATION

The 330 mL can

A drinks can holds 330 mL. The manufacturer wants a taller can of the same shape holding 500 mL, keeping the same proportions.

Can	Height (mm)	Diameter (mm)	Volume (mL)
Current	115	66	330
Proposed	<	<	500

1. Find the scale factor for volume between the two cans, and use it to find the linear scale factor.
2. Calculate the proposed can's height and diameter, giving each to the nearest millimetre.
3. Calculate both surface areas and find the percentage increase in material needed.
4. Explain why the material does not increase in the same proportion as the volume, and what that means for the cost per millilitre.
5. The manufacturer instead chooses to keep the height the same and increase the diameter. Explain why the can is no longer similar to the original, and why that matters for the surface-area calculation.

CRITERION C COMMUNICATING

A method a field team could use

Write a one-page field method for measuring the height of a building using only a tape measure and a clinometer, for a team with no mathematical training beyond this chapter.

1. Explain, with a diagram, why the method works, using similar triangles or a trigonometric ratio.
2. Give the procedure step by step, including what to record and in what units.
3. Explain how to calculate the height from the readings, with one worked example.
4. State the assumptions, and explain what the team should do if the ground slopes or the building is not vertical.

1–2 The procedure is given but the geometry is not explained or is wrong.

3–4 The geometry is correct and the procedure is usable, but the assumptions are not stated.

5–6 The diagram, procedure and calculation are correct and clear, and the assumptions are identified.

7–8 As above, and the team is told how to recognise and correct a violated assumption.

Chapter summary and review

Key ideas

- Similar triangles have equal angles and proportional sides.
- Two equal angles are enough to establish similarity.
- Pythagoras: $a^2 + b^2 = c^2$ in a right triangle.
- $\sin = opp/hyp$, $\cos = adj/hyp$, $\tan = opp/adj$.
- Trigonometric ratios depend only on the angle, because of similarity.
- An angle of depression equals the corresponding angle of elevation.
- In similar solids, lengths scale as k, areas as k^2, volumes as k^3.
- Report a calculated measurement to the precision the measurements support.

Vocabulary

similar
Same shape, possibly different size.

scale factor
The constant ratio between corresponding lengths.

hypotenuse
The side opposite the right angle.

opposite
The side across from the angle in use.

adjacent
The side next to the angle in use, other than the hypotenuse.

sine
Opposite divided by hypotenuse.

cosine
Adjacent divided by hypotenuse.

tangent
Opposite divided by adjacent.

angle of elevation
The angle measured upward from the horizontal.

angle of depression
The angle measured downward from the horizontal.

What students often think	What is actually true
AAA proves congruence.	It proves similarity; the size is not determined.
Opposite and adjacent are fixed labels.	They depend on which angle you are using.
A scale factor of 2 doubles the volume.	Volume is multiplied by 8.

Misconceptions worth confronting directly.

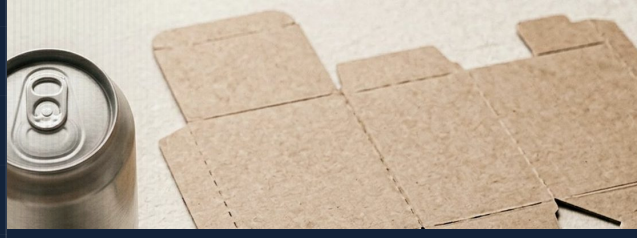
What students often think	What is actually true
The angle of depression is measured from the vertical.	It is measured from the horizontal.

Revision checklist

- ☐ I can establish similarity and find the scale factor.
- ☐ I can find unknown lengths using ratios.
- ☐ I can use Pythagoras in both directions.
- ☐ I can choose the correct trigonometric ratio from the information given.
- ☐ I can find an unknown angle with an inverse function.
- ☐ I can solve an elevation or depression problem with a diagram.
- ☐ I can apply the k, k^2, k^3 rule to similar solids.
- ☐ I can state the assumptions in a measurement method.

REFLECT

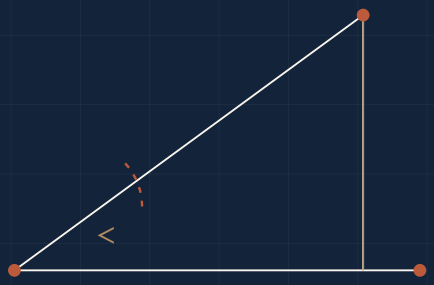
Write three problems of your own: one that needs sine, one cosine and one tangent. Exchange with a partner and check that each problem can be solved with the ratio intended.



8

SHAPE, SPACE AND REASONING

Solids, measurement and accuracy



A solid is described by its faces, its surface and the space it encloses. Measuring a solid is not one question but three, and each needs its own formula and its own units.

KEY CONCEPT FORM | RELATED SPACE | RELATED QUANTITY

CONTEXT SCIENTIFIC AND TECHNICAL INNOVATION

INQUIRY

How do we measure a three-dimensional object, and how do we know when a measurement is precise enough to be useful?

ATL SELF-MANAGEMENT AND THINKING SKILLS

Find a cylindrical or prism-shaped container in your home. Measure it with the most accurate instrument available, calculate its capacity, and compare with the capacity printed on the label. Explain any difference in terms of measurement error rather than in terms of the manufacturer being wrong.

MYP UNIT PLANNER THE WHOLE CHAPTER IN ONE VIEW

Framing the inquiry

Unit 8 approximately 17 hours of class time ATL focus: self-management and thinking

KEY AND RELATED CONCEPTS

Form. The formula for a solid's volume follows from its form: a prism is a constant cross-section swept through a length; a pyramid is a third of the prism that contains it.

Space. Surface area is a two-dimensional measure of a three-dimensional object. Nets make the relationship visible, and the units differ from those of volume.

Quantity. Every measurement has a precision, and a calculated volume inherits the precision of its least precise input. Accuracy is a claim about the measurement, not about the arithmetic.

GLOBAL CONTEXT AND EXPLORATION

Scientific and technical innovation. Packaging, engineering tolerances and drug dosing all depend on volume and surface area calculated to a stated precision. The difference between 330 mL and 331 mL is trivial in a drinks can and critical in a medical dose; the mathematics is the same, and knowing which precision matters is professional judgement built on this chapter.

STATEMENT OF INQUIRY

In order for a student to measure and describe a three-dimensional object reliably, students must be able to select and apply the correct formula for volume and surface area, use nets to explain the formulas, and report a calculated quantity with a precision the measurements support.

FACTUAL

<What is the difference between volume and capacity?

<How do you find the surface area of a cylinder?

<What is a net?

CONCEPTUAL

<Why does a pyramid have one third of the volume of the prism with the same base and height?

DEBATABLE

<Should packaging be designed to minimise material, or to be convenient < and what does mathematics have to say about the choice?

Criterion	Strands	Evidence in this chapter
A Knowing and understanding	i, ii, iii	Calculate volumes and surface areas of prisms, cylinders, pyramids, cones and spheres; convert units.
B Investigating patterns	i, ii, iii, iv	Investigate which cylinder of a fixed volume uses least material; generalise and justify.
C Communicating	i, ii, iv	Use nets to explain a formula; report results with correct units and precision.
D Applying mathematics	i, ii, iii, v	Solve a packaging problem, compare designs, and justify the recommendation.

Each teaching section closes with a graded practice set. The investigation addresses Criterion B; the application addresses Criterion D; the criterion task asks for extended written communication (Criterion C) about knowledge and understanding (Criterion A).

Learning intentions and success criteria

The formulas in this chapter are not a list to memorise. Each one can be derived from a net or from a simple solid, and knowing where a formula comes from is what lets you use it with confidence on a shape you have never seen.

YOU ARE LEARNING TO

- **Know** the formulas for the volume and surface area of prisms, cylinders, pyramids, cones and spheres.
- **Understand** why the volume of a pyramid is one third of the enclosing prism.
- **Be able to** use a net to derive a surface-area formula.
- **Be able to** convert between units of length, area and volume.
- **Be able to** calculate the capacity of a container and compare it with a stated value.
- **Be able to** report a calculated volume with an appropriate precision.

SUCCESS CRITERIA I CAN

- ☐ I can calculate the volume of a prism and state the units.
- ☐ I can calculate the surface area of a cylinder without a formula sheet.
- ☐ I can explain why the volume of a cone is $\frac{1}{3}\pi r^2 h$.
- ☐ I can convert 1.5 m to litres.
- ☐ I can find the capacity of a container and compare it with the labelled value.
- ☐ I can state how precisely a calculated volume should be reported.

PREREQUISITE CHECK 10 MINUTES, CALCULATOR

1. Calculate the area of a circle with radius 5 cm.
2. Calculate 6^3.
3. Convert 2.5 m to cm.
4. Convert 1 m to cm
5. Calculate the area of a triangle with base 8 cm and height 5 cm.
6. State the value of π to three significant figures.

Answers are in the chapter answer section. If more than two are incorrect, review area of plane figures, powers and unit conversion before continuing.



A can and its flattened packaging net. The net turns a three-dimensional problem into a two-dimensional one: every face of the solid appears once, with its true shape.
Original photograph made for this edition.

Explore: which box holds more

Two cuboid boxes have the same surface area of 216 cm< One is 6 □ 6 □ 6 cm; the other is 9 □ 4 □ 6 cm. Before calculating, decide which you expect to hold more.

YOUR TASK ≣BEFORE YOU ARE TAUGHT

1. Calculate both volumes, showing each step.
2. Calculate both surface areas to confirm they really are equal.
3. Record both results in a table with a column for volume □ surface area.
4. Find one more cuboid with surface area 216 cm<that is neither of the above, and record its volume.
5. Plot volume against one dimension for the boxes you have, and describe the shape of the relationship.
6. Write a conjecture about which shape maximises volume for a fixed surface area, and state what evidence would support it.

Section 8.2 explains why the cube wins, and the investigation at the end of the chapter applies the same reasoning to a cylindrical can.

BEFORE YEAR 3	YEAR 3 THIS CHAPTER	LATER
▪ Area of rectangles, triangles and circles. ▪ Units of length and simple conversions. ▪ Naming 3D shapes and counting faces, edges and vertices.	▪ Volume and surface area of prisms and cylinders. ▪ Volume of pyramids, cones and spheres. ▪ Nets and the derivation of surface-area formulas. ▪ Units of volume and capacity; precision of measurement.	▪ Similar solids and the cube rule. ▪ Surface area and volume in calculus. ▪ Optimisation problems in engineering.

Where this chapter sits in the sequence.

Inquiry pathway for this chapter

	Stage	What you do
1	Context	A 330 mL drinks can and a shoebox.
2	Question	How much material does a container need, and how much can it hold?
3	Concept	Volume, surface area, capacity, net, precision.
4	Investigation	Find the cylinder of a given volume that uses least material.
5	Communication	Report a capacity with units and an honest precision.
6	Reflection	Why is the cheapest can not always the best one?

8.1 Prisms and cylinders

CRITERION A • VOLUME • CROSS-SECTION

A prism is a shape with a constant cross-section. Its volume is the area of that section multiplied by the distance it is swept through.

A **prism** has the same cross-section along its whole length. Its volume is $A \times l$, where A is the cross-sectional area and l the length. A **cylinder** is a prism whose cross-section is a circle, so its volume is $\pi r^2 h$. Its surface area is the two circular ends plus the rectangle that unrolls from the curved surface: $2\pi r^2 + 2\pi rh$.

DEFINITION

Volume of a prism $= A \times l$. Volume of a cylinder $= \pi r^2 h$. Surface area of a cylinder $= 2\pi r^2 + 2\pi rh$. **Capacity** is the volume of space inside a container, usually given in litres or millilitres.

KEY IDEA

Unroll the surface

The curved surface of a cylinder unrolls into a rectangle whose width is the circumference $2\pi r$ and whose height is h. That is the whole derivation of the surface-area formula, and it is worth being able to reproduce it rather than recall it.

WORKED EXAMPLE 8.1.1 A TRIANGULAR PRISM

PROBLEM

A prism has a right-triangular cross-section with legs 6 cm and 8 cm, and a length of 15 cm. Find its volume.

1. Find the cross-sectional area: $\tfrac{1}{2} \times 6 \times 8 = 24$ cm^2.
2. Multiply by the length: $24 \times 15 = 360$.
3. State the units: the answer is in cm^3.

ANSWER 360 cm^3

The prism is equivalent to a cuboid of base area 24 cm^2 and height 15 cm, which gives the same volume.

WORKED EXAMPLE 8.1.2 A CYLINDER

PROBLEM

A can has radius 3.3 cm and height 11.5 cm. Find its volume and its total surface area.

1. Volume: $V = \pi r^2 h$.
2. Compute: $V \approx 10.89 \times 11.5 \approx 393$ cm^3.
3. Surface area: $2\pi(3.3)^2 + 2\pi(3.3)(11.5)$.
4. Compute: $68.4 + 238.4 \approx 307$ cm^2.

ANSWER volume about 393 cm^3; surface area about 307 cm^2.

393 cm^3 is 393 mL, close to the 330 mL label — the difference is the headspace and the can's own thickness.

COMMON ERROR

Radius and diameter confused

Using the diameter in place of the radius. Every circle formula is written in terms of r; halve the diameter first, and the error of a factor of 4 disappears.

COMMON ERROR

The wrong power in the units

Reporting a volume in cm^3. Volume is always cubic: cm^3, m^3, mL.

PRACTICE 8.1 PRISMS AND CYLINDERS

SUPPORT 1–4 CORE 1–7 EXTENSION 8–9

1. Find the volume of a cuboid 7 cm by 4 cm by 3 cm. F
2. Find the volume of a triangular prism with cross-section base 9 cm, height 4 cm and length 12 cm. F
3. Find the volume of a cylinder with radius 4 cm and height 10 cm, to 3 significant figures. F
4. Find the total surface area of a cylinder with radius 5 cm and height 12 cm. F
5. A pipe has an outer radius of 4 cm and an inner radius of 3 cm, and a length of 2 m. Find the volume of material in it. R
6. Explain why the volume of a cylinder is $\pi r^2 h$ and not $2\pi rh$. R
7. A water tank is a cylinder of radius 1.2 m and height 2.5 m. Find its capacity in litres. P
8. A label wraps exactly once around a can of radius 3.3 cm. Find the label's width and explain the calculation. P
9. An open-topped cylinder of volume 500 cm<has height equal to its radius. Find the radius to 3 significant figures. E

8.2 Pyramids, cones and spheres

CRITERION A I • VOLUME • DERIVATION

A pyramid is one third of the prism that contains it. That single fact explains the cones, and the sphere's formula follows from a different argument altogether.

For a pyramid or cone with base area A and perpendicular height h, the volume is $\frac{1}{3}Ah$. For a cone with base radius r this becomes $\frac{1}{3}\pi r^2 h$. The volume of a sphere is $\frac{4}{3}\pi r^3$, and its surface area is $4\pi r^2$ — the surface area being exactly four times the area of its greatest circle.

KEY IDEA

Three cones make a cylinder

Why one third? Fill a cone and pour it into the cylinder of the same base and height: it takes three cones to fill it. The result can be proved by slicing, but the physical demonstration is the reason the formula is memorable, and the reason it is not a coincidence.

Solid	Volume	Surface area
Prism	$A\,l$	sum of faces
Cylinder	$\pi r^2 h$	$2\pi r^2 + 2\pi rh$
Pyramid	$\frac{1}{3}\,A\,h$	sum of faces
Cone	$\frac{1}{3}\pi r^2 h$	$\pi r^2 + \pi rl$
Sphere	$\frac{4}{3}\pi r^3$	$4\pi r^2$

In the cone's surface area, l is the slant height: $l = \sqrt{(r^2 + h^2)}$.

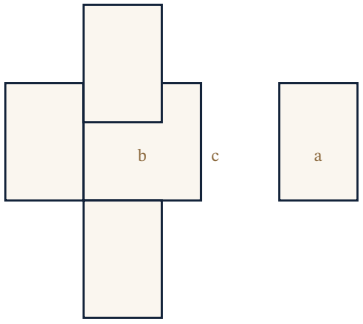


Figure 8.1 A net of a cuboid. Every face appears once with its true shape, which is why the net is the honest way to derive a surface-area formula.

WORKED EXAMPLE 8.2.1 A CONE

PROBLEM

A cone has radius 6 cm and height 8 cm. Find its volume and its curved surface area.

1. Volume: $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \times 36 \times 8$.
2. Compute: $V = 96\pi \approx 302$ cm³.
3. Slant height: $l = \sqrt{(6^2 + 8^2)} = \sqrt{100} = 10$ cm.
4. Curved surface area: $A = \pi r l = \pi \times 6 \times 10 = 60\pi \approx 188$ cm².

ANSWER volume about 302 cm³, curved area about 188 cm²

The slant height 10 comes from the 6–8–10 triangle, which is a scaled 3–4–5.

WORKED EXAMPLE 8.2.2 A SPHERE

PROBLEM

A ball has radius 11 cm. Find its volume and surface area.

1. Volume: $V = \frac{4}{3}\pi r^3$.
2. Compute: $V = \frac{4}{3}\pi \times 1331 \approx 5575$ cm³.
3. Surface area: $A = 4\pi r^2$.
4. Compute: $A = 4\pi \times 121 = 484\pi \approx 1521$ cm².

ANSWER volume about 5 580 cm³, surface area about 1 520 cm²

A sphere's surface area equals four great circles: $4\pi \times 121 = 484\pi \approx 1521$, agreeing.

COMMON ERROR

Height used for slant height

Using the height instead of the slant height in the cone's curved surface area. πrl needs l, the length along the surface, not h.

PRACTICE 8.2 PYRAMIDS, CONES AND SPHERES

SUPPORT 1–4 CORE 1–6 EXTENSION 7–9

1. Find the volume of a square pyramid with base 6 cm and height 10 cm. F
2. Find the volume of a cone with radius 5 cm and height 9 cm. F
3. Find the volume of a sphere with radius 3 cm, in terms of π. F
4. Find the surface area of a sphere with diameter 14 cm. F
5. Show that the volume of a cone is one third of the cylinder with the same base and height. R
6. Explain why the surface area of a sphere is four times the area of its greatest circle. R
7. An ice-cream cone has radius 3 cm, height 10 cm and is filled level with the rim, plus a hemisphere of ice cream on top. Find the total volume. P
8. A hemisphere has volume 100 cm< Find its radius. E
9. Two cones have the same volume but the second has twice the radius. Find the ratio of their heights. E

8.3 Nets and surface area

CRITERION A ≣ II • NET • SURFACE

A net is a flat pattern that folds into a solid. It is how you see, rather than remember, where a surface-area formula comes from.

Every face of a solid appears exactly once on its net, undistorted. The surface area is therefore the sum of the areas of the shapes on the net. For a cuboid the net has six rectangles; for a cylinder, two circles and one rectangle; for a cone, one circle and one sector.

KEY IDEA

Flat and solid are different measurements

A net also settles arguments about units. A flat pattern is measured in square units; the solid it folds into is measured in cubic units. The two numbers are never interchangeable, and seeing one folded into the other makes the distinction concrete.

WORKED EXAMPLE 8.3.1 SURFACE AREA FROM A NET

PROBLEM

A cuboid measures 8 cm by 5 cm by 4 cm. Find its surface area using a net.

1. The net has three pairs of matching rectangles.
2. Two faces are $8 \times 5 = 40$; two are $8 \times 4 = 32$; two are $5 \times 4 = 20$.
3. Total: $2(40 + 32 + 20) = 2 \times 92 = 184$.
4. The surface area is 184 cm<

ANSWER 184 cm<

Summing all six faces individually gives the same total, which is what the net shows.

COMMON ERROR

Counting three faces instead of six

Forgetting that each pair of opposite faces appears twice. A cuboid has six faces, not three, and the factor 2 in $2(lw + lh + wh)$ is the step most often omitted.

PRACTICE 8.3 NETS AND SURFACE AREA

SUPPORT 1–3 CORE 1–6 EXTENSION 7–8

1. Find the surface area of a cuboid 10 cm by 6 cm by 3 cm. F
2. Sketch a net of a cube of edge 4 cm and find its surface area. F
3. Find the surface area of a cylinder with radius 7 cm and height 2 cm, in terms of π. F
4. A cube of edge 5 cm is painted and then cut into 1 cm cubes. Explain how many small cubes have exactly two painted faces. R
5. Explain why the net of a cone contains a sector rather than a triangle. R
6. A box with a lid is made from a net of area 500 cm< If the wastage is 8 per cent, find the area of card needed. P
7. An open box is 20 cm by 15 cm by 10 cm. Find the area of card needed, and explain why it is less than the closed box. P
8. Design a net for a cuboid with volume 60 cm<and surface area 94 cm< Show that your dimensions satisfy both. E

8.4 Units of volume and precision

CRITERION A≣V • UNITS • ACCURACY

One cubic metre is a thousand litres, because each of the three dimensions converts by ten. Getting the units right is half of getting the answer right.

Length, area and volume convert differently: 1 m = 100 cm, so 1 m<= 10 000 cm<and 1 m<= 1 000 000 cm< For capacity, 1 cm<= 1 mL, 1 000 cm<= 1 litre and 1 m<= 1 000 litres.

From	To	Multiply by
m	cm	100
m<	cm<	10 000
m<	cm<	1 000 000
m<	litres	1 000
litres	mL	1 000

The exponent in the unit is the exponent in the conversion factor.

WORKED EXAMPLE 8.4.1 CONVERTING A VOLUME

PROBLEM

A tank has volume 0.75 m< Express this in litres and in cm<

1. 1 m<= 1 000 litres, so 0.75 □ 1 000 = 750 litres.
2. 1 m<= 1 000 000 cm< so 0.75 □ 1 000 000 = 750 000 cm<
3. Check consistency: 750 litres = 750 □ 1 000 = 750 000 mL = 750 000 cm<

ANSWER 750 litres, or 750 000 cm<

Both conversions agree, which confirms the factors.

WORKED EXAMPLE 8.4.2 PRECISION OF A CALCULATED

PROBLEM

A cuboid measures 4.2 cm by 3.1 cm by 2.6 cm. Calculate its volume and state an appropriate precision.

1. Multiply: 4.2 × 3.1 × 2.6 = 33.852.
2. The least precise input, 4.2 cm, has two significant figures.
3. A product cannot exceed the precision of its least precise factor.
4. Report the volume to two significant figures.

ANSWER about 34 cm<

If each measurement is out by 0.05 cm, the volume can vary by about 1.5 cm<< so the second digit is already uncertain.

COMMON ERROR

A precision the measurements do not support

Reporting 33.852 cm<from measurements given to two significant figures. The arithmetic is exact but the claim of precision is false.

PRACTICE 8.4 UNITS AND PRECISION

SUPPORT 1 CORE 1 EXTENSION 7

1. Convert: (a) 3.5 m<to litres (b) 450 mL to cm<(c) 2 500 cm<to litres. F
2. Convert: (a) 1.2 m<to cm<(b) 0.5 m<to cm< F
3. A carton measures 24 cm by 16 cm by 9 cm. Find its capacity in litres. F
4. A cuboid measures 7.5 cm by 4.2 cm by 3.1 cm. Calculate the volume and state an appropriate precision. R
5. Explain why 1 m<is a thousand litres and not a hundred. R
6. A swimming pool is 25 m by 12 m and holds 600 m< Find its average depth. P
7. A bottle is labelled 750 mL. Its measured internal dimensions give 748 mL. Calculate the percentage difference and explain whether the label is wrong. P
8. A manufacturer states a tin's capacity as 400 mL ± 2 per cent. Find the range of acceptable volumes and decide whether a tin measuring 407 mL passes inspection. E

An investigation is not a longer exercise. It is a short piece of mathematics you do yourself: you gather evidence, look for a pattern, say what you think is true, and then test it until you can defend it.

INVESTIGATION

The 330 mL can

A cylindrical can must hold 330 mL. The manufacturer wants the design that uses the least metal. Your task is to find the dimensions, and to decide whether the design that uses least metal is the one that should be made.

Radius (cm)	Height for 330 mL (cm)	Surface area (cm²)
3.0	11.67	277
3.3	9.65	268
3.7	7.67	265
4.0	6.57	265
4.5	5.19	274

1. **Compute.** Using $h = 330/(\pi r^2)$, complete the table for $r = 2.5$ and $r = 5.0$.
2. **Plot.** Draw a graph of surface area against radius. Describe its shape and locate the minimum approximately.
3. **Describe.** As the radius increases, the height falls. Explain in words why the surface area first falls and then rises.
4. **Extend.** Find the radius that minimises surface area to one decimal place, and state the corresponding height.
5. **Compare.** The real can has radius about 3.3 cm. Explain why a manufacturer might choose a design that is not the mathematical minimum.
6. **Conclude.** Recommend a design, justifying it in terms of material, stability and handling, and say which consideration is mathematical and which is not.

Present the completed table, the graph, the estimated minimum with its radius and height, and a recommendation that separates the mathematics from the manufacturing judgement.

APPLICATION

Packaging a school award

A school needs 240 small cuboid boxes for awards. Each box must hold a cylinder of diameter 6 cm and height 5 cm, with 0.5 cm clearance on each side.

Design	Dimensions (cm)	Card per box (cm²)	Total card (m²)
Tight fit	7 × 7 × 6	266	6.38
Slightly larger	8 × 7 × 6.5	295	7.08
Cube	7 × 7 × 7	294	7.06

1. Calculate the minimum internal dimensions that give the required clearance, and explain each addition.
2. Calculate the card needed for the tight-fit design, including the lid, and compare with the table.
3. Suggest a reason the school might choose the slightly larger design despite the extra card.
4. The card costs 4.5 units per m<and is sold in whole square metres. Calculate the cost for 240 boxes of your recommended design.
5. Explain how the design would change if the boxes had to be posted, referring to both surface area and volume in your answer.

CRITERION C COMMUNICATING

Explaining a formula you cannot see

A younger student asks why the volume of a cone is one third of the cylinder around it. Write an explanation they could follow, without calculus.

1. Describe an experiment with a cone and a cylinder of the same base and height that would convince them, and say what it shows and what it does not prove.
2. Give a second argument using a stack of thin discs, and explain why the discs approximate the cone better as they get thinner.
3. Explain why the same one-third factor appears for every pyramid, not only the square-based one.
4. State clearly which parts of your explanation are evidence and which are proof.

1–2 States the formula; the explanation is assertion only.
3–4 Gives one convincing argument and states what it shows.
5–6 Gives two arguments and distinguishes evidence from proof.
7–8 As above, and explains why the factor is one third independently of the shape of the base.

Chapter summary and review

Key ideas

- Volume of a prism is cross-sectional area times length.
- A cylinder's curved surface unrolls into a rectangle.
- Pyramid and cone volumes are one third of the enclosing prism.
- Sphere: volume $\frac{4}{3}\pi r^3$, surface $4\pi r^2$.
- A net shows the true shape of every face.
- Units of area and volume convert by the square and cube of the length factor.
- 1 cm<= 1 mL and 1 m<= 1 000 litres.
- A calculated volume cannot be more precise than its least precise measurement.

Vocabulary

prism
A solid with a constant cross-section.

cross-section
The shape cut by a plane through a solid.

volume
The space a solid occupies, in cubic units.

capacity
The volume a container can hold, usually in litres.

surface area
The total area of a solid's faces, in square units.

net
A flat pattern that folds into a solid.

slant height
The distance along a cone's surface from rim to apex.

hemisphere
Half of a sphere.

tolerance
The permitted variation in a manufactured measurement.

What students often think	What is actually true
Using the diameter as the radius.	Halve it first; the error is a factor of four in area and volume.
A cone's curved area is πrh.	It is πrl, with the slant height.
1 m<= 100 litres.	It is 1 000 litres, because each dimension converts by 10.

Misconceptions worth confronting directly.

What students often think	What is actually true
A calculated volume is exact.	It inherits the precision of the measurements that produced it.

Revision checklist

- ☐ I can calculate the volume of any prism.
- ☐ I can derive the surface area of a cylinder from its net.
- ☐ I can use the pyramid and cone volume formulas.
- ☐ I can use the sphere formulas.
- ☐ I can build a net and find a surface area from it.
- ☐ I can convert between m< litres and cm<
- ☐ I can state an appropriate precision for a calculated volume.
- ☐ I can compare a calculated capacity with a labelled value and explain differences.

REFLECT

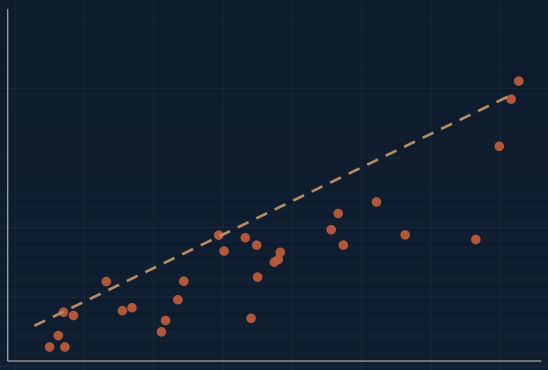
Choose three containers in your home. Estimate each capacity, then calculate it from measurements, then compare with the label. Rank the three by how close your estimate was, and explain the ranking.

IV

PART IV

Data, chance and evidence

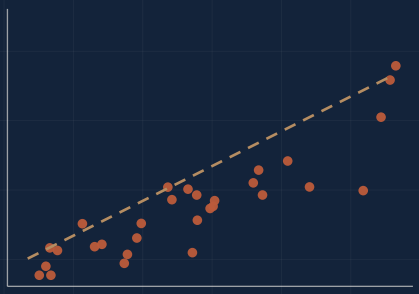
Most of the numbers you meet outside school are measurements, and most of the claims made from them are arguments. This part teaches the mathematics of association, probability and sampling so that a claim can be judged rather than merely believed.



09 Bivariate data and correlation

10 Probability and risk





9

EVIDENCE, RISK AND DECISION

Bivariate data and correlation

Two measurements taken from the same thing are called bivariate data. The question is always whether they move together, how strongly, in which direction < and, most importantly, whether one causes the other.

KEY CONCEPT RELATIONSHIPS | RELATED CHANGE | RELATED REPRESENTATION | CONTEXT GLOBALISATION AND SUSTAINABILITY

INQUIRY

What can a scatter graph tell us about the relationship between two variables, and what can it never tell us?

ATL RESEARCH AND CRITICAL THINKING SKILLS

Find a published claim of the form <X is linked to Y<in a newspaper or on a government data site. Identify the two variables, state the direction of the claimed association, and name one third variable that could produce the same pattern without either causing the other.

MYP UNIT PLANNER ≣THE WHOLE CHAPTER IN ONE VIEW

Framing the inquiry

Unit 9 ?approximately 16 hours of class time ?ATL focus: research and critical thinking

KEY AND RELATED CONCEPTS

Relationships. An association between two variables is described by its direction, its strength and its form. Correlation measures strength of a linear association only.

Change. A correlation coefficient is a summary, not a story. It says that the variables change together at a certain rate, not which one moves first or why.

Representation. The same data can be shown as a table, a scatter graph or a single coefficient. Each representation hides something the others show, and the coefficient hides the most.

GLOBAL CONTEXT AND EXPLORATION

Globalisation and sustainability. International data sets invite comparison: emissions against income, life expectancy against health spending, temperature against latitude. Every such comparison is bivariate, and every one is vulnerable to the same three errors < confounding, selection and extrapolation. Reading them responsibly is part of arguing about sustainability with evidence rather than assertion.

STATEMENT OF INQUIRY

In order for a student to evaluate a claim that two quantities are related, students must be able to construct and interpret a scatter graph, describe the association in terms of direction, strength and form, and distinguish a statistical association from a causal claim.

FACTUAL

<What is a scatter graph?

<What does a correlation coefficient measure?

<What is an outlier?

CONCEPTUAL

<Why can a strong correlation be consistent with no causal relationship at all?

DEBATABLE

<If two variables are strongly correlated and the mechanism is unknown, should policy act on the correlation?

Criterion	Strands	Evidence in this chapter
A ?Knowing and understanding	i, ii, iii	Construct scatter graphs; describe association; interpret a line of best fit and a correlation coefficient.
B ?Investigating patterns	i, ii, iii, iv	Investigate a bivariate data set, justify the model, and evaluate its predictions.
C ?Communicating	i, ii, iii, iv	Report an association accurately without overclaiming; explain confounding.
D ?Applying mathematics	i, ii, iii, v	Use a fitted model to make and criticise a prediction in a sustainability context.

Each teaching section closes with a graded practice set. The investigation addresses Criterion B; the application addresses Criterion D; the criterion task asks for extended written communication (Criterion C) about knowledge and understanding (Criterion A).

Learning intentions and success criteria

The mathematics of this chapter is straightforward. The judgement is not: deciding what an association does and does not license you to say is the real work.

YOU ARE LEARNING TO

- **Understand** what bivariate data are and how a scatter graph displays them.
- **Be able to** describe an association by direction, strength and form.
- **Be able to** draw and interpret a line of best fit.
- **Understand** what a correlation coefficient measures and what it does not.
- **Be able to** distinguish correlation from causation and identify confounding variables.
- **Be able to** evaluate a prediction made from bivariate data.

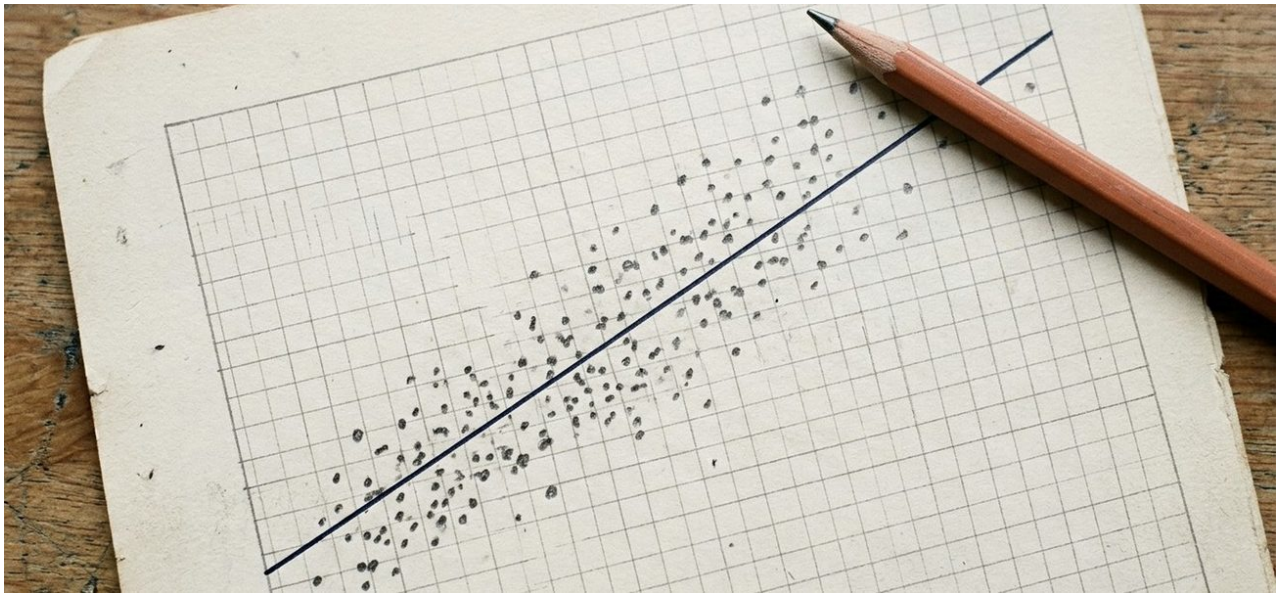
SUCCESS CRITERIA ≣ I CAN

- ☐ I can plot a scatter graph with correctly labelled and scaled axes.
- ☐ I can describe an association as positive or negative, strong or weak, linear or non-linear.
- ☐ I can draw a line of best fit and use it to interpolate a value.
- ☐ I can interpret a correlation coefficient of 0.85 and one of -0.3.
- ☐ I can give an example of confounding and explain how it produces a correlation.
- ☐ I can say why a prediction beyond the data is unreliable.

PREREQUISITE CHECK ≣10 MINUTES, GRAPH PAPER

1. Find the mean of 4, 7, 9, 10.
2. Plot (2, 5) and (6, 9) and find the gradient of the line through them.
3. State the median of 3, 8, 8, 11, 15.
4. Calculate (7 - 4)/(10 - 2).
5. Explain what a negative gradient means on a graph.
6. State one difference between a table and a graph of the same data.

Answers are in the chapter answer section. If more than two are incorrect, review plotting coordinates, gradient and the mean before continuing.



A scatter graph drawn by hand, with a line of best fit. The line describes the trend; the scatter around it is the information about how reliable that trend is.
Original photograph made for this edition.

Explore: arm span and height

Before any formula, collect data. You will test the claim that a person's arm span is roughly equal to their height, and decide how strong the evidence is.

YOUR TASK BEFORE YOU ARE TAUGHT

1. Measure the height and arm span of ten people, to the nearest centimetre. Record the pairs in a table with a column for the difference.
2. Plot the ten points with height on the horizontal axis. Add the line $y = x$ and count how many points lie above and below it.
3. Describe the pattern in words: is it rising or falling, tight or loose, straight or curved?
4. Identify any point that lies far from the others and decide whether to include it, giving a reason.
5. Calculate the mean difference between height and arm span across your sample, and interpret it.
6. Write a claim your data supports and a stronger claim it does not support.

Section 9.2 formalises the description, and section 9.3 explains why a correlation coefficient alone cannot settle the claim.

BEFORE YEAR 3	YEAR 3 THIS CHAPTER	LATER
▪ Reading and plotting coordinates. ▪ Calculating the mean, median and range. ▪ Drawing and interpreting bar charts and line graphs.	▪ Scatter graphs and the description of association. ▪ Lines of best fit and interpolation. ▪ Correlation coefficients and their limitations. ▪ Correlation, causation and confounding.	▪ Regression and residual analysis. ▪ Hypothesis testing for correlation. ▪ Experimental design and control groups.

Where this chapter sits in the sequence.

Inquiry pathway for this chapter

	Stage	What you do
1	Context	Arm span and height; a claim about screen time.
2	Question	When two quantities move together, why is that not enough?
3	Concept	Association, correlation, causation, confounding, residual.
4	Investigation	Test whether an arm-span claim survives measurement.
5	Communication	Report an association without overclaiming.
6	Reflection	What evidence would you need before acting on a correlation?

9.1 Scatter graphs and association

CRITERION A • ASSOCIATION • GRAPH

A scatter graph plots one measurement against another for the same individuals. Its value is that it shows the shape of the relationship before any single number summarises it.

In bivariate data each observation supplies two values. One variable is placed on the horizontal axis and the other on the vertical; the choice matters, because the two axes are not interchangeable when you fit a line. An association is described by **direction** (positive or negative), **strength** (strong, moderate, weak) and **form** (linear or curved).

DEFINITION

Bivariate data consist of pairs of measurements of the same individuals. An **outlier** is an observation that lies far from the overall pattern; it should be investigated, not automatically deleted.

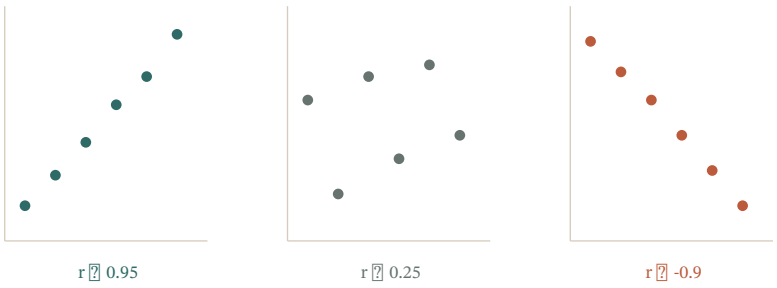


Figure 9.1 Four scatter graphs: strong positive, weak positive, strong negative and no association. Direction and strength are separate descriptions, and neither says anything about cause.

KEY IDEA

The axes carry meaning

The axes must be chosen deliberately and labelled with units. Plotting the explanatory variable horizontally is a convention, not a law, but swapping the axes changes the line of best fit and therefore the prediction. In a report, state which variable you placed where.

WORKED EXAMPLE 9.1.1 DESCRIBING AN ASSOCIATION

PROBLEM

A scatter graph of daily temperature against hot-drink sales shows points rising from left to right, tightly clustered around a straight path, with one point far below the others.

1. Direction: as temperature rises, sales rise, so the association is positive.
2. Strength: the points are tightly clustered, so the association is strong.
3. Form: the trend is close to a straight line, so it is linear.
4. The isolated low point is a possible outlier; check whether it was a public holiday before deciding what to do with it.

ANSWER strong positive linear association, with one possible outlier

Compare against a negative example: a graph of temperature against soup sales would fall, and the description would begin with direction.

COMMON ERROR

Joined-up scatter

Joining the points in the order they appear in the table. A scatter graph shows the relationship, not the sequence; joining the dots implies an ordering that the data may not have.

COMMON ERROR

Strength confused with steepness

Calling an association strong because the gradient is steep. Strength is about how closely the points follow the trend, not about how steep it is.

PRACTICE 9.1 SCATTER GRAPHS AND ASSOCIATION

SUPPORT 1 CORE 1 EXTENSION 7

1. Describe the association in each case: (a) hours of study and test score (b) age of a car and its value (c) shoe size and reading age in a class of 12-year-olds. F
2. State whether each association is likely to be strong or weak: (a) height and weight of adults (b) number of letters in a name and month of birth. F
3. Explain why a scatter graph should not join its points. R
4. A data set has one point far from the rest. Describe two different explanations for it and what you would check in each case. R

PRACTICE 9.1 SCATTER GRAPHS AND ASSOCIATION (CONTINUED)

SUPPORT 1 CORE 1 EXTENSION 7

1. Sketch a scatter graph showing a strong negative association with one outlier. P
2. The same data are plotted with the axes swapped. Explain what changes and what does not. P
3. A student claims <the association is strong because the points are high<. Explain the error. E
4. Sketch two data sets with the same mean values but different strengths of association, and explain what distinguishes them. E

9.2 Lines of best fit and prediction

CRITERION A I • MODEL • PREDICTION

A line of best fit summarises the trend in a scatter graph. Its gradient is a rate and its intercept a starting value, exactly as in chapter 4, but both are now estimates from imperfect data.

The line of best fit passes through the middle of the cloud of points, with roughly equal numbers above and below and as little total vertical distance as possible. It need not pass through any data point and should not be forced through the origin. Once drawn, its equation allows a value to be **interpolated** (inside the data) or **extrapolated** (outside it).

KEY IDEA

The cloud matters more than the line

The scatter around the line is the measure of how much to trust it. Two data sets can give the same gradient and wildly different reliability; the width of the cloud, not the line, tells you which prediction to believe.

WORKED EXAMPLE 9.2.1 FITTING AND USING A LINE

PROBLEM

Data for a plant's height (cm) against weeks give a line of best fit $h = 3.5w + 4$. Interpret both numbers and predict the height at 10 weeks, given that the data cover weeks 1 to 6.

1. The gradient 3.5 means the plant grows about 3.5 cm per week.
2. The intercept 4 means the modelled height at week 0 is 4 cm, which is the height when measurement began.
3. At $w = 10$: $h = 3.5(10) + 4 = 39$ cm.
4. Week 10 lies outside the data, so the prediction is an extrapolation and assumes the growth rate is unchanged.

ANSWER 39 cm, an extrapolation four weeks beyond the data

The value is plausible (plants do not reach 39 cm in six weeks at this rate: the model gives 25 cm at week 6), but its reliability is lower than an interpolated value.

WORKED EXAMPLE 9.2.2 READING A PREDICTION HONESTLY

PROBLEM

Using the same model, predict the height at 40 weeks.

1. Substitute: $h = 3.5(40) + 4 = 144$ cm.
2. Note that 144 cm is taller than most house plants and that the data end at week 6.
3. Growth is unlikely to remain linear for 40 weeks: it usually slows.
4. Report the arithmetic, then state why the number should not be used.

ANSWER the model gives 144 cm, which is not a credible prediction

A good answer reports the model's output and the reason to distrust it; a bad answer reports only the output.

COMMON ERROR

Prediction without a warrant

Extrapolating far beyond the data and presenting the result as a prediction. The arithmetic is valid; the assumption that the trend continues is not supported by anything in the data.

PRACTICE 9.2 LINES OF BEST FIT AND PREDICTION

SUPPORT 1–3 CORE 1–6 EXTENSION 7–8

1. A line of best fit is $y = 2.4x + 7$ with data for x from 1 to 8. Interpret the gradient and intercept. F
2. Using the model in question 1, predict y at $x = 5$ and at $x = 20$. Say which is an extrapolation. F
3. Explain why a line of best fit need not pass through any of the data points. R
4. Two models fit the same data: $y = 3x + 1$ and $y = 3.1x + 0.8$. Explain how you would choose between them. R
5. Data show a car's value falling from 12 000 to 7 500 units over 4 years. Fit a linear model and predict the value after 6 years, stating one reason the prediction may fail. P
6. Explain why extrapolating a linear model of a plant's growth to 40 weeks is unreasonable, using a biological reason. P
7. A line fitted to data for x between 10 and 20 is used to predict at $x = 12$. Explain why this is interpolation and why it is more reliable than a prediction at $x = 45$. E
8. Two scatter graphs have the same line of best fit but one cloud is much narrower. Explain which prediction you would trust and why. E

9.3 Correlation and its limits

CRITERION A III • CORRELATION • CAUSATION

The correlation coefficient is a single number between -1 and 1 that measures how closely the points follow a straight line. It is useful, easily misread, and says nothing whatever about cause.

A coefficient near 1 indicates a strong positive linear association, near -1 a strong negative one, and near 0 a weak or absent linear association. Values in between describe intermediate strength. The coefficient is unaffected by the units used and by which variable is on which axis, but it can be misleading when the relationship is curved or when an outlier dominates the data.

Coefficient	Reading	Caution
0.95	strong positive linear	check the scatter; a curve can produce this too
0.45	moderate positive linear	much scatter remains unexplained
-0.80	strong negative linear	direction does not imply cause
0.02	no linear association	a strong curved relationship can give this

Correlation is not causation. Two variables can move together because one causes the other, because the other causes the first, because a third variable causes both (**confounding**), or by coincidence in a small sample. Only a controlled experiment, or a well-understood mechanism, supports a causal claim.

WORKED EXAMPLE 9.3.1 A CORRELATION WITH AN OBVIOUS

PROBLEM

Sales of ice cream and the number of drowning incidents are strongly correlated across the year. A newspaper reports that ice cream increases the risk of drowning. Evaluate the claim.

1. The correlation is real and positive: both rise in the same months.
2. A third variable, hot weather, plausibly causes both: more ice cream is bought and more people swim.
3. No mechanism links eating ice cream to drowning risk.
4. The causal claim is unsupported; the association is explained better by confounding.

ANSWER the correlation is real, the causal claim is not supported

If the claim were causal, eating ice cream in winter would raise the risk; there is no evidence for that, and the confounder explains the pattern more simply.

WORKED EXAMPLE 9.3.2 A STUDY THAT DOES SUPPORT A

PROBLEM

A trial gives 200 volunteers either a new fertiliser or a placebo, assigned at random, and compares plant growth. The treated group grows significantly more. Why is this evidence of cause?

1. Random assignment makes the two groups similar in every other respect, on average.
2. The only systematic difference between them is the fertiliser.
3. So a difference in growth is attributable to the fertiliser.
4. The conclusion is causal within the conditions tested, and generalising beyond them still needs care.

ANSWER randomisation removes confounding, so the difference supports a causal claim

Compare with the ice-cream case: there, no variable was manipulated, so no causal attribution was possible.

COMMON ERROR

No linear correlation, no relationship

Reading a correlation coefficient of 0.02 as proof that the variables are unrelated. A strong curved relationship < for example, plant growth against temperature, which rises then falls < can give a coefficient near zero.

ATL

ATL ≠Critical thinking

Critical thinking. When you meet a correlation in the news, ask three questions in order: what are the two variables, how were they measured, and what third variable could produce the same pattern? Most weak claims fail the third question.

PRACTICE 9.3 CORRELATION AND ITS LIMITS

SUPPORT 1 CORE 1 EXTENSION 7

1. Interpret each coefficient: (a) 0.88 (b) -0.35 (c) -0.97 (d) 0.06. F
2. State whether each pair is likely to be positively correlated, negatively correlated or uncorrelated: (a) altitude and air pressure (b) hours of exercise and resting heart rate (c) shoe size and height in adults. F
3. Explain the difference between correlation and causation in your own words, with an example. R
4. A study finds that towns with more hospitals have higher death rates. Explain why this does not show hospitals cause deaths, and name the confounder. R
5. Design a study that could establish whether a new revision programme improves test scores, and explain what makes it convincing. P
6. A data set of 12 points has one extreme outlier. Explain how the outlier could change the coefficient from 0.2 to 0.8. P
7. Explain why a coefficient of 0.02 does not establish independence, using a curved example. E
8. Two variables are correlated at -0.9. Describe two different causal explanations and one non-causal explanation that are all consistent with the data. E

9.4 Presenting and criticising data

CRITERION C V • REPRESENTATION • EVIDENCE

A graph can be accurate in every number and still mislead in its impression. Learning to spot the gap between the two is a mathematical skill, not a cynical one.

Common distortions do not require dishonesty: a truncated vertical axis exaggerates a change, unequal intervals on an axis distort a trend, areas used to represent lengths exaggerate differences, and a selective time window can reverse a conclusion. Each has a remedy: start a vertical axis at zero for magnitudes, label the scale, use one dimension for one quantity, and show the whole relevant period.

KEY IDEA

Criticism must be checkable

A criticism should be specific and checkable. <This graph is misleading<is an opinion; <the vertical axis starts at 40, which triples the apparent size of the change<is a mathematical statement that anyone can verify.

WORKED EXAMPLE 9.4.1 CRITICISING A GRAPH

PROBLEM

A chart shows a school's recycling rising from 40 kg to 60 kg with the vertical axis running from 40 to 60. The headline says <recycling up 50 per cent< Assess the chart.

1. The percentage is correct: (60 - 40)/40 = 0.5, so 50 per cent.
2. The axis starts at 40, so the bar for 60 kg appears infinitely taller than the bar for 40 kg, which is impossible.
3. Starting a magnitude axis at zero would show a rise of half the original height, which is the honest picture.
4. The claim is accurate; the visual impression exaggerates it.

ANSWER the number is right, the axis exaggerates the change

The remedy is to redraw with the axis from 0 to 70 and compare; the change is then visibly substantial but not overwhelming.

COMMON ERROR

One rule applied too broadly

Assuming any non-zero axis is dishonest. For some quantities < temperature in degrees Celsius, for instance < zero is not a meaningful baseline, and a scaled axis is legitimate provided it is labelled.

PRACTICE 9.4 PRESENTING AND CRITICISING DATA

SUPPORT 1–3 CORE 1–5 EXTENSION 6–7

1. A graph shows sales rising from 1 000 to 1 100 units with the axis from 950 to 1 100. Describe the impression given and the honest description. F
2. Explain why using a picture of a 3D bar twice as tall as another exaggerates the difference. R
3. Find an example of a misleading graph in a newspaper or online, and describe precisely what is misleading about it. P
4. Redraw a misleading graph you have found so that it is honest, and explain the changes you made. P
5. Explain why a percentage change and an absolute change can tell opposite stories about the same data. R
6. A report gives the mean income of a town but not the median. Explain which is more informative and why. E
7. Write three specific, checkable criticisms of a graph of your choice, each in one sentence. E

An investigation is not a longer exercise. It is a short piece of mathematics you do yourself: you gather evidence, look for a pattern, say what you think is true, and then test it until you can defend it.

INVESTIGATION

Arm span and height

The claim that arm span equals height is old and widely repeated. You will test it on your own data and report how strong the evidence is.

Person	Height (cm)	Arm span (cm)	Difference (cm)
1	168	170	+2
2	155	152	−3
3	175	176	+1
4	161	163	+2
5	182	180	−2

1. **Collect.** Measure at least fifteen people, recording both values to the nearest centimetre. Note how you ensured the measurements were comparable.
2. **Plot.** Draw a scatter graph with height on the horizontal axis and add the line $y = x$. Count points above and below.
3. **Describe.** Describe the association by direction, strength and form, and identify any outliers.
4. **Fit.** Draw a line of best fit and find its equation, using two points on your line.
5. **Test.** Use your line to predict the arm span of someone 170 cm tall, and compare with the claim *arm span = height*.
6. **Conclude.** State whether your data support the claim, and explain what a larger sample would and would not change about your conclusion.

Present the raw table, the scatter graph with both lines, the fitted equation, the comparison with the claim, and a conclusion that distinguishes what your sample shows from what it cannot show.

APPLICATION

CRITERION D

The screen-time claim

A newspaper reports that students who use screens for more than four hours a day score 8 per cent lower in mathematics, based on a survey of 2 000 students.

Screen time (hours/day)	Mean score	Number of students
0<1	72	410
1<2	70	620
2<4	66	610
4+	63	360

1. Plot the four group means against the midpoint of each screen-time band, and describe the association.
2. Calculate the percentage difference between the lowest and highest bands and check the newspaper's figure.
3. Identify two confounding variables that could explain part of the difference, and explain how each would produce the pattern.
4. Explain why a survey cannot establish that screen time causes lower scores, and describe the study you would need.
5. Write a headline that is accurate to the evidence, and explain what you changed and why.

CRITERION C COMMUNICATING

Reading a correlation responsibly

Write a one-page guide for younger students on how to read a claim of the form <X is linked to Y<in the news.

1. Explain what a correlation coefficient measures, in language a 12-year-old could follow, using a diagram.
2. Give three questions a reader should ask, in order, and explain what each one rules out.
3. Give one example of a correlation that supports a causal claim and one that does not, and explain the difference between them.
4. Explain what would change your mind about a claim you currently believe.

1–2 Defines correlation but the explanation of cause is confused.

3–4 Explains correlation and gives one example of confounding.

5–6 Explains the coefficient, gives the three questions with reasons, and distinguishes the two examples.

7–8 As above, and explains what evidence would be required to move from association to cause.

Chapter summary and review

Key ideas

- Bivariate data are pairs of measurements of the same individuals.
- An association is described by direction, strength and form.
- A line of best fit summarises a linear trend and need not pass through any point.
- The scatter around the line is the measure of reliability.
- Interpolation is inside the data; extrapolation is beyond it.
- A correlation coefficient measures linear association only, from -1 to 1.
- Correlation does not establish causation; confounding is the usual explanation.
- Only randomisation or a known mechanism supports a causal claim.
- A graph can be numerically accurate and visually misleading.

Vocabulary

bivariate data
Pairs of measurements of the same individuals.

scatter graph
A plot of one variable against another.

association
A systematic relationship between two variables.

outlier
An observation far from the general pattern.

line of best fit
The line summarising a linear trend.

interpolation
Estimating inside the range of the data.

extrapolation
Estimating outside the range of the data.

correlation coefficient
A number from -1 to 1 measuring linear association.

confounding variable
A third variable influencing both others.

causation
A relationship in which one variable produces a change in another.

What students often think	What is actually true
A steep line of best fit means a strong association.	Strength is how closely the points follow the line.
A correlation of 0 means the variables are unrelated.	It means no *linear* relationship; a curve can have a coefficient near zero.

Misconceptions worth confronting directly.

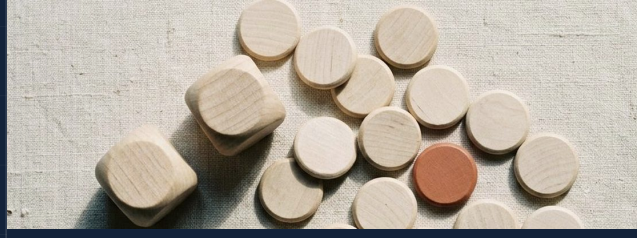
What students often think	What is actually true
A strong correlation shows that one variable causes the other.	Confounding or coincidence can produce the same correlation.
Any axis not starting at zero is dishonest.	It is legitimate when zero is not a meaningful baseline, provided it is labelled.

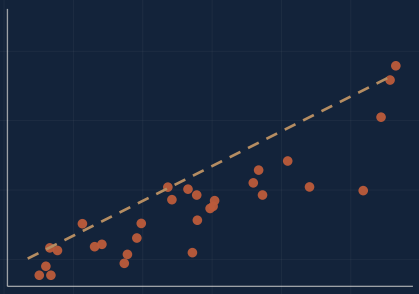
Revision checklist

- ☐ I can construct a scatter graph with labelled, scaled axes.
- ☐ I can describe an association by direction, strength and form.
- ☐ I can draw a line of best fit and find its equation.
- ☐ I can distinguish interpolation from extrapolation.
- ☐ I can interpret a correlation coefficient correctly.
- ☐ I can identify confounding variables in a claim.
- ☐ I can explain why randomisation supports a causal claim.
- ☐ I can criticise a graph with specific, checkable statements.

REFLECT

Find one correlation reported in the news this week. Write four sentences: the two variables, the direction and strength, one possible confounder, and what evidence would make the causal claim credible.





10

EVIDENCE, RISK AND DECISION

Probability and risk

Probability is a number between 0 and 1 that measures how strongly the evidence supports a claim about what will happen. It does not remove uncertainty; it makes uncertainty comparable.

KEY CONCEPT ▪ CHANGE

RELATED ▪ QUANTITY

RELATED ▪ SYSTEMS

CONTEXT ▪ FAIRNESS AND DEVELOPMENT

> INQUIRY
>
> How can uncertainty be measured, and when is a probability a fact about the world rather than a fact about our knowledge?

ATL ▪ THINKING AND COMMUNICATION SKILLS

Find a risk stated as a probability in a news report < a medical risk, a weather warning or a safety notice. Rewrite it as a frequency out of 1 000, and decide whether the rephrasing changes how alarming it sounds.

MYP UNIT PLANNER — THE WHOLE CHAPTER IN ONE VIEW

Framing the inquiry

Unit 10 — approximately 16 hours of class time — ATL focus: thinking and communication

KEY AND RELATED CONCEPTS

Change. Probability describes how a quantity is expected to vary. A distribution is the pattern of that variation across outcomes.

Quantity. A probability is a number with rules: between 0 and 1, summing to 1 over all outcomes, and combining predictably for independent events.

Systems. Repeated trials produce relative frequencies that settle towards the theoretical probability. That convergence is what makes probability a statement about the world rather than about a single throw.

GLOBAL CONTEXT AND EXPLORATION

Fairness and development. Insurance, screening programmes and public health advice are all decisions about risk, and all of them distribute costs and benefits unevenly. A screening test with a high false-positive rate imposes anxiety and cost on the healthy; a test with a high false-negative rate shifts risk onto everyone else. Probability is how these trade-offs can be argued with numbers.

STATEMENT OF INQUIRY

In order for a student to make a defensible decision under uncertainty, students must be able to calculate theoretical and experimental probabilities, combine them for independent and mutually exclusive events, and interpret a risk stated as a probability in terms of frequencies.

FACTUAL

- What is the probability scale?
- What does mutually exclusive mean?
- How do you calculate the probability of two independent events?

CONCEPTUAL

- Why does a run of heads not make tails more likely on the next toss?

DEBATABLE

- If a screening test has a 5 per cent false-positive rate, should everyone be screened?

Criterion	Strands	Evidence in this chapter
A — Knowing and understanding	i, ii, iii	Calculate theoretical and experimental probabilities; use the addition and multiplication rules; draw tree diagrams and two-way tables.
B — Investigating patterns	i, ii, iii, iv	Investigate relative frequency over many trials and establish the convergence.
C — Communicating	i, ii, iv	Interpret a probability as a frequency; explain a risk to a non-specialist.
D — Applying mathematics	i, ii, iii, v	Evaluate a screening or sampling decision and justify a recommendation.

Each teaching section closes with a graded practice set. The investigation addresses Criterion B; the application addresses Criterion D; the criterion task asks for extended written communication (Criterion C) about knowledge and understanding (Criterion A).

Learning intentions and success criteria

Most errors with probability are not arithmetic errors. They are errors about what the numbers mean, and this chapter is arranged to confront them directly.

YOU ARE LEARNING TO

- **Understand** the probability scale and that probabilities of all outcomes sum to 1.
- **Be able to** calculate theoretical probabilities from equally likely outcomes.
- **Be able to** calculate experimental probabilities and compare them with theory.
- **Know** and apply the addition rule for mutually exclusive events.
- **Know** and apply the multiplication rule for independent events.
- **Be able to** use tree diagrams and two-way tables for combined events.
- **Be able to** interpret a risk as a frequency out of a stated number.

SUCCESS CRITERIA — I CAN

- ☐ I can calculate the probability of each total when two dice are thrown.
- ☐ I can explain why the probabilities of all outcomes sum to 1.
- ☐ I can compute a relative frequency and compare it with a theoretical probability.
- ☐ I can use $P(A \text{ or } B) = P(A) + P(B)$ when the events cannot both happen.
- ☐ I can use $P(A \text{ and } B) = P(A) \times P(B)$ when the events are independent.
- ☐ I can complete a tree diagram for two events without replacement.
- ☐ I can rewrite ‘a 1 in 250 risk’ as 4 people in 1 000.

PREREQUISITE CHECK — 10 MINUTES, CALCULATOR

1. Write 1/4 as a decimal and a percentage.
2. Simplify 18/24.
3. List the outcomes when a coin is tossed twice.
4. Calculate 0.4×0.3.
5. Calculate $\frac{1}{2} + \frac{1}{2}$.
6. State the probability of an impossible event.

Answers are in the chapter answer section. If more than two are incorrect, review fractions, decimals and percentages, and listing outcomes before continuing.



Counters sorted into a bag. Which counter is drawn is chance; how often each colour is drawn over many trials is a frequency, and the two are not the same claim. This chapter keeps them apart.
Original photograph made for this edition.

Explore: how many throws

You will test whether a die is fair. Fairness is a claim about the long run, so the experiment has to be long enough to say anything.

YOUR TASK BEFORE YOU ARE TAUGHT

1. Predict how many sixes you would expect in 60 throws, and state the probability you used.
2. Throw a die 60 times, record the frequency of each face in a table, and calculate the relative frequency of a six.
3. Combine your results with the class's to reach at least 600 throws. Recalculate the relative frequency of a six.
4. Plot relative frequency of a six against number of throws on a graph and describe what happens as the number of throws increases.
5. Decide whether the class's die is fair. State the evidence you are using and how confident you are.
6. Explain why 60 throws is not enough to detect a small bias, and what you would need to do to detect one.

Section 10.2 explains why relative frequency settles down, and what ≪settles down≪does and does not guarantee.

BEFORE YEAR 3	YEAR 3 THIS CHAPTER	LATER
▪ The probability scale from 0 to 1.	▪ Theoretical and experimental probability.	▪ Conditional probability and Bayes' theorem.
▪ Listing outcomes of simple experiments.	▪ Addition and multiplication rules.	▪ Binomial and normal distributions.
▪ Converting between fractions, decimals and percentages.	▪ Tree diagrams and two-way tables.	▪ Statistical inference and significance.
	▪ Risk, sampling and screening.	

Where this chapter sits in the sequence.

Inquiry pathway for this chapter

	Stage	What you do
1	Context	A game of chance and a medical screening programme.
2	Question	What does a probability claim about the world?
3	Concept	Sample space, relative frequency, independence, risk.
4	Investigation	Test whether a die is fair, and how many throws that needs.
5	Communication	Explain a 1-in-250 risk so that it is understood.
6	Reflection	Why is a rare disease test more often wrong than right?

10.1 The probability scale and sample spaces

CRITERION A • SAMPLE SPACE • SCALE

Every probability lies between 0 and 1, and the probabilities of all possible outcomes add to 1. Those two facts do most of the work.

The **sample space** is the set of all possible outcomes of an experiment. For equally likely outcomes, the probability of an event is the number of outcomes in the event divided by the number in the sample space. Listing the sample space systematically < in a table for two dice, for instance < prevents both omissions and double counting.

DEFINITION

For equally likely outcomes, $P(A) = n(A)/n(S)$. The complement rule is $P(\text{not } A) = 1 - P(A)$. Probabilities are written as fractions, decimals or percentages, but never as numbers outside 0 to 1.

Total	Outcomes	Probability
2	1	1/36
7	6	6/36 = 1/6
12	1	1/36
all totals	36	1

Two dice: thirty-six equally likely ordered pairs. Seven is the most likely total because it can be made in six ways.

WORKED EXAMPLE 10.1.1 TWO DICE

PROBLEM

Two fair dice are thrown and the totals added. Find the probability that the total is 7 and the probability that it is greater than 9.

1. There are $6 \times 6 = 36$ equally likely ordered outcomes.
2. Totals of 7: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1), six outcomes.
3. So $P(7) = 6/36 = 1/6$.
4. Totals over 9: 10 (3 ways), 11 (2 ways) and 12 (1 way), six outcomes.
5. So $P(\textit{more than } 9) = 6/36 = 1/6$.

ANSWER $P(7) = 1/6$; $P(>9) = 1/6$

The two probabilities happen to be equal, but the events are different: one peaks in the middle of the table, the other sits in the corner.

WORKED EXAMPLE 10.1.2 THE COMPLEMENT RULE

PROBLEM

A bag contains 5 red, 3 blue and 2 green counters. One is taken at random. Find the probability it is not red.

1. There are $5 + 3 + 2 = 10$ counters.
2. $P(\textit{red}) = 5/10 = 0.5$.
3. $P(\text{not } \textit{red}) = 1 - 0.5 = 0.5$.
4. Check directly: 5 of the 10 counters are not red, so $5/10 = 0.5$.

ANSWER 0.5

The complement rule and the direct count agree, as they must.

COMMON ERROR

Totals treated as equally likely

Listing the totals of two dice as if there were eleven outcomes and they were equally likely. There are thirty-six equally likely outcomes; the totals are not equally likely, which is why 7 is six times as probable as 2.

PRACTICE 10.1 PROBABILITY AND SAMPLE SPACES

SUPPORT 1–4 CORE 1–7 EXTENSION 8–9

1. A fair die is thrown. Find: (a) $P(4)$ (b) $P(\textit{even})$ (c) $P(\text{not } 4)$. F

PRACTICE 10.1 PROBABILITY AND SAMPLE SPACES (CONTINUED)

SUPPORT 1–4 CORE 1–7 EXTENSION 8–9

1. A bag holds 4 red, 5 blue and 6 green balls. Find: (a) $P(red)$ (b) $P(\text{not } green)$ (c) $P(red \text{ or } blue)$. F
2. Two coins are tossed. List the sample space and find $P(two\ heads)$. F
3. Two dice are thrown. Find $P(total\ 5)$ and $P(total\ 12)$. F
4. Explain why the probabilities of all outcomes in a sample space must sum to 1. R
5. Explain why $P(A) = 0.6$ and $P(B) = 0.7$ cannot describe two mutually exclusive events. R
6. A spinner has five equal sectors labelled 1 to 5. It is spun twice. Find $P(same\ number\ twice)$. P
7. Two dice are thrown. Find $P(at\ least\ one\ six)$ by using the complement, and check by counting. P
8. A bag has n red and 8 blue balls, and $P(red) = 0.4$. Find n. E

10.2 Experimental probability and relative frequency

CRITERION A • FREQUENCY • CONVERGENCE

Relative frequency is the probability you measure rather than calculate. Over many trials it approaches the theoretical value, which is why probability has anything to say about the world.

Relative frequency is the number of times an event occurred divided by the number of trials. It is an estimate of the probability, and its accuracy improves as the number of trials increases < but not monotonically, and never with a guarantee for any finite number of trials.

DEFINITION

Relative frequency = *frequency of the event* ? *number of trials*. As the number of trials increases, relative frequency tends to the theoretical probability.

KEY IDE A

The coin has no memory

This convergence is an empirical regularity, not a promise about the next trial. A coin that has produced ten heads is not <due< for tails: the coin has no memory, and the probability of heads on the next toss is still one half.

WORKED EXAMPLE 10.2.1 COMPARING THEORY AND

PROBLEM

A die is thrown 200 times and a six appears 28 times. Compare the relative frequency with the theoretical probability, and decide whether the die looks fair.

1. The theoretical probability is $1/6 \approx 0.167$.
2. The relative frequency is $28/200 = 0.14$.
3. Expected number of sixes in 200 throws is $200 \times 6 \approx 33.3$, so 28 is about 5 below the expected count.
4. In 200 throws the standard spread is around 5, so 28 is within ordinary variation.
5. There is no evidence that the die is unfair.

ANSWER relative frequency 0.14 against a theoretical 0.167; no evidence of bias

A much larger number of throws would be needed to detect a small bias; 200 throws cannot distinguish 0.14 from 0.167 reliably.

COMMON ERROR

Too few trials

Concluding that a coin is biased after five tosses. With five tosses, getting four or five heads happens often by chance; a claim about bias needs many more trials.

COMMON ERROR

The gambler's fallacy

Believing that after a run of heads, tails becomes more likely. Independence means the past does not change the next probability; the long-run average corrects itself by dilution, not by compensation.

PRACTICE 10.2 EXPERIMENTAL PROBABILITY

SUPPORT 1 CORE 1 EXTENSION 7

1. A coin is tossed 80 times and lands heads 44 times. Find the relative frequency of heads. F
2. A spinner lands on red 27 times in 150 spins. Estimate the probability of red. F
3. A die is thrown 300 times and a 3 appears 62 times. Compare the relative frequency with 1/6. F

PRACTICE 10.2 EXPERIMENTAL PROBABILITY (CONTINUED)

SUPPORT 1–3 CORE 1–6 EXTENSION 7–8

1. Explain why 20 throws cannot establish whether a die is fair. R
2. Explain the gambler's fallacy in your own words, using a coin. R
3. A factory tests 500 components and 17 are faulty. Estimate the probability that a component is faulty, and predict the number faulty in a batch of 12 000. P
4. Design an experiment to test whether a spinner is fair, stating how many spins you would use and why. P
5. Two students estimate the probability of a drawing pin landing point-up: one gets 0.6 from 20 trials, the other 0.45 from 400. Explain which is more trustworthy and why. E

10.3 Combining events

CRITERION A II • INDEPENDENCE • RULE

Two rules cover most combined events: add the probabilities when the events cannot both happen, multiply them when they are independent. Knowing which applies is the mathematical decision.

Events are **mutually exclusive** when they cannot occur together; then $P(A \text{ or } B) = P(A) + P(B)$. Events are **independent** when one occurring does not change the probability of the other; then $P(A \text{ and } B) = P(A) \times P(B)$. Drawing with replacement keeps events independent; drawing without replacement does not.

KEY IDEA

A built-in check

The wrong rule gives an answer that is often obviously impossible. If two probabilities of 0.6 are added for events that can both happen, the total exceeds 1 < which cannot be a probability. Checking that the answer lies between 0 and 1 catches many errors.

WORKED EXAMPLE 10.3.1 INDEPENDENT EVENTS

PROBLEM

A fair coin is tossed and a fair die is thrown. Find the probability of a head and a six.

1. The coin and the die do not influence each other, so the events are independent.
2. $P(\text{head}) = 1/2$ and $P(\text{six}) = 1/6$.
3. Multiply: $1/2 \times 1/6 = 1/12$.
4. Check by listing: of the twelve equally likely combinations, one is (head, 6).

ANSWER 1/12

The listing confirms the multiplication rule, which is where the rule comes from.

WORKED EXAMPLE 10.3.2 WITHOUT REPLACEMENT

PROBLEM

A bag contains 4 red and 6 blue counters. Two are taken without replacement. Find the probability that both are red.

1. First counter: $P(\text{red}) = 4/10$.
2. After a red is removed, 3 red remain among 9 counters: $P(\text{red second}) = 3/9$.
3. The events are not independent, so multiply the successive probabilities: $4/10 \times 3/9$.
4. Compute: $12/90 = 2/15$.

ANSWER $2/15$

The probability is less than $4/10 \times 4/10 = 0.16$, which is right: removing a red makes a second red less likely.

COMMON ERROR

Adding when the events can overlap

Adding probabilities for events that can happen together. Two independent events with $P = 0.5$ each have $P(\text{both}) = 0.25$, not 1. Addition is only for mutually exclusive events.

COMMON ERROR

Independence assumed where it does not hold

Treating successive draws without replacement as independent. The denominator and often the numerator change after the first draw.

PRACTICE 10.3 COMBINING EVENTS

SUPPORT 1–4 CORE 1–7 EXTENSION 8–9

1. A die is thrown twice. Find $P(\text{two sixes})$. F
2. A coin is tossed three times. Find $P(\text{three heads})$. F
3. A bag holds 3 red and 7 green balls. Two are drawn without replacement. Find $P(\text{both red})$. F
4. From the same bag, find $P(\text{one red and one green})$ in either order. F
5. Explain the difference between mutually exclusive and independent events, with an example of each. R
6. Explain why $P(A) + P(B)$ can exceed 1 for events that are not mutually exclusive. R

PRACTICE 10.3 ▤ COMBINING EVENTS (CONTINUED)

SUPPORT 1▤4 ▤ CORE 1▤7 ▤ EXTENSION 8▤9

1. Two fair dice are thrown. Find $P(\textit{total } 7 \text{ or } \textit{total } 11)$. P
2. A bag holds 5 red and 5 blue. Two are drawn without replacement. Find $P(\textit{both the same colour})$. P
3. A test is 95 per cent accurate in each of two independent readings. Find the probability that both readings are correct, and explain what that says about repeating a test. E

10.4 Tree diagrams and two-way tables

CRITERION A V • REPRESENTATION • CONDITIONAL

Two representations handle most multi-stage probability problems: the tree diagram, which multiplies along branches, and the two-way table, which counts within categories.

On a **tree diagram** the probability of a path is the product of the probabilities along it, and the probabilities of all paths sum to 1. Where the second stage depends on the first, the branch labels change, which is the diagram's way of showing conditional probability. A **two-way table** is often easier when the data are counts rather than probabilities.

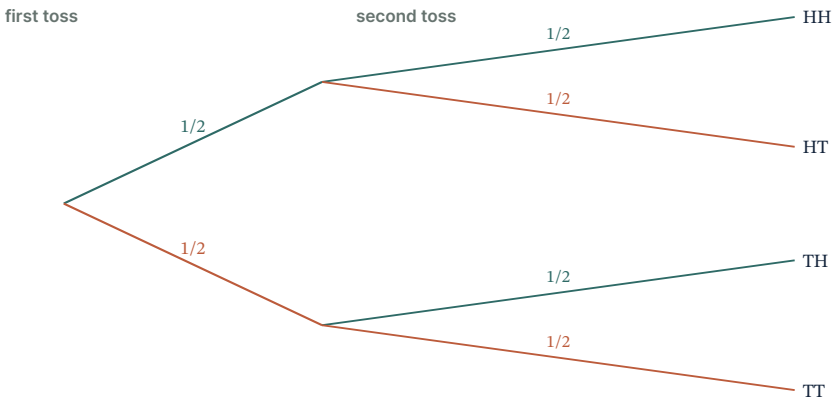


Figure 10.1 A tree for two draws without replacement. Multiplying along the branches gives the probability of each path; adding the paths that satisfy an event gives its probability.

WORKED EXAMPLE 10.4.1 A TREE WITHOUT REPLACEMENT

PROBLEM

A bag holds 3 red and 2 blue counters. Two are drawn without replacement. Find the probability of exactly one red.

1. First draw: $P(R) = 3/5$, $P(B) = 2/5$.
2. If red first, 2 red and 2 blue remain: $P(R \text{ then } B) = 3/5 \times 2/4 = 6/20$.
3. If blue first, 3 red and 1 blue remain: $P(B \text{ then } R) = 2/5 \times 3/4 = 6/20$.
4. Exactly one red occurs on either path: $6/20 + 6/20 = 12/20 = 3/5$.

ANSWER 3/5

$P(\text{both red}) = 3/5 \times 2/4 = 3/10$ and $P(\text{both blue}) = 2/5 \times 1/4 = 1/10$; together with $3/5$ these sum to 1, as they must.

WORKED EXAMPLE 10.4.2 A TWO-WAY TABLE

PROBLEM

In a school of 500 students, 300 study French, 250 study Spanish and 150 study both. Find the probability that a student chosen at random studies French but not Spanish.

1. Fill the table: French and Spanish 150; French only 300 - 150 = 150; Spanish only 250 - 150 = 100.
2. Neither language: 500 - 150 - 150 - 100 = 100.
3. French only is 150 out of 500.
4. So $P(\text{French only}) = 150/500 = 0.3$.

ANSWER 0.3

The four cells sum to 500: $150 + 150 + 100 + 100 = 500$.

COMMON ERROR

Only one path counted

Multiplying branch probabilities and forgetting that two different paths may satisfy the event. ‹Exactly one red‹arises from red-then-blue **and** blue-then-red, and both paths must be added.

PRACTICE 10.4 TREE DIAGRAMS AND TWO-WAY TABLES

SUPPORT 1 CORE 1 EXTENSION 7

1. Draw a tree for two tosses of a coin and find $P(\text{exactly one head})$. F

PRACTICE 10.4 TREE DIAGRAMS AND TWO-WAY TABLES (CONTINUED)

SUPPORT 13 CORE 16 EXTENSION 79

1. A bag holds 4 red and 6 blue. Draw a tree for two draws without replacement and find $P(\text{both blue})$. F
2. In a class of 40, 24 play football, 18 play tennis and 10 play both. Find $P(\text{plays neither})$. F
3. A bag holds 5 white and 3 black. Two are drawn without replacement. Find $P(\text{at least one black})$. R
4. Explain why the branch probabilities on the second level of a tree change when drawing without replacement. R
5. Explain when a two-way table is easier to use than a tree diagram. R
6. A factory has two machines: A makes 60 per cent of output with a 2 per cent defect rate, B makes 40 per cent with a 5 per cent defect rate. Find $P(\text{a random item is defective})$. P
7. In the factory above, a defective item is found. Find the probability it came from B, and explain why it is not simply 0.05. E
8. Two events have $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \text{ and } B) = 0.2$. Decide whether they are independent, showing your reasoning. E

10.5 Risk, sampling and decisions

CRITERION D II, IV • RISK • EVIDENCE

A risk is a probability with consequences attached. Stating it as a frequency out of a round number is usually the clearest way to make it comparable.

A probability of 0.004 is hard to feel; ⩽4 people in 1 000<is not. Converting between the two is a communication skill with mathematical content: multiply by a convenient round number and round sensibly. The comparison of two risks is often more useful than either alone, and it is usually expressed as a ratio or a difference.

DEFINITION

Risk is the probability of an adverse outcome. **Relative risk** compares two risks as a ratio; **absolute risk** compares them as a difference. The same change can be a large relative risk and a small absolute one.

Statement	As a probability	As a frequency
1 in 250	0.004	4 in 1 000
1 in 20	0.05	50 in 1 000
5 per cent	0.05	1 in 20
twice as likely	<	2 in 1 000 instead of 1

The frequency form is usually understood more accurately than the probability form.

WORKED EXAMPLE 10.5.1 A SCREENING TEST

PROBLEM

A disease affects 1 in 1 000 people. A test correctly identifies 95 per cent of cases and wrongly flags 5 per cent of healthy people. In a population of 10 000, how many positive results are true cases?

1. Of 10 000 people, about 10 have the disease.
2. Of those 10, about 95 per cent, that is 9.5, test positive.
3. About 9 990 are healthy; 5 per cent of them, about 500, test positive wrongly.
4. So about 9.5 + 500 ≈ 510 positives occur, of which only about 10 are genuine.

ANSWER about 10 of 510 positives, that is roughly 2 per cent

This surprises most people: a 95 per cent accurate test is usually wrong when the disease is rare, because there are so many more healthy people to produce false positives.

WORKED EXAMPLE 10.5.2 COMPARING TWO RISKS

PROBLEM

A treatment reduces a risk from 2 in 1 000 to 1 in 1 000. Express the benefit in absolute and relative terms.

1. Absolute reduction: 0.002 - 0.001 = 0.001, that is 1 in 1 000.
2. Relative reduction: (0.002 - 0.001)/0.002 = 0.5, that is 50 per cent.
3. Both statements are true, and they support different decisions.
4. For an individual, the absolute risk matters; for a health service planning costs across a population, the relative reduction may matter more.

ANSWER absolute 1 in 1 000; relative 50 per cent

A 50 per cent relative reduction sounds larger than a 0.1 percentage-point absolute reduction because it is a different measure, not a different fact.

COMMON ERROR

Relative risk without absolute risk

Quoting a relative risk reduction without the absolute figures. ‹Halves the risk‹is compatible with a change from 2 in a million to 1 in a million, which may be negligible for any individual.

ATL

ATL ≠Communication

Communication. When you report a probability, give it twice: once as a number and once as a frequency out of a round figure. If the two forms seem to say different things, you have learned something about the number.

PRACTICE 10.5 ≣RISK AND DECISIONS

SUPPORT 1≣3 ≣CORE 1≣6 ≣EXTENSION 7≣3

1. Rewrite as a frequency out of 1 000: (a) 0.03 (b) 1 in 500 (c) 12 per cent. F
2. Rewrite as a probability: (a) 7 in 1 000 (b) 1 in 25 (c) 3 in 10 000. F
3. A risk falls from 6 in 1 000 to 4 in 1 000. Find the absolute and relative reductions. F
4. Explain why <a 50 per cent reduction in risk<can be misleading without the absolute figures. R
5. Explain why a test that is 99 per cent accurate can still be wrong most of the time when the condition is very rare. R
6. A disease affects 2 in 1 000. A test detects 90 per cent of cases and falsely flags 2 per cent of healthy people. In 5 000 people, estimate the number of positives and how many are genuine. P
7. A weather forecast gives a 30 per cent chance of rain. Explain what that statement does and does not mean. P
8. Two screening programmes are proposed: one detects a rare disease with many false positives, the other a common disease with few. Discuss which to fund first, stating the criteria you use. E

An investigation is not a longer exercise. It is a short piece of mathematics you do yourself: you gather evidence, look for a pattern, say what you think is true, and then test it until you can defend it.

INVESTIGATION

A sample is meant to represent a population. You will test how well a small sample does, and how much the answer changes with the size of the sample.

How representative is a sample?

Sample size	Proportion of reds (example)	Typical error
10	0.40	large
30	0.33	moderate
100	0.31	small
300	0.30	very small

1. **Set up.** Put 30 red and 70 blue counters in a bag, so the true proportion of red is 0.3. Do not look inside while sampling.
2. **Sample.** Take samples of 10, 30, 100 and 300 with replacement, recording the proportion of reds each time.
3. **Repeat.** Repeat each sample size ten times and record the range of the ten proportions for each size.
4. **Describe.** Plot the range against sample size and describe the relationship.
5. **Explain.** Explain why a larger sample gives a more reliable estimate, using your own ranges as evidence.
6. **Conclude.** State which sample size you would use to estimate the proportion to within 0.05, and justify your choice with your data.

Present a table of ranges by sample size, a graph of range against sample size, and a conclusion that uses your data rather than a general statement.

APPLICATION

The screening test

A health service is considering screening 50 000 people for a condition that affects 1 in 2 000 of them. The test detects 98 per cent of true cases and falsely flags 6 per cent of healthy people.

Group	Number of people	Test positive	Test negative
With condition	25	about 25	about 0
Without condition	49 975	about 2 999	about 46 976

1. Check the table's figures and correct any that are wrong, showing your working.
2. Calculate the proportion of positive results that are genuine, to the nearest per cent.
3. Explain, in one sentence, why this proportion is so low despite the test being 98 per cent accurate.
4. Estimate how many people will experience unnecessary anxiety or further testing, and how many true cases would be missed if the programme were not run.
5. Recommend whether the programme should proceed, stating the criteria you use and one further piece of information you would want before deciding.

CRITERION C COMMUNICATING

Explaining a 1-in-250 risk

A safety notice states that a product carries a 1-in-250 risk of a serious fault. Write a one-page explanation for a general audience.

1. Express the risk as a probability, as a percentage and as a frequency out of 10 000.
2. Explain what the figure does and does not tell a reader, including what it says about any individual product.
3. Explain how a reader should interpret the figure if the risk is halved, using both absolute and relative language.
4. Write the explanation so that a reader who is anxious about the product is informed rather than alarmed or reassured without reason.

1–2 Converts the figure but offers no interpretation.

3–4 Converts correctly and explains the meaning in one form.

5–6 Explains the figure in several forms and distinguishes absolute from relative change.

7–8 As above, and addresses what the number cannot say about an individual case, in language suited to the reader.

Chapter summary and review

Key ideas

- Probabilities lie between 0 and 1 and sum to 1 over a sample space.
- For equally likely outcomes, $P(A) = n(A)/n(S)$.
- Relative frequency estimates probability and converges as trials increase.
- The coin has no memory: past results do not change the next probability.
- Mutually exclusive events: add the probabilities.
- Independent events: multiply the probabilities.
- On a tree, multiply along branches and add across paths.
- Risk is clearest as a frequency out of a round number.
- A rare condition makes false positives dominate a screening result.

Vocabulary

sample space
The set of all possible outcomes.

event
A set of outcomes of interest.

relative frequency
The observed proportion of trials in which an event occurred.

mutually exclusive
Events that cannot both occur.

independent
Events where one does not affect the other's probability.

tree diagram
A branching diagram showing successive events.

two-way table
A table of counts by two categories.

risk
The probability of an adverse outcome.

false positive
A positive result when the condition is absent.

relative risk
The ratio of two risks.

What students often think	What is actually true
After several heads, tails is more likely.	The coin has no memory; the probability is still one half.
A correlation of 0 means no relationship.	It means no linear relationship.

Misconceptions worth confronting directly.

What students often think	What is actually true
A 99 per cent accurate test means a positive result is 99 per cent likely to be correct.	When the condition is rare, most positives are false.
Relative and absolute risk reductions are the same.	A 50 per cent relative reduction may be a tiny absolute change.

Revision checklist

- ☐ I can list a sample space systematically.
- ☐ I can calculate a theoretical probability.
- ☐ I can calculate and interpret a relative frequency.
- ☐ I can use the complement rule.
- ☐ I can apply the addition rule correctly.
- ☐ I can apply the multiplication rule, with and without replacement.
- ☐ I can complete a tree diagram or two-way table.
- ☐ I can interpret a risk as a frequency and compare two risks.

REFLECT

Write a probability question of your own for each of these: a mutually exclusive pair, an independent pair, a without-replacement pair, and a risk statement. Solve each and explain which rule you used.

Cumulative review and examination practice

A course is not a list of chapters but a set of connected tools. These papers ask you to choose a method when nobody names it for you < which is what an examination does, and what most real problems do.

> **HOW TO USE THESE PAPERS**
>
> Work each paper under timed conditions, then mark it against the answers. Paper A checks knowledge, B investigates a pattern, C communicates a proof, D applies mathematics; the exit task is a capstone that draws on all four.

PAPER A ▤ KNOWING AND UNDERSTANDING

Knowing and understanding

Criterion A ▯ suggested time 45 minutes ▯ no calculator except where indicated

1. Classify each number as rational or irrational, giving a reason: ▯49, ▯50, 0.121212<, ▯/2.
2. Write ▯108 in the form k▯m and state the value of $k + m$.
3. Evaluate without a calculator: (a) $(2/3)^{-2}$ (b) $16^{3/4}$ (c) $(3 \times 10^{5}) \div (4 \times 10^{-8})$ in scientific notation.
4. Expand and simplify: (a) $(2x - 3)(x + 5)$ (b) $(x + 4)^2 - (x - 4)^2$.
5. Factorise fully: (a) $6x^2 - 15x$ (b) $x^2 - 7x + 12$.
6. Solve $3(2x - 1) = 4x + 11$ and check your answer.
7. Solve the system $2x + 3y = 13$, $x - y = 1$.
8. Find the equation of the line through $(2, -1)$ and $(6, 7)$.
9. A cylinder has radius 4 cm and height 9 cm. Find its volume and total surface area, to three significant figures.
10. The probability that a machine produces a faulty part is 0.03. In a batch of 400 parts, how many would you expect to be faulty? Explain why the actual number may differ.

PAPER B INVESTIGATING PATTERNS

Investigating patterns

Criterion B suggested time 30 minutes no calculator except where indicated

1. A sequence of patterns is made from square tiles: figure 1 has 5 tiles, figure 2 has 9, figure 3 has 13. (a) Write the number of tiles in figure n. (b) Explain why the pattern is linear. (c) Find the figure with 97 tiles.
2. The table shows n and $n^2 + n$: 1 2, 2 6, 3 12, 4 20. (a) Describe the pattern in words. (b) Write a formula. (c) Test your formula on $n = 10$ by direct calculation.
3. A student claims that $n^2 + n + 41$ is prime for every whole number n. (a) Test $n = 1$ to $n = 5$. (b) Find a value of n for which the claim fails. (c) Explain what your counterexample does and does not show.
4. Investigate the sum of the first n odd numbers. Build a table for $n = 1$ to 6, describe the pattern, state a general rule, and test it for $n = 12$.

PAPER C ▤ COMMUNICATING

Communicating

Criterion C □ suggested time 35 minutes □ no calculator except where indicated

1. Prove that the sum of any three consecutive integers is divisible by 3. State the form you use for the integers and justify each step.
2. Prove that □3 is irrational. Your proof must state the assumption, use the fact that 3 is prime, and identify the contradiction.
3. Explain, with a diagram and with reference to the ratio of corresponding sides, why two triangles with equal corresponding angles are similar.
4. A classmate writes: <Correlation is 0.85, so screen time causes poor sleep.<Write a corrected version of the sentence that reports the same evidence without the error, and explain what evidence would be needed for the causal claim.
5. Describe, in no more than 120 words, the difference between an exact answer and an approximation, using □ and one other example.

PAPER D APPLYING MATHEMATICS IN REAL-LIFE CONTEXTS

Applying mathematics in real-life contexts

Criterion D suggested time 45 minutes no calculator except where indicated

1. A community hall is 12 m by 8 m with walls 3.2 m high. Paint covers 11 m per litre and is sold in 5 L tins. Allow for two coats and for 30 m of doors and windows that are not painted. How many tins must be bought, and what does the leftover paint cost at 24 units per tin?
2. A water tank is a cylinder of diameter 1.6 m and height 2.4 m, filled to 85% of capacity. A household uses 320 litres a day. For how many whole days will the tank last? State two assumptions in your model.
3. A phone contract costs 12 units a month plus 0.04 units per megabyte. A second costs 20 units a month plus 0.01 units per megabyte. Find the monthly data use at which the costs are equal, and recommend a contract for a user of 150 MB a month and for a user of 600 MB a month, with reasons.
4. A ramp must rise 0.9 m over a horizontal distance of 7 m. Find the angle of inclination and the length of the ramp. A regulation requires an angle of no more than 8 does this ramp comply?
5. A survey of 240 students finds that 156 walk to school. (a) Estimate the proportion for the whole school of 1 400 students. (b) Name one source of bias in the survey and explain the direction in which it might move the estimate.

The exit task

Year 3 exit task ≠ Does the revision app work?

A school trials a revision app with two Year 3 classes. Class A uses the app for six weeks; Class B continues as normal. At the end of the trial both classes sit the same test. The mean score in Class A is 68% and in Class B is 61%. The app's maker publishes the headline seven percentage points better

1. State two reasons why the comparison may not be fair, and explain how each would affect the conclusion.
2. Design a better trial. Say what you would randomise, what you would keep the same, and what you would measure.
3. The class sizes were 24 and 26. Calculate the mean score across both classes, and explain why the headline figure is not the same as the combined mean.
4. Write a recommendation to the head teacher. It must state the strength of the evidence, the decision you recommend, and one condition under which you would change your recommendation.

Paper	Target	Band	Evidence	Next action

Record the band you reached for each criterion, the evidence that shows it, and the next action you will take.

Assessment rubric for the cumulative papers

Criterion	1▤	3▤	5▤	7▤
A □ Knowing and understanding	1<2: recalls some facts and procedures	3<4: applies appropriate procedures in familiar situations	5<6: applies procedures accurately and selects correctly	7<8: applies and justifies the selection of methods
B □ Investigating patterns	1<2: describes a pattern with help	3<4: describes and continues a pattern	5<6: states a general rule and tests it	7<8: justifies the rule and its range of validity
C □ Communicating	1<2: uses some notation, with support	3<4: uses notation and organises reasoning	5<6: communicates coherently for the audience	7<8: communicates completely and concisely, interpreting the result
D □ Applying mathematics	1<2: identifies some relevant elements	3<4: applies a model to a situation	5<6: selects and applies a model, checking its adequacy	7<8: evaluates the model and justifies a decision in context

Answers and worked solutions

Full solutions, not only final values. A wrong answer should tell you which step to look at, and a right answer should tell you whether your reasoning would have worked in a slightly different case.

CHAPTER 1 THE REAL NUMBER SYSTEM

PREREQUISITE CHECK

1. 18/24 = 3/4 (divide by 6). 0.35 = 35/100 = 7/20. $7^2 = 49$ and $(-7)^2 = 49$. 4/3 = 1.333< , 7/5 = 1.4, so 7/5 is greater. 2.5 × 4 = 10. 1, 4, 9, 16, 25.

PRACTICE 1.1 RATIONAL NUMBERS AND DECIMAL FORM

1. (a) integer (b) rational (c) rational (d) natural (and integer, rational).
2. (a) 0.35 (b) 1.625 (c) 0.18.
3. (a) 0.666< (b) 0.454545< (c) 0.58333< .
4. Reduce first: (a) 1/8, terminates (b) 1/10, terminates (c) 11/30, repeats (denominator has a factor 3) (d) 9/64, terminates. A fraction terminates exactly when the reduced denominator has no prime factor other than 2 and 5.
5. (a) 5/9 (b) 3/11 (c) 5/12.
6. 7/24 lies between 1/4 = 6/24 and 1/3 = 8/24; then 13/48 lies between 1/4 = 12/48 and 7/24 = 14/48.
7. Let $x = 0.999<$. Then $10x = 9.999<$, so $10x - x = 9$, giving $9x = 9$ and $x = 1$. The two decimals are the same number written twice.
8. 1/6 = 0.1666< repeats, and 1/3, 1/9, 1/7 also repeat. The denominator 6 has a prime factor 3, which no power of 10 can divide.

PRACTICE 1.2 IRRATIONAL NUMBERS AND SURDS

1. (a) rational, $\sqrt{9} = 3$ (b) irrational (c) rational (d) irrational (e) rational.
2. (a) $5\sqrt{2}$ (b) $4\sqrt{3}$ (c) $10\sqrt{2}$ (d) 2.
3. (a) $2\sqrt{2}$ (b) $2\sqrt{5}$ (c) $\sqrt{3}$.
4. (a) 6 and 7 (b) 8 and 9 (c) -5 and -4.
5. $\sqrt{3} \times \sqrt{3} = 3$ because a square root is defined as the number whose square is the radicand. Adding gives $2\sqrt{3} \approx 3.46$, which is not 3: addition does not undo a square root, only squaring does.
6. Suppose $\sqrt{5} = p/q$ in lowest terms. Then $5q^2 = p^2$, so 5 divides p^2 and hence 5 divides p. Write $p = 5k$; then $5q^2 = 25k^2$, so $q^2 = 5k^2$ and 5 divides q. That contradicts p and q having no common factor. Primality of 5 is used when concluding that 5 divides p from 5 dividing p^2.
7. $2\sqrt{2}$ is exact; 2.82 is a rounded decimal. Writing = 2.82 claims an equality that is false, because $2\sqrt{2} = 2.8284<$. The correct statement is $\sqrt{8} = 2\sqrt{2} \approx 2.83$.
8. $\sqrt{2} + \sqrt{8} = \sqrt{2} + 2\sqrt{2} = 3\sqrt{2}$. $\sqrt{2} + \sqrt{3}$ cannot be a single surd: if $\sqrt{2} + \sqrt{3} = \sqrt{n}$ then squaring gives $5 + 2\sqrt{6} = n$, so $\sqrt{6}$ would be rational, which it is not.

PRACTICE 1.3 ORDER, ABSOLUTE VALUE AND INTERVALS

1. (a) $\sqrt{6} \approx 2.449$, so < (b) $\sqrt{11} \approx 3.317$, so > (c) $\sqrt{2}/2 \approx 0.707$, so >.
2. (a) [-1, 4) (b) (6, ∞) (c) (-∞, -2].
3. (a) $x = 7$ or $x = 1$ (b) $-3 < x < 1$ (c) $x \geq 4$ or $x \leq 6$.

4. $|x - 3| < 0.1$.
5. □10 □ 3.1623, □□ 3.1416, 22/7 □ 3.1429, $3.1 <$ □ $< 22/7 <$ □10.
6. For positive values squaring preserves order, because $a < b$ gives $b^2 - a^2 = (b - a)(b + a) > 0$. For negative values the factor $b + a$ is negative, so the inequality reverses: $-5 < -3$ but $25 > 9$.
7. Acceptable lengths run from 249.5 mm to 250.5 mm, that is |L - 250| □ 0.5.
8. $|n - 7| < 3$ means $4 < n < 10$, so the integers are 5, 6, 7, 8 and 9. The open interval excludes 4 and 10.

PRACTICE 1.4 ▤ WORKING WITH SURDS

1. (a) 2□3 + 3□3 = 5□3 (b) 3□5 - 2□5 = □5 (c) 4□2 + 2□2 - 3□2 = 3□2.
2. (a) 5 - 1 = 4 (b) 4 □ 3 = 12 (c) 3 - □2 + 3□2 - 2 = 1 + 2□2.
3. (a) 2 - □3 (b) (5(□7 + □2))/5 = □7 + □2 (c) (8 + 4□3)/4 = 2 + □3.
4. □8/□2 = □(8/2) = □4 = 2, and □8/□2 = 2□2/□2 = 2. *Both are valid because* □a/□b = □(a/b).
5. □2 □ □18 = □36 = 6, an integer. Adding gives □2 + 3□2 = 4□2, which is irrational.
6. Area (3 + □2)(3 - □2) = 9 - 2 = 7 cm≤ perimeter 2(3 + □2 + 3 - □2) = 12 cm.
7. 1/(□2 + 1) = (□2 - 1)/((□2 + 1)(□2 - 1)) = □2 - 1. Similarly 1/(□2 - 1) = □2 + 1. The pair are reciprocals, so each rationalises to the other.
8. $(3 + □2)^2 = 9 + 6□2 + 2 = 11 + 6□2$, so $a = 11$, $b = 6$. $(3 + □2)^3 = (11 + 6□2)(3 + □2) = 33 + 11□2 + 18□2 + 12 = 45 + 29□2$.

PRACTICE 1.5 ▤ SETS, LOGIC AND PROOF

1. (a) □ (b) □ (c) □ (d) □.
2. (a) universal; one square that is not a rectangle disproves it (b) existential; showing no square is a rectangle disproves it.
3. 9 is odd but 9 = 3 □ 3, not prime; 15 and 21 also fail.
4. Let the odd numbers be $2m + 1$ and $2n + 1$. Their product is $4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1$, which has the form $2k + 1$ and is therefore odd.
5. One example in which the product happens to be rational does not establish the general claim. A counterexample is □2 □ □3 = □6, which is irrational.
6. If $n = 2k + 1$ then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, of the form $2m + 1$, so n^2 is odd.
7. Suppose r is the largest rational less than 1. Then $(r + 1)/2$ is rational and lies strictly between r and 1, contradicting the assumption.
8. Every rational can be written in lowest terms by cancelling common factors, so the assumption loses no generality. Without it, the contradiction at the end would not follow: p and q would share a factor that could simply be cancelled.

INVESTIGATION AND APPLICATION

1. $T = 0.1 □ 2^n$ mm; $n = 1$ gives 0.2 mm and $n = 10$ gives 102.4 mm, matching the table.
2. The graph is nearly flat at first because adding one to n doubles a very small number; the same doubling only becomes visible once the value is large.
3. Each fold doubles the previous thickness, so the factor 2 is applied n times: 2^n, not $0.1n$.
4. $0.1 □ 2^n > 3.84 □ 10^8$ mm requires $2^n > 3.84 □ 10^9$, so $n = 32$ (since $2^{31} □ 2.15 □ 10^9$ and $2^{32} □ 4.29 □ 10^9$).
5. At $n = 32$ the thickness is $4.29 □ 10^8$ mm = $4.29 □ 10^5$ m, about 429 km < well beyond the Moon's distance of 384 400 km; at $n = 31$ it is about 215 km.
6. Physical paper cannot be folded 32 times, so the model's prediction cannot be tested by folding. The mathematics is still correct as a statement about doubling; what fails is the assumption that the physical process can be continued.
7. 1 m represents $4.5 □ 10^9 □ 40 = 1.125 □ 10^8$ km.
8. $1.39 □ 10^6 □ 1.125 □ 10^8 = 1.24 □ 10^{-2}$ m □ 12.4 mm.

9. At 0.11 mm the Earth is smaller than a grain of sand and cannot be seen from any distance at which Neptune is in view. One response is to draw the planets at an exaggerated size and say so in a note.
10. For the Earth to be 2 mm, the scale is $1.27 \times 10^4 \times 2 = 6.35 \times 10^3$ km per mm, so the corridor would need $4.5 \times 10^9 \times 6.35 \times 10^6 = 709$ m of length.
11. Any defensible recommendation with a stated trade-off: the 40 m scale shows distance well and size badly; a size-exaggerated scale shows both but misrepresents the ratio of diameter to distance.

CHAPTER 2 POWERS AND SCIENTIFIC NOTATION

PREREQUISITE CHECK

1. $3^4 = 81$. $\pm 81 = 9$; $\sqrt{(1/4)} = 1/2$. $10\,000 = 10^4$. $x^2 \times x^3 = x^5$. 4.68. 3.5 m = 3500 mm.

PRACTICE 2.1 INTEGER EXPONENTS AND THE EXPONENT LAWS

1. (a) a^{10} (b) b^5 (c) c^{15} (d) $8d^3$.
2. (a) 1 (b) 1/25 (c) 125/8 (d) 1/10.
3. (a) $4x^4$ (b) $9a^4b^2$ (c) x^2y^2.
4. (a) 2^3 (b) 2^{-4} (c) 2^4.
5. $a^n \times a^n = a^{n-n} = a^0$, and the same quotient by direct cancellation is 1, so a^0 must be 1.
6. With $x = 2$: $(3 \times 2)^2 = 36$ but $3 \times 2^2 = 12$. The correct expansion is $(3x)^2 = 9x^2$.
7. $(2m^3n^2)^2 = 4m^6n^4$ (power of a product, power of a power); $4m^6n^4 \times 3mn^4 = 12m^7n^8$ (multiplication law).
8. Raise a^m to the power n by multiplying exponents in either order: $mn = nm$. The order in which the two powers are applied does not affect the result.
9. $x = 5$ and $y = 4$, so $x + y = 9$. $2^5 \times 3^4 = 32 \times 81 = 2592$.

PRACTICE 2.2 ROOTS AND RATIONAL EXPONENTS

1. (a) 5 (b) 2 (c) 3 (d) -3.
2. (a) 4 (b) 125 (c) 1/4 (d) 1/16.
3. (a) $a^{1/2} \times a^{1/4} = a^{3/4}$ (b) $a^{3/2 - 1/2} = a^1$ (c) a^3.
4. (a) not real (b) -5 (c) not real.
5. The square root is defined to be non-negative. For $a = -3$, $\sqrt{((-3)^2)} = \sqrt{9} = 3 = |\{-3\}|$, not -3.
6. The order of the operations does not matter because the exponents multiply: $(2/3) \times 2 = 4/3$ in either order.
7. Edge $= 125^{1/3} = 5$ cm.
8. $x = 27^{2/3}$; $27^{1/3} = 3$ and $3^2 = 9$, so $x = 9$. Check: $9^{3/2} = 27$.
9. $\sqrt{2} \times \sqrt{2} = 2^{1/2} \times 2^{1/3} = 2^{5/6}$, which is a single power of 2, and not an integer.

PRACTICE 2.3 SCIENTIFIC NOTATION

1. (a) 4.7×10^4 (b) 8.1×10^{-4} (c) 6.3×10^6 (d) 5×10^{-7}.
2. (a) 27 000 (b) 0.009 03 (c) 1 000 000.
3. (a) 8×10^8 (b) 3×10^{-6}.
4. (a) 1.24×10^5 (b) 3.7×10^{-3}.
5. Estimate $(6 \times 10^6) \div (2 \times 10^2) = 3 \times 10^4$; exact value 2.81×10^4. The estimate is within 10 per cent.
6. $600 \times 90 = 54\,000$ g = 54 kg, so $54 \div 2 \times 10^{-5} = 2.7 \times 10^6$ grains.
7. $1 \times 10^{-4} = 0.0001$ and $1 \times 10^{-3} = 0.001$, so every mantissa between 1 and 10 times 10^{-4} lies between 0.0001 and 0.001.
8. $1 \times 1.67 \times 10^{-27} = 5.99 \times 10^{26}$, so about 6.0×10^{26} protons.

PRACTICE 2.4 ORDERS OF MAGNITUDE AND ESTIMATION

1. (a) 8×10^3, order 3 (b) 3×10^{-2}, order -2 (c) 6×10^7, order 7.
2. (a) $30 \times 40 = 1200$ (b) $(6 \times 10^5) \times (2 \times 10^2) = 3 \times 10^3$ (c) $5 \times 10^{-3} \times 2 \times 10^{-3} = 10^{-5}$.
3. $7 \times 2 \times 10^2 = 3.5 \times 10^2$, so about 350 times, or 4×10^2 to one significant figure.
4. Report 1.2×10^5: two significant figures, matching the inputs. The remaining digits are artefacts of the arithmetic, not information.
5. Assume a grain has mass 2×10^{-5} kg; $1 \times 2 \times 10^{-5} = 5 \times 10^4$ grains, of the order of 10^5.
6. Estimate the volume as 8 m × 6 m × 3 m = 144 m<and the density of air as about 1.2 kg/m< giving about 170 kg, of the order of 10^2 kg.
7. Twice as large is a factor of 2; one order of magnitude larger is a factor of about 10. The second is five times the change of the first.
8. 3×10^6 and $8 \times 10^5 = 0.8 \times 10^6$ differ by a factor of about 3.75, which is less than one order of magnitude, so the two estimates are consistent to within their stated precision.

PRACTICE 2.5 SIGNIFICANT FIGURES AND PRECISION

1. (a) 2 (b) 4 (c) 2 (ambiguous; 1.2×10^3 would make it 2) (d) 3.
2. (a) 0.088 (b) 46 (c) 10 (d) 0.0051.
3. (a) 12 300 (b) 0.123 (c) 1.01.
4. 33.6 cm; the inputs are given to tenths of a centimetre, so the total is reported to 0.1 cm.
5. In 1.5×10^3 the mantissa shows which digits are significant. The plain 1500 does not say whether the zeros are measured or merely place-holders.
6. $42.0 \times 1.6 = 26.25$, but 1.6 has two significant figures, so report 26 km/h.
7. Yes: doubling an exact count does not reduce the precision of Avogadro's constant. Both values carry three significant figures.
8. Areas: $3.4 \times 5.6 = 19.04 \times 19$ cm<(two significant figures); $3.40 \times 5.60 = 19.04 \times 19.0$ cm<(three significant figures). The second measurement is finer, so the result may be reported to three significant figures.

INVESTIGATION AND APPLICATION

1. $T = 0.1 \times 2^n$ mm; $n = 1$ gives 0.2 mm, $n = 10$ gives 102.4 mm.
2. The curve is nearly flat for small n because the values are tiny; doubling a small number produces a small change.
3. Folding n times doubles the thickness n times, so it multiplies by 2^n. Adding a fixed 0.1 mm per fold would give $0.1(n + 1)$ mm, a straight line rather than a curve.
4. Using the formula in metres, thickness $= 10^{-4} \times 2^n$ m. Solving $10^{-4} \times 2^n > 3.84 \times 10^8$ gives $2^n > 3.84 \times 10^{12}$, so the smallest whole n is 42.
5. At $n = 42$, $2^{42} = 4.398 \times 10^{12}$, so the thickness is 4.398×10^8 m; at $n = 41$ it is 2.199×10^8 m, below the Moon's distance. The step from 41 to 42 crosses it.
6. The formula predicts a thickness no sheet could reach; the failure is physical, not mathematical.
7. Scale: 40 m represents 4.5×10^9 km, so 1 m represents 1.125×10^8 km.
8. Sun: $1.39 \times 10^6 \times 1.125 \times 10^8 = 1.24 \times 10^{-2}$ m = 12.4 mm.
9. At 0.11 mm the Earth is invisible in practice; the model can show distance or size, not both.
10. For a 2 mm Earth the scale is 6.35×10^3 km per mm, needing about 709 m.
11. Any justified recommendation; the essential point is that a scale model must choose which quantity it represents faithfully.

CHAPTER 3 FUNCTIONS AND ALGEBRAIC REASONING

PREREQUISITE CHECK

1. $3x + 5x - 2x = 6x$. $8x - 12$. $2(-3) + 1 = -5$. $x = 5$. 14, 17, 20. $(-2)^2 = 4$; $-2^2 = -4$.

PRACTICE 3.1 VARIABLES, EXPRESSIONS AND SUBSTITUTION

1. Three terms, coefficient -3, constant term 8.
2. (a) 15 (b) 0 (c) -2/3.
3. (a) $6y$ (b) $3m + 5n$ (c) $2p^2 + 2p$.
4. With $x = 3$: $-x^2 = -9$ and $(-x)^2 = 9$; the bracket decides whether the sign is squared.
5. With $x = 2$: $3x = 6$ but $x^3 = 8$. Different variable parts never combine.
6. $2.50 + 1.80 \times 7 = 15.10$; the constant term is the fixed charge for hiring the taxi at all.
7. $x^2 > x^3$ when $0 < x < 1$, for example $x = 1/2$; $x^2 < x^3$ when $x > 1$, for example $x = 2$. Higher powers are not always larger.
8. $2x + 3$ is even when $x = 1/2$ (its value is 4), so the claim is false. Testing only whole numbers hides the counterexample.

PRACTICE 3.2 FUNCTIONS, DOMAIN AND RANGE

1. (a) 7 (b) 1 (c) 8.
2. (a) 5 (b) 5 (c) $x = 2$ or $x = -2$.
3. (a) all real x (b) $x \neq -4$ (c) $x \neq -2$.
4. $x = 4$ and $x = 0$.
5. Two outputs for one input would mean the vertical line at that input met the graph twice.
6. $C(40) = 720$; $C(n) = 420$ gives $n = 20$, the number of people the budget covers.
7. Minimum at $x = 0$ is 1; endpoints give $f(-2) = 5$ and $f(3) = 10$; range $1 \leq f(x) \leq 10$.
8. $f(0) = 1$ but $g(0) = 2$; the bracket changes the order of operations, so the two rules differ for every x.

PRACTICE 3.3 EXPANDING AND FACTORISING

1. (a) $6x - 15$ (b) $-2x - 8$ (c) $x^2 - 7x$.
2. (a) $x^2 + 6x + 8$ (b) $x^2 - 5x - 6$ (c) $2x^2 - 5x - 3$.
3. (a) $x^2 + 10x + 25$ (b) $x^2 - 8x + 16$ (c) $x^2 - 49$.
4. (a) $4(2x + 3)$ (b) $x(x + 6)$ (c) $3x(x - 3)$.
5. (a) $(x + 3)(x + 5)$ (b) $(x - 2)(x - 3)$ (c) $(x - 4)(x + 3)$.
6. Draw a rectangle x by x, two rectangles x by a and x by b, and one a by b; the four areas sum to the expansion.
7. $(x + 3)(x + 4)$; for positive side lengths $x > -3$.
8. $(x + 3)^2 - (x - 3)^2 = (x^2 + 6x + 9) - (x^2 - 6x + 9) = 12x$. The difference of the two squares grows linearly with x.
9. $2x^2 + 6x + x + 3 = 2x(x + 3) + (x + 3) = (2x + 1)(x + 3)$. The split reproduces the middle term as the sum of the two products.

PRACTICE 3.4 REARRANGING FORMULAS

1. (a) $x = y - 7$ (b) $x = y/4$ (c) $x = (y + 2)/5$.
2. (a) $t = (v - u)/a$ (b) $t = \pm(s/5)$ (c) $t = P/2 - l$.
3. $a = 2A/b$.
4. $h = V/(\pi r^2)$.
5. A radius is a length, so the negative root is rejected; the formula describes a positive quantity.

6. $d = (C - 45)/0.6$; with $C = 120$, $d = 125$ km.
7. $x = (3y + 1)/(2 - y)$; $y \neq 2$.
8. The error is dividing 12 and $3x$ by 3 but leaving $2y$ undivided: the correct form is $x = (12 - 2y)/3$. With $y = 3$, $x = 2$; the student's version gives $x = -2$.
9. With $d = 0$, $S = (n/2)(2a) = an$, so $n = S/a$; the number of terms is the total divided by the first term.

PRACTICE 3.5 SEQUENCES, PATTERNS AND GENERALISATION

1. (a) 18, 22 (b) 11, 8 (c) 30, 42.
2. (a) $3n + 1$ (b) $5n + 4$ (c) $34 - 4n$.
3. 1, 4, 7, 10, 13; $3n - 2 = 43$ gives $n = 15$.
4. $n(n + 1)/2$; the differences 2, 3, 4, 5 increase by 1, so the rule is quadratic and fits the triangular numbers.
5. Consecutive terms differ by d, and adding d once per step is exactly what multiplication by n does, so the general term is $dn + c$.
6. $n^3 - 6n^2 + 12n - 6$ also begins 2, 4, 8, 16 but gives 46 at $n = 5$. Finitely many terms cannot determine a rule.
7. $T = 3n + 2$; $3n + 2 = 62$ gives $n = 20$.
8. $(n + 1)^2 + 2 - (n^2 + 2) = 2n + 1$, which is odd because it is one more than an even number.
9. $1 + 3 + < + (2n - 1) = n^2$; for $n = 8$ the sum is 64.

INVESTIGATION AND APPLICATION

1. Counts: 3, 5, 7, 9 for figures 1 to 4.
2. Each new triangle shares one side with the previous one, so $3 - 1 = 2$ matches are added each time.
3. $M = 2n + 1$: one match for the shared side of each triangle plus the two outer sides of the last, or $3 + 2(n - 1)$.
4. Figure 12 needs 25 matches and figure 50 needs 101.
5. Sharing two sides adds only one match each time: $M = n + 2$.
6. The structure of the construction fixes the number added at each step, so the rule holds for every n, not only for the figures drawn.
7. $C = 9(3 + A/14) = 27 + (9/14)A$; the constant 27 is the three hours of preparation.
8. $A = 14(C/9 - 3)$; with $C = 108$, $A = 14(12 - 3) = 126$ m<
9. The rate changes the coefficient of A from 9/14 to 9/10; the constant term is unchanged.
10. The model treats the volunteer's time as free. For a decision about cost only, that is defensible; for a decision about time, it is not.
11. Any recommendation that identifies the objective being optimised and the assumption it relies on.

CHAPTER 4 LINEAR RELATIONSHIPS

PREREQUISITE CHECK

1. The points lie on a straight line rising 2 for every 1 across. x = 4. y = 5. y = 7 - 2x. 2. 2 for 1 has gradient 2; 3 for 2 has gradient 1.5, so the first line is steeper.

PRACTICE 4.1 GRADIENT AND RATE OF CHANGE

1. (a) $m = 3$ (b) $m = -1$ (c) $m = 2$.
2. (a) negative, falling (b) zero, horizontal (c) positive, gently rising.
3. $45 \div 3 = 15$ km per hour.
4. +15 units per month; positive.
5. A vertical line has x<- x<= 0, and dividing by zero is undefined, so no gradient exists.

6. The product of the gradients is 2 × (-0.5) = -1, so the lines meet at right angles.
7. Rate = (72 - 96)/6 = -4 °C per minute; 40 = 96 - 4t gives t = 14 minutes.
8. During the stop the gradient is zero: the distance does not change while time passes.

PRACTICE 4.2 THE EQUATION OF A STRAIGHT LINE

1. (a) m = 2, c = 5 (b) m = -1, c = 1 (c) m = 0.5, c = -3.
2. (a) y = 6 - x (b) y = 9 - 2x (c) y = 2x - 5.
3. y = 4x - 2.
4. (a) y = 2x + 1 (gradient 2, intercept 1) (b) m = (1 - 7)/(4 - 1) = -2, so y = -2x + 9.
5. (a) parallel (b) not parallel (c) parallel, since the first is y = 2x + 0.5.
6. The fixed charge for hiring the taxi, before any distance is travelled.
7. y = 3x - 7.
8. y = 4, a horizontal line, because the y-value does not change.
9. m = (-7 - 5)/(4 + 2) = -2, so y = -2x + 1. It crosses the y-axis at 1 and the x-axis at x = 0.5.

PRACTICE 4.3 LINEAR MODELS AND SIMULTANEOUS LINES

1. (a) x = 2, y = 5 (b) 3x - 2 = -x + 6 gives x = 2, y = 4.
2. x-axis at (-2, 0); y-axis at (0, 4).
3. 15 + 3h = 30 + 1.5h gives 1.5h = 15, so h = 10 hours.
4. 2x - 1 = 6 - x gives x = 7/3 ≈ 2.33, y = 3.67; the reading from the graph should be checked against this.
5. Parallel lines never meet, so two tariffs with the same rate but different fixed charges are never equal: one is always cheaper.
6. 40 + 18m and 100 + 8m; equal when 10m = 60, so m = 6 months. Over two years the second is cheaper: 40 + 432 = 472 against 100 + 192 = 292.
7. h = 24 - 1.5t; h = 0 at t = 16 hours. Valid for 0 ≤ t ≤ 16.
8. (a) parallel when m = 3 and c ≠ 2 (b) (1, 5) on y = mx + 2 gives m = 3; on y = 3x + c gives c = 2.

PRACTICE 4.4 LINEAR MODELS FROM DATA

1. Gradient about 2.0: the points rise roughly 2 units for each 1 across.
2. (12 000 - 9 000)/3 = 1 000 units per year.
3. h(60) = 24.0 - 8.4 = 15.6 cm. At t = 200 the model gives a negative height, which is impossible, so the model has been used past its domain.
4. Interpolation estimates between known values; extrapolation estimates beyond them and assumes the trend continues.
5. Two points fix a line but ignore the rest of the data; if either endpoint is unusual, the line misrepresents the other points.
6. B = 100 - (38/90)t; B = 0 at t ≈ 237 minutes, so about 147 minutes remain, assuming the drain rate is constant (it usually is not near zero).
7. Richer towns may afford more libraries; income is the third variable driving both.
8. The model predicts 1200 + 630 = 1830 but the output is 1500, about 18 per cent below prediction < perhaps demand changed or the rate of increase slowed.

PRACTICE 4.5 INVERSE PROPORTION AND NON-LINEAR RELATIONS

1. (a) direct (b) inverse (c) neither (area grows as the square).
2. k = 20, so y = 2.
3. 4 × 30 = 120 tap-minutes; 120 ÷ 6 = 20 minutes.
4. A curve falling from (2, 10) to (10, 2).

5. As x grows the value approaches zero but never reaches it; as x approaches zero the value grows without limit.
6. 80 ? 3 = 240 km; at 60 km/h the time is 4 hours. It **is** 4 hours, because time is inversely proportional to speed.
7. $k = 90$; with 9 workers, $t = 10$ days.
8. Too many workers get in each other's way and the work cannot be subdivided indefinitely, so the inverse model holds only for a limited range.

INVESTIGATION AND APPLICATION

1. The points lie close to a straight line.
2. Gradient about -2.8/20 = -0.14 cm per minute; intercept about 24 cm.
3. The model predicts 24 - 0.14(60) = 15.6 cm against a measured 15.3 cm; the difference of 0.3 cm is within reading error.
4. The gradient is the rate at which the candle shortens; it is negative because the height falls.
5. $0 = 24 - 0.14t$ gives t ? 171 minutes, about 100 minutes beyond the last reading.
6. The model is adequate: the differences are consistent with measurement error and the points show no systematic curve.
7. X: $C = 12 + 1.4u$; Y: $C = 2.6u$. The intercept of X is the fixed quarterly charge; Y has none.
8. $12 + 1.4u = 2.6u$ gives $1.2u = 12$, so $u = 10$ m<
9. 18 m< X costs 37.20, Y costs 46.80, so X. 30 m< X 54, Y 78, so X. 6 m< X 20.40, Y 15.60, so Y.
10. At 4 m< Y costs 10.40 against X's 17.60. Fairness is not the same as cheapness: X's fixed charge makes low users pay for infrastructure they barely use, which a committee may consider fairer as a contribution to the network.
11. $20 + 1.4u = 2.6u$ gives $u = 16.7$ m< so the break-even rises. The difference between the tariffs at $u = 4$ rises from 7.20 to 15.20 units, so low users are hit hardest.

CHAPTER 5 EQUATIONS AND INEQUALITIES

PREREQUISITE CHECK

1. $x = 5$. ?$x = 7$. ?$x = 7$. ?$4x - 12$. ?$x = -4$. ?$2x$.

PRACTICE 5.1 SOLVING LINEAR EQUATIONS

1. (a) $x = 4$ (b) $x = -3$ (c) $x = 15$.
2. (a) $x = 4$ (b) $x = 6$ (c) $x = -2$.
3. (a) $x = 10$ (b) $x = 11$ (c) $x = 13$.
4. $7 - 3x + 6 = 2x + 1$ gives $13 - 3x = 2x + 1$, so $x = 12/5 = 2.4$. Check: $7 - 3(0.4) = 5.8$ and $2(2.4) + 1 = 5.8$.
5. Multiplying by zero destroys information: $2x = 6$ becomes $0 = 0$, which is true for every x, so the solution set is no longer the same.
6. Dividing only the left side by 3 gives $x + 6 = 12$. The correct step divides both sides: $x + 2 = 4$, so $x = 2$.
7. Width w, length $w + 5$: $2(w + w + 5) = 46$ gives $w = 9$, so 9 cm by 14 cm.
8. Both sides simplify to $-5 = 4$, which is false; the equation has no solution.
9. $kx + 5 = 3x + 5$ becomes $(k - 3)x = 0$. If $k = 3$ every x satisfies it; if $k = 4$ the only solution is $x = 0$. No value of k makes it have no solution, because the constants match; changing it to $kx + 5 = 3x + 4$ makes $k = 3$ give no solution.

PRACTICE 5.2 FORMING EQUATIONS FROM PROBLEMS

1. 32 and 33.
2. $4n - 9 = 31$ gives $n = 10$.
3. Width 8 cm, length 24 cm.
4. Son 6, father 24 now; in 12 years 18 and 36.

5. Because a count of people must be a non-negative whole number; an algebraic solution outside that set is rejected by the context, not by the algebra.
6. $15a + 9c = 3660$ and $a + c = 320$; substituting gives $6a = 780$, so 130 adult and 190 child tickets.
7. 41 and 55.
8. $3s + 6l = 138$ and $s + l = 30$ gives $3s + 180 - 6s = 138$, so $s = 14$, $l = 16$. With 139 kg the solution is not a whole number, which tells you no combination of whole boxes gives that weight.

PRACTICE 5.3 LINEAR INEQUALITIES

1. (a) $x > -3$ (b) $x \leqslant 5$ (c) $x < 8$.
2. (a) $x \geqslant -4$ (b) $x < 3$ (c) $x > -3$.
3. (a) $2x < 8$, so $x < 4$ (b) $2x + 2 \leqslant 5x - 4$ gives $6 \leqslant 3x$, so $x \geqslant 2$.
4. (a) $s \leqslant 50$ (b) $n \geqslant 12$ (c) $T < 4$.
5. $2 < 3$ but $-2 > -3$: multiplying by -1 reflects the number line, so the order reverses. Test $-2x < 6$: at $x = 4$, $-8 < 6$ is true and 4 is greater than -3, so the correct solution is $x > -3$.
6. $480 \div 68 = 7.06$, so at most 7 people; the solution set is $\{0, 1, <, 7\}$.
7. $4.5x + 15 \leqslant 60$ gives $x \leqslant 10$, so at most 10 notebooks.
8. $2 < 3x - 1 \leqslant 11$ adds 1: $3 < 3x \leqslant 12$, so $1 < x \leqslant 4$. As a set of integers this is $\{2, 3, 4\}$; as an interval, $(1, 4]$.
9. Taking square roots of an inequality is not reversible: $x^2 > 9$ is satisfied by $x = -4$, which does not satisfy $x > 3$. The solution is $x > 3$ or $x < -3$.

PRACTICE 5.4 SIMULTANEOUS EQUATIONS

1. (a) $x = 5, y = 2$ (b) $x = 4, y = 3$.
2. (a) $x = 4, y = 2$ (b) $x = 3, y = 3$.
3. (a) $x = 3, y = 7$ (b) $y = 3, x = 9$.
4. Multiply the first by 2 and the second by 3: $4x + 6y = 14$ and $9x - 6y = 12$. Adding gives $13x = 26$, so $x = 2$ and $y = 1$.
5. Substituting into the equation you rearranged just reproduces that equation; the second equation's information is never used.
6. $c + 2t = 12$ and $2c + 3t = 21$; from the first $c = 12 - 2t$, so $24 - 4t + 3t = 21$, giving $t = 3$ and $c = 6$. A coffee costs 6 units and a tea 3 units.
7. The second equation is twice the first, so they describe the same line; every point on it satisfies both.
8. The gradients are equal and the intercepts differ, so the lines are parallel and never meet.
9. $y = 5 - kx$ and $y = 5 - 2x$ are parallel when $k = 2$; then the lines are distinct and there is no solution.

PRACTICE 5.5 SYSTEMS, CONSTRAINTS AND FEASIBILITY

1. (a) $2 + 6 = 8 \leqslant 10$, yes (b) $6 + 6 = 12 > 10$, no (c) $0 + 10 = 10 \leqslant 10$, yes.
2. A triangle with vertices at (0, 0), (6, 0) and (0, 6), shaded inside.
3. The region below the line $y = 2x - 1$; (0, 2) satisfies $2 \geqslant -1$.
4. A vertex is a plan that uses all of at least two resources exactly; for a linear objective the best plan is always at one of them.
5. $b + r \leqslant 200$ and $2b + r \leqslant 320$, with $b, r \geqslant 0$.
6. Vertices (0, 0), (5, 0), (5, 3), (3, 5), (0, 5). Values of P: 0, 15, 21, 19, 10. Maximum 21 at (5, 3).
7. Along an edge one variable changes at a constant rate, so P changes linearly; a linear function on a segment attains its extremes at the endpoints, which are vertices.
8. Each project is a decision variable, the budget gives one constraint and any minimum benefit a second; the objective is total benefit. The solution identifies the combination that maximises benefit within the budget, and shows which constraint binds.

INVESTIGATION AND APPLICATION

1. Eliminating the variable gives a true statement ($0 = 0$) when the lines coincide and a false one ($1 = 4$) when they are parallel and distinct.
2. Equal gradients and equal intercepts give infinitely many solutions; equal gradients and different intercepts give none.
3. Different gradients always give exactly one solution.
4. $3x - y = 4$ and $6x - 2y = 9$ have equal gradients (3 and 3) and different intercepts, so no solution. Solving confirms $0 = 1$.
5. Coincident lines, parallel distinct lines, and lines crossing once.
6. The variable vanishes because the coefficients are proportional; what remains is a statement about the constants, and its truth or falsehood decides the case.
7. $90w + 150k \leq 1200$, $w \geq 4$, $0 \leq k \leq 8$, $w \geq 0$.
8. A quadrilateral with vertices at (4, 0), (13.3, 0), (4, 5.6) and along the lines given, subject to $k \leq 8$.
9. (4, 5), (6, 4) and (8, 3) are feasible; (12, 1) costs 1230 and breaks the budget.
10. Reach $= 25w + 60k$; testing vertices gives the maximum at the largest feasible k with $w \geq 4$, namely (4, 5) with $100 + 300 = 400$, or (4, 5.6) rounded to (4, 5) in whole kits.
11. Any recommendation with a stated judgement: for example, prioritising workshops because they reach more young people per unit cost is a value judgement about breadth versus depth that the arithmetic cannot settle.

CHAPTER 6 TRANSFORMATIONS AND CONGRUENCE

PREREQUISITE CHECK

1. Distance 3 from the y-axis. (2, -2). Equilateral triangle. 60° each. 12. 360°

PRACTICE 6.1 TRANSLATIONS, ROTATIONS AND REFLECTIONS

1. (5, 1).
2. (a) (3, -2) (b) (-3, 2) (c) (2, 3).
3. (2, 0) rotated 90° anticlockwise is (0, 2); rotated a further 180° it is (0, -2).
4. (-2, 1), (2, 1), (-2, 4).
5. The reflection exchanges left and right, so a sequence read clockwise in the object is read anticlockwise in the image.
6. Reflection in the line $y = x$.
7. A 90° clockwise rotation then a reflection in the x-axis is equivalent to a single reflection in the line $y = -x$.
8. Each reflection reverses the direction perpendicular to its mirror line; two parallel mirrors reverse it twice, leaving a translation whose length is twice the distance between the lines.
9. (-1, 5): the midpoint (2, 3) is the centre, so each coordinate is reflected through it.

PRACTICE 6.2 ENLARGEMENT AND SCALE

1. (6, 9).
2. 7.5 cm by 12 cm; area 90 cm<
3. 2.5.
4. $6 \times 25\,000 = 150\,000$ cm $= 1.5$ km.
5. Enlargement multiplies every length by the same factor, so the ratios of sides are unchanged and the angles are unchanged.
6. Nine times as much, since area scales as the square of the linear factor.
7. $250 \times 24^2 = 144\,000$ cm<= 14.4 m<
8. The image is the object rotated 180° about the centre; a rotation is an isometry, so the triangles are congruent.

PRACTICE 6.3 CONGRUENCE AND THE TRIANGLE CRITERIA

1. (a) SSS (b) SAS.
2. Yes, AAS (the third angle is forced, and the side matches a corresponding pair of angles).
3. Equal angles fix the shape but not the size, so triangles with equal angles can be any size.
4. $AB = CD$ and $BC = DA$ (opposite sides of a parallelogram), and AC is common; the triangles are congruent by SSS.
5. $AB = AC$, $BD = DC$ and AD is common, so triangles ABD and ACD are congruent (SSS). Hence angle ADB = angle ADC; they are also supplementary, so each is 90°.
6. Similar figures have equal angles and proportional sides; congruent figures have equal sides as well. A 3–4–5 triangle and a 6–8–10 triangle are similar but not congruent.
7. Each diagonal splits the rectangle into two right triangles with legs equal to the sides; both are congruent by SAS to the same pair, so the diagonals are equal.
8. Looking the same is not a criterion; a proof needs matching sides, angles or hypotenuse and leg with reasons.
9. Six lines (three through opposite vertices, three through opposite edge midpoints) and rotational order 6, because each rotation by 60° maps the hexagon onto itself.

PRACTICE 6.4 SYMMETRY AND TESSELLATION

1. (a) 3 lines, order 3 (b) 4 lines, order 4 (c) 8 lines, order 8 (d) 1 line, order 1.
2. A line of symmetry would pair equal sides; a scalene triangle has no pair of equal sides, so no reflection maps it onto itself.
3. (a) yes, 360 ÷ 120 = 3 (b) no, 360 ÷ 108 is not a whole number (c) yes, opposite angles sum to 360°.
4. Interior angle = (5 - 2) × 180 ÷ 5 = 108°; three pentagons give 324°, leaving a 36° gap that no pentagon can fill.
5. A pinwheel or a swastika-like figure with four arms of equal length and no mirror symmetry.
6. Any quadrilateral tiles because its four angles sum to 360°, so four copies (one of each vertex) meet at every point.
7. Any triangle tiles because two copies form a parallelogram whose angles alternate to 360° at a point; replicated, this fills the plane.
8. Symmetry is a property of the arrangement of a figure, area of its extent; a scalene triangle and an equilateral triangle of equal area differ in symmetry.

PRACTICE 6.5 CONGRUENCE, COORDINATES AND PROOF

1. $\sqrt{(3^2 + 4^2)} = 5$.
2. (1, 2).
3. $AB = \sqrt{(9 + 16)} = 5$ and $BC = \sqrt{(16 + 9)} = 5$, so two sides are equal.
4. Sides 4 and 2 with right angles, so perimeter 12.
5. One example shows the statement holds in that case; other parallelograms might behave differently, so a general argument is needed.
6. Place the right angle at the origin with vertices (0, 0), (a, 0) and (0, b). The midpoint of the hypotenuse is (a/2, b/2), whose distance to each vertex is $< \sqrt{(a^2 + b^2)}$, the same for all three.
7. Let the diagonals meet at M, the midpoint of both. Then triangles AMB and CMD are congruent (SAS, vertically opposite angles), so $AB = CD$ and AB is parallel to CD; similarly for the other pair, so the figure is a parallelogram.
8. A rhombus with vertices (0, 0), (2, 0), (3, 1), (1, 1) has all sides equal but its angles are not right angles.

INVESTIGATION AND APPLICATION

1. The repeating unit should be marked and its boundary justified by matching edges on opposite sides.
2. The translation vector connects equivalent points in adjacent units.
3. Applying the vector to boundary points must land on matching boundary points of the next unit.

4. Rotational symmetry is described by a centre and an order; reflective symmetry by the mirror lines.
5. Smaller translations fail because the pattern does not repeat at a shorter interval.
6. Every candidate shorter vector maps some feature onto a non-matching position, which can be shown by naming the feature and where it lands.
7. 4.2 ÷ 0.3 = 14 and 3.6 ÷ 0.3 = 12; whole numbers because the room dimensions are exact multiples of the tile.
8. A design symmetric under a half turn about the centre maps each tile onto the tile diametrically opposite.
9. Each quadrant is the image of the first under a rotation of 90°, 180° and 270° about the centre.
10. 168 × 1.05 = 176.4, so 15 boxes of 12 (180 tiles).
11. The border removes tiles from the rotational design unless it is itself symmetric; the inner 12 × 10 area is 120 tiles and the border adds 48, giving the same 168 but a different symmetry.

CHAPTER 7 SIMILARITY AND TRIGONOMETRY

PREREQUISITE CHECK

1. 180°. 9 + 16 = 25, root 5. x = 6. The hypotenuse. 38.4. 0.5.

PRACTICE 7.1 SIMILAR TRIANGLES

1. (a) similar (AA) (b) similar (SSS with ratio 2) (c) not similar.
2. Scale factor 2.
3. $EF = 15$.
4. $h = 1.8 \times 12/2.4 = 9$ m.
5. Two equal angles force the third, so all three angles match and the triangles are similar.
6. AAA fixes shape but not size; congruence also requires equal sides.
7. New height 37.5 cm; the areas are in the ratio $2.5^2 = 6.25$.
8. $AD/DB = AE/EC$ gives $3/5 = 4/EC$, so $EC = 20/3 \approx 6.67$.
9. Areas are in ratio 36 : 81 = 4 : 9, so the sides are in ratio 2 : 3.

PRACTICE 7.2 PYTHAGORAS' THEOREM

1. (a) 5 (b) 13 (c) 17.
2. (a) 12 (b) 8.
3. (a) yes (b) no, since 25 + 36 = 61 ≠ 64 (c) yes.
4. 13 cm.
5. In a non-right triangle the relationship is the cosine rule, $c^2 = a^2 + b^2 - 2ab\cos C$, which reduces to Pythagoras when $C = 90°$.
6. √(36 - 30.25) = √5.75 ≈ 2.4 m.
7. Face diagonal 4√2; space diagonal √(16 + 32) = 4√3 cm.
8. $(n^2 - 1)^2 + (2n)^2 = n^4 - 2n^2 + 1 + 4n^2 = (n^2 + 1)^2$. For $n = 2$: 3, 4, 5; for $n = 3$: 8, 15, 17.
9. 5 + 5 = 10 < 11, so the two short sides cannot reach each other; the triangle inequality fails before Pythagoras is relevant.

PRACTICE 7.3 THE TRIGONOMETRIC RATIOS

1. sin = 3/5, cos = 4/5, tan = 3/4.
2. (a) 0.743 (b) 0.326 (c) 0.268.
3. (a) 30° (b) 60° (c) 45°
4. Opposite = 20 × sin 40° ≈ 12.9 cm; adjacent = 20 × cos 40° ≈ 15.3 cm.

5. An equilateral triangle of side 2 split in half gives a right triangle with hypotenuse 2 and opposite 1, so sin 30 = 1/2 exactly.
6. The tangent is opposite over adjacent, and the opposite side can be longer than the adjacent; the sine is a side over the hypotenuse, which is always the longest side.
7. tan = 1.2/8.5, so 8.0.
8. With sin = a/c and cos = b/c, $\sin^2 + \cos^2 = (a^2 + b^2)/c^2 = c^2/c^2 = 1$.
9. sin 45 = 1/2, cos 45 = 1/2, tan 45 = 1.

PRACTICE 7.4 SOLVING RIGHT TRIANGLES

1. 15 sin 35 8.6 cm.
2. $\tan^{-1}(4.5/7.2)$ 32
3. 9 tan 48 10 m.
4. 50 tan 31 30 m.
5. The two angles are alternate angles between the horizontal at each end and the line of sight.
6. Height 4.5 sin 68 4.2 m; foot 4.5 cos 68 1.7 m from the wall.
7. tan = 1200/3500, so 19
8. The balloon's height fixes each observer's horizontal distance; only if the elevations were equal would it be above the midpoint. Solve h/tan 34 + h/tan 52 = 80.
9. $h = 30 \tan$; at 45 $h = 30$, at 46 h 31.1, at 44 h 29.0, so the height could be out by about 1 m.

PRACTICE 7.5 SIMILARITY IN THREE DIMENSIONS

1. Areas 1 : 4; volumes 1 : 8.
2. Lengths 20, areas 400, volumes 8000.
3. (2000 250) = 8 = 2, so the heights are in the ratio 2 : 1.
4. Sixteen times as much ink, since area scales as the square.
5. Volume grows as the cube of the linear scale while the cross-section of the legs grows only as the square, so the stress per unit area rises with size.
6. 300 8^3 = 153 600 g 154 kg.
7. Radii in ratio 3 : 4; surface areas in ratio 9 : 16.
8. Weight scales as the cube while the strength of the material scales as the cross-section, so a model is proportionally much stronger than the full-size structure.

INVESTIGATION AND APPLICATION

1. Both methods should be planned with a labelled diagram showing what each measurement corresponds to.
2. Repeat readings and record the spread; the precision of the instrument sets the precision of the result.
3. The shadow method uses $h/H = s/S$; the clinometer method uses $h = d \tan$.
4. The two results should be reported with their difference as a percentage of the mean.
5. Shadow: rays are parallel. Clinometer: ground level and vertical object. Photograph: equal distance.
6. The method with fewer violated assumptions, judged on the evidence collected, should be preferred.
7. Volume ratio 500/330 1.515, linear factor 1.515 1.148.
8. Height 115 1.148 132 mm; diameter 66 1.148 76 mm.
9. Surface area of the current can 2 $(33)^2$ + 2 (33)(115) 30 680 mm< of the new 40 500 mm< an increase of about 32 per cent.
10. Material scales as the square while volume scales as the cube, so the cost per millilitre falls as the can grows.
11. Keeping the height and widening the base gives a different shape, so the scale factor does not apply and the surface area must be recalculated from the formula.

CHAPTER 8 SOLIDS AND MEASUREMENT

PREREQUISITE CHECK

1. 25 78.5 cm< 216. 250 cm. 10 000 cm< 20 cm< 3.14.

PRACTICE 8.1 PRISMS AND CYLINDERS

1. 84 cm<
2. < 9 4 = 18; 18 12 = 216 cm<
3. 16 10 503 cm<
4. 2(25) + 2(5)(12) = 50 + 120 = 170 534 cm<
5. (16 - 9) 200 = 1400 4398 cm<
6. r^2h is the area of the circular cross-section swept through the height; $2rh$ is the area of the curved surface, which is a different quantity.
7. 1.44 2.5 11.3 m<= 11 300 litres.
8. The label's width equals the circumference 3.3 20.7 cm.
9. $r^3 = 500$, so $r =$ (159.2) 5.42 cm.

PRACTICE 8.2 PYRAMIDS, CONES AND SPHERES

1. < 36 10 = 120 cm<
2. < 25 9 236 cm<
3. 36 cm<
4. 4 49 616 cm<
5. Filling the cone three times fills the cylinder: the algebra gives < Ah against Ah.
6. Unrolling the sphere's surface is not possible, but a classical argument shows it equals four great circles.
7. Cone < 9 10 94.2 cm< hemisphere < 27 56.5 cm< total about 151 cm<
8. < $r^3 = 100$ gives r^3 47.7, so r 3.6 cm.
9. r^2h is constant, so doubling r makes the height one quarter.

PRACTICE 8.3 NETS AND SURFACE AREA

1. 2(60 + 30 + 18) = 216 cm<
2. Six faces of 16 cm< so 96 cm<
3. 2(49) + 2(7)(2) = 98 + 28 = 126 396 cm<
4. The 12 edge cubes not at a corner have exactly two painted faces.
5. The curved surface of a cone unrolls into a sector of a circle whose radius is the slant height, not into a triangle.
6. 500 1.08 = 540 cm<
7. Base 20 15 = 300, sides 2(20 10) + 2(15 10) = 700; total 1000 cm< less than the closed box because the lid is missing.
8. For example 5 cm by 4 cm by 3 cm: volume 60 cm< surface area 2(20 + 15 + 12) = 94 cm<

PRACTICE 8.4 UNITS OF VOLUME AND PRECISION

1. (a) 3 500 litres (b) 450 cm<(c) 2.5 litres.
2. (a) 12 000 cm<(b) 500 000 cm<
3. 24 16 9 = 3456 cm< 3.46 litres.

4. Volume ≈ 97.7 cm³; the inputs have two significant figures, so report 98 cm³.
5. Each of the three dimensions converts by 100, so the volume converts by $100^3 = 1\,000\,000$ cm³ and 1 000 cm³ is a litre.
6. $600 \div (25 \times 12) = 2$ m.
7. Difference 2/748 ≈ 0.27 per cent, well within manufacturing tolerance and measurement error; the label is not wrong.
8. Acceptable range 392 to 408 mL; 407 mL passes.

INVESTIGATION AND APPLICATION

1. $r = 2.5$: $h \approx 16.8$ cm, surface ≈ 303 cm²; $r = 5.0$: $h \approx 4.2$ cm, surface ≈ 289 cm²
2. The graph falls to a minimum near $r = 3.8$ and rises again.
3. For small radii the height, and therefore the curved surface, is very large; for large radii the two circular ends dominate.
4. About $r = 3.7$ cm with $h \approx 7.7$ cm, giving a surface area of about 265 cm²
5. A height close to the diameter is easier to hold and to stack, and the can must fit standard shelves and vending machines.
6. Any recommendation that separates the mathematical minimum from the practical constraints of handling, display and manufacture.
7. Internal dimensions at least 7 cm by 7 cm by 6 cm: 6 cm diameter plus 0.5 cm clearance each side, and 5 cm height plus clearance.
8. Surface area $2(49 + 42 + 42) = 266$ cm²
9. Extra room protects the award, allows padding, and tolerates manufacturing variation.
10. $240 \times 266 = 63\,840$ cm² = 6.384 m², so 7 m² at 4.5 units, costing 31.5 units.
11. Posting rewards a compact shape: the cube reduces surface area for a given volume, but padding adds volume without protecting the award better than a close fit.

CHAPTER 9 BIVARIATE DATA

PREREQUISITE CHECK

1. 7.5. $m = 1$. 8. 3/8. The quantity falls as the horizontal variable increases. A graph shows the shape of the relationship; a table gives exact values but hides the shape.

PRACTICE 9.1 SCATTER GRAPHS AND ASSOCIATION

1. (a) positive, strong, linear (b) negative, strong, linear (c) probably none: reading age and shoe size are not related at a fixed age.
2. (a) strong (b) weak or none.
3. Joining the points implies an order and a continuity that the data need not have.
4. Measurement error, or a genuinely different case (a different group, a recording mistake); check the source and re-measure.
5. Points falling from left to right, tightly clustered, with one point far above the trend.
6. The line of best fit and the gradient change, because the axes are not interchangeable; the strength of association does not.
7. "High" describes the values, not how closely they follow the trend.
8. The two sets have different spreads about their trends, so one is more predictable even with the same mean values.

PRACTICE 9.2 LINES OF BEST FIT AND PREDICTION

1. Gradient 2.4 units of y per unit of x; intercept 7 is the value at $x = 0$.
2. At $x = 5$, $y = 19$; at $x = 20$, $y = 55$, which is an extrapolation.
3. The line describes the trend, not the individual points, so it passes through the middle of the cloud.
4. Compare the total vertical deviations from each line; the one with smaller deviations fits better.

5. Gradient □ -1125 units per year, so $v = 12\,000 - 1125t$; at $t = 6$, $v = 5250$. Depreciation usually slows, so the prediction is probably too low.
6. Plant growth slows and eventually stops, so a linear model cannot hold for 40 weeks.
7. $x = 12$ lies within the data range, so the trend has been observed there; 45 lies outside it.
8. The narrower cloud, because the line's position is better determined and the prediction is less uncertain.

PRACTICE 9.3 ▤CORRELATION AND ITS LIMITS

1. (a) strong positive linear (b) weak negative (c) very strong negative (d) no linear association.
2. (a) negative (b) negative (c) positive.
3. Correlation is a statistical association; causation means one variable produces a change in the other. Ice-cream sales and drowning are correlated without either causing the other.
4. Very ill people are more likely to be in hospital and more likely to die; severity of illness is the confounder.
5. Randomly assign students to the programme or a control, and compare score changes; randomisation removes confounding.
6. The outlier can dominate the calculation, pulling the coefficient towards its own direction; recompute without it and report both.
7. A strong curved relationship, such as growth against temperature, has a coefficient near zero because positive and negative deviations cancel.
8. Causal: high temperature raises both. Reverse causal: people buy less heating. Non-causal: both driven by season.

PRACTICE 9.4 ▤PRESENTING AND CRITICISING DATA

1. The rise looks like most of the chart; honestly, it is a 10 per cent increase.
2. All three dimensions double, so the volume, and the visual weight, is eight times larger rather than twice.
3. A specific example should be described with the axis, the scale and the feature that misleads.
4. Redrawing with the axis from zero, one dimension per quantity, and the whole period shown.
5. A small absolute change can be a large percentage if the starting value is small.
6. The median, because a few very high incomes distort the mean and describe almost nobody in the town.
7. Each criticism should name the feature, the effect and the remedy.

INVESTIGATION AND APPLICATION

1. Measurements should be consistent: same posture, same instrument, same definition of arm span.
2. Points should cluster around $y = x$; count above and below.
3. Typically a strong positive linear association with a few outliers.
4. The line's equation should be close to $y = x$ but not identical.
5. The prediction should be compared with the claim: a difference of 2<3 cm is within measurement error.
6. A larger sample narrows the uncertainty in the mean difference but cannot turn an association into a causal claim.
7. Means 72, 70, 66, 63 against midpoints 0.5, 1.5, 3 and 5+; the association is negative and roughly linear.
8. (72 - 63)/72 = 12.5 per cent, so the report's 8 per cent understates the difference between the extreme bands.
9. Sleep, homework time, or device access; each could reduce scores independently of screen time.
10. A survey observes rather than manipulates; a randomised trial would be needed, and even then the effect size would need to be separated from substitution effects.
11. For example <Scores fall as screen time rises, but the cause is not established<

CHAPTER 10 PROBABILITY AND RISK

PREREQUISITE CHECK

1. 0.25, 25 per cent. 3/4. HH, HT, TH, TT. 0.12. 5/6. 0.

PRACTICE 10.1 THE PROBABILITY SCALE AND SAMPLE SPACES

1. (a) 1/6 (b) 1/2 (c) 5/6.
2. (a) 4/15 (b) 9/15 = 3/5 (c) 9/15 = 3/5.
3. HH, HT, TH, TT; $P(HH) = 1/4$.
4. $P(5) = 4/36 = 1/9$; $P(12) = 1/36$.
5. The outcomes are exhaustive and mutually exclusive, so their probabilities account for certainty, which is 1.
6. Mutually exclusive probabilities cannot sum to more than 1, and 0.6 + 0.7 = 1.3.
7. Five matching pairs out of 25, so 1/5.
8. $P(\text{no six}) = (5/6)^2 = 25/36$, so $P(\text{at least one six}) = 11/36$.
9. $n/(n + 8) = 0.4$ gives $n = 5.33$; since n must be a whole number, the closest workable values are $n = 5$ (0.385) or $n = 6$ (0.429), so the question has no exact whole-number answer and $n = 5$ is closest.

PRACTICE 10.2 EXPERIMENTAL PROBABILITY AND RELATIVE FREQUENCY

1. 44/80 = 0.55.
2. 27/150 = 0.18.
3. Relative frequency 62/300 ≈ 0.207 against 1/6 ≈ 0.167; the difference is about 2.4 in the expected count of 50, which is within ordinary variation.
4. Twenty throws give a very wide range of plausible relative frequencies, so bias cannot be distinguished from chance.
5. After a run of heads, the next toss is still equally likely; the average corrects by dilution over many future tosses, not by compensation.
6. 17/500 = 0.034; expect about 408 faulty in 12 000.
7. Many spins, at least several hundred, so that chance variation is small relative to any real bias.
8. The 400-trial estimate, because its sampling error is much smaller.

PRACTICE 10.3 COMBINING EVENTS

1. 1/36.
2. 1/8.
3. 3/10 × 2/9 = 6/90 = 1/15.
4. 2 × (3/10 × 7/9) = 42/90 = 7/15.
5. Mutually exclusive events cannot both happen (rolling a 2 and a 5); independent events do not influence each other (two tosses of a coin).
6. Because the same outcomes can be counted twice when the events overlap.
7. 6/36 + 2/36 = 8/36 = 2/9.
8. Either both red (5/10 × 4/9 = 2/9) or both blue (2/9), so 4/9.
9. $0.95^2 = 0.9025$; a single reading being wrong is not corrected by repeating, but repeating makes a wrong pair less likely < which is why tests are repeated.

PRACTICE 10.4 TREE DIAGRAMS AND TWO-WAY TABLES

1. *P*(*exactly one head*) = 2 × < × < = < .
2. 6/10 × 5/9 = 30/90 = 1/3.
3. Neither: 40 - 24 - 18 + 10 = 8, so 8/40 = 1/5.
4. *P*(*at least one black*) = 1 - *P*(*both white*) = 1 - (5/8 × 4/7) = 1 - 20/56 = 9/14.
5. Removing a counter changes the number remaining and often the number of that colour.
6. When the problem gives counts in categories rather than probabilities in stages.
7. 0.6 × 0.02 + 0.4 × 0.05 = 0.012 + 0.02 = 0.032.
8. *P*(*B* | *defective*) = 0.02/0.032 = 0.625, because B produces fewer items but has the higher defect rate.
9. *P*(*A*) × *P*(*B*) = 0.5 × 0.4 = 0.2 = *P*(*A* and *B*), so they are independent.

PRACTICE 10.5 RISK, SAMPLING AND DECISIONS

1. (a) 30 in 1 000 (b) 2 in 1 000 (c) 120 in 1 000.
2. (a) 0.007 (b) 0.04 (c) 0.0003.
3. Absolute 2/1000 = 0.002; relative 0.002/0.006 × 33 per cent.
4. The relative figure can be large while the absolute change is tiny, so it can make a negligible benefit sound substantial.
5. Healthy people vastly outnumber cases, so even a small false-positive rate produces more false positives than true ones.
6. 2/1000 × 5000 = 10 genuine cases; 90 per cent detected, so 9 true positives; 4990 × 0.02 × 100 false positives; about 109 positives of which 9 are genuine, roughly 8 per cent.
7. It means that in 10 comparable forecasts, rain is expected in about 3; it does not mean rain for 30 per cent of the day or 30 per cent of the area.
8. Any recommendation that states criteria (number of cases detected, harm from false positives, cost) and weighs them explicitly.

INVESTIGATION AND APPLICATION

1. The true proportion is 0.3; samples are drawn with replacement so each draw is independent.
2. Record the proportion of reds in each sample.
3. Ranges narrow as sample size increases.
4. The range of proportions falls as the sample size rises.
5. Larger samples are less affected by chance variation, which is visible in the narrower ranges.
6. The largest size available gives the smallest error; justify with the observed ranges.
7. 25 people with the condition (about), 24 detected; 49 975 without, of whom about 2 999 test positive. The table's first row should read about 25 positives rather than 25 exactly, and the second row's positive figure is correct.
8. 24/(24 + 2999) × 0.008, so about 1 per cent.
9. Because the condition is rare, the healthy majority generate far more false positives than the sick generate true ones.
10. About 3 000 people face unnecessary anxiety or further testing; about 1 true case is missed.
11. Any recommendation that states the criteria and names the missing information, for example the cost per case detected or the harm of a false positive.

Formula bank

A compact reminder of the relations used in this course. A formula you cannot derive is a formula you will misuse; the derivation of each of these is in the chapter named beside it.

NUMBER

Relation	Form
Rational form	p/q, $q \neq 0$, p, q integers
Terminating decimal	denominator has only prime factors 2 and 5
Repeating decimal	$0.a\dot{a}_k = (a\dot{a}_k)/99\dot{9}$
Exponent laws	$a^m \times a^n = a^{m+n}$; $a^m \div a^n = a^{m-n}$; $(a^m)^n = a^{mn}$; $a^0 = 1$; $a^{-n} = 1/a^n$
Rational exponents	$a^{1/n} = \sqrt[n]{a}$; $a^{m/n} = \sqrt[n]{(a^m)}$
Scientific notation	$a \times 10^n$ with $1 \leq a < 10$
Absolute value	$\lvert x\rvert = x$ if $x \geq 0$; $\lvert x\rvert = -x$ if $x < 0$
Interval	$\lvert x - a\rvert \leq r \Leftrightarrow a - r \leq x \leq a + r$
Surds	$\sqrt{(ab)} = \sqrt{a}\sqrt{b}$; $(\sqrt{a})^2 = a$; $(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = a - b$

ALGEBRA

Relation	Form
Expansion	$(a + b)^2 = a^2 + 2ab + b^2$; $(a - b)^2 = a^2 - 2ab + b^2$; $(a + b)(a - b) = a^2 - b^2$
Quadratic factorisation	$x^2 + (p+q)x + pq = (x+p)(x+q)$
Line	$y = mx + c$; $m = (y_2 - y_1)/(x_2 - x_1)$; $y - y_1 = m(x - x_1)$
Direct proportion	$y = kx$; graph is a straight line through the origin
Simultaneous equations	substitution, or elimination by adding multiples
nth term, linear	$u_n = a + (n - 1)d$

GEOMETRY AND MEASURE

Relation	Form
Angles	straight line 180°; at a point 360°; triangle 180°; polygon $(n - 2) \times 180^\circ$
Pythagoras	$a^2 + b^2 = c^2$ in a right triangle
Trigonometry	$\sin\theta = opp/hyp$; $\cos\theta = adj/hyp$; $\tan\theta = opp/adj$

Relation	Form
Similarity	corresponding lengths in ratio k; areas in ratio k^2; volumes in ratio k^3
Perimeter and area	circle $C = 2\pi r$, $A = \pi r^2$; triangle $A = \frac{1}{2}bh$; trapezium $A = \frac{1}{2}(a+b)h$
Prisms	$V = Ah$; $S = 2A + Ph$
Cylinder	$V = \pi r^2 h$; curved area $2\pi rh$; total $2\pi r^2 + 2\pi rh$
Pyramid and cone	$V = \frac{1}{3}Ah$; cone $V = \frac{1}{3}\pi r^2 h$, curved area πrl
Sphere	$V = \frac{4}{3}\pi r^3$; $S = 4\pi r^2$
Density	$\rho = m/V$

DATA AND PROBABILITY

Relation	Form
Mean	$\bar{x} = \Sigma x / n$; grouped: $\bar{x} = \Sigma fx / \Sigma f$
Median and quartiles	order the data; Q1, Q2, Q3 at the quartile positions
Probability	$P(E) = n(E)/n(S)$ for equally likely outcomes; $P(\text{not } E) = 1 - P(E)$
Addition	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Independent events	$P(A \text{ and } B) = P(A) \times P(B)$
Experimental probability	$P(E) \approx$ *frequency of E / number of trials*
Percentage change	(*new - old*)/*old* × 100%; percentage *points* is a difference of percentages

Glossary

Working definitions of the terms used in this book. Where a term has a precise mathematical meaning, the definition given here is the one the course uses.

absolute value
The distance of a number from zero on the number line; written $|x|$.

adjacent
The side next to the angle in use, other than the hypotenuse.

ambiguous
A situation in which a construction (for example SSA in triangles) can produce more than one valid figure.

angle of depression
The angle measured downwards from the horizontal to an object below.

angle of elevation
The angle measured upwards from the horizontal to an object above.

association
A systematic relationship between two variables.

balance principle
Whatever is done to one side must be done to the other.

base
In a^n, the number a that is multiplied repeatedly.

bivariate data
Data in which two variables are recorded for each individual.

capacity
The volume a container can hold, usually in litres.

causation
A relationship in which one variable produces a change in another.

census
A survey of every member of a population.

centre of rotation
The fixed point about which a rotation turns a figure.

coefficient
The numerical factor of a term.

compound event
An event described by combining two or more simple events.

confounding variable
A third variable influencing both others.

congruent
Identical in shape and size; the symbol is $\cong$.

conjecture
A statement believed to be true on the basis of evidence but not yet proved.

constant term
A term with no variable part.

constraint
A condition a solution must satisfy.

converse
The statement obtained by exchanging the hypothesis and conclusion of a conditional statement.

correlation
A measure of the strength and direction of a linear association between two variables.

correlation coefficient
A number from -1 to 1 measuring linear association.

cosine
Adjacent divided by hypotenuse.

counterexample
A single case that makes a universal statement false.

cross-section
The shape cut by a plane through a solid.

density
Mass per unit volume; also, in number theory, the property that another number lies between any two distinct numbers.

dependent variable
The variable whose value is determined by another; plotted on the vertical axis.

direct proportion
$y = kx$: the ratio is constant.

domain
The set of input values for which a function is defined.

domain of validity
The range over which a model can reasonably be trusted.

elimination
Solving simultaneous equations by adding or subtracting multiples so that one variable disappears.

enlargement
A transformation that changes size but not shape; a scale factor applied about a centre.

equation
A statement that two expressions are equal.

equivalent
Having exactly the same solutions.

estimate
A value obtained by rounding inputs to convenient sizes.

event
A set of outcomes of interest.

expand
Multiply out brackets to remove them.

exponent
The number n in a^n, counting the factors.

expression
A sum of terms; it has no equals sign.

extrapolation
Using a model outside the range of the data from which it was built.

factorise
Write an expression as a product of factors.

false positive
A positive result when the condition is absent.

feasible region
The set of points satisfying every constraint in a system of inequalities.

function
A rule that assigns exactly one output to each input in its domain.

gradient
The change in the vertical quantity per unit change in the horizontal one.

hemisphere
Half of a sphere.

hypotenuse
The side opposite the right angle in a right triangle.

independent
Events where one does not affect the other's probability.

independent events
Events for which the occurrence of one does not change the probability of the other.

inequality
A statement that one quantity is less than, greater than, or not equal to another.

intercept
Where a line crosses an axis; the y-intercept is the value when $x = 0$.

interpolation
Estimating a value between known data points.

interval
A connected set of real numbers described by its endpoints.

inverse proportion
$y = k/x$: the product is constant.

irrational number
A real number that cannot be written as a quotient of integers.

isometry
A transformation that preserves distance.

line of best fit
A line that summarises the trend in a scatter plot.

line of symmetry
A mirror line mapping a figure onto itself.

locus
The set of all points satisfying a given condition.

mantissa
The a part of $a \times 10^n$.

mutually exclusive
Events that cannot both occur.

net
A flat pattern that folds to make a solid.

non-response bias
Bias arising because those who do not answer differ systematically from those who do.

objective
The quantity to be maximised or minimised.

opposite
The side across from the angle in use.

order of magnitude
The exponent in scientific notation, expressing size as a power of ten.

outlier
A value that lies far from the pattern of the rest of the data.

pi
The ratio of a circle's circumference to its diameter; $\pi \approx 3.14159$.

population
The whole group about which information is wanted.

precision
The fineness of a measurement, indicated by the number of significant figures or decimal places.

prism
A solid with a constant cross-section; its volume is the cross-sectional area times the length.

probability
A number between 0 and 1 expressing how likely an event is.

proof by contradiction
A proof that assumes the negation of the claim and derives an impossibility.

proportional
Having a constant ratio; the graph is a straight line through the origin.

Pythagoras' theorem
In a right triangle, the square on the hypotenuse equals the sum of the squares on the other two sides.

radical
A root sign, or an expression containing one.

radicand
The number or expression under a radical sign.

range
The set of output values of a function; also the difference between the largest and smallest values in a data set.

rate of change
How quickly one quantity changes relative to another.

rational number
A number that can be written as p/q with integers p, q and $q \neq 0$.

rationalise
Rewrite a quotient so no surd remains in the denominator.

reciprocal
The result of dividing 1 by a number; $1/a$.

reflection
A flip in a mirror line.

relative frequency
The proportion of trials in which an event occurred; an estimate of probability.

relative risk
The ratio of two risks.

repeating decimal
A decimal with a block of digits that repeats forever.

representative sample
A sample whose composition reflects the population.

residual
The vertical distance between a data point and a line of best fit.

risk
The probability of an adverse outcome.

rotation
A turn about a centre through a given angle.

rotational symmetry
The property of looking unchanged after a rotation of less than 360°.

sample space
The set of all possible outcomes of an experiment.

scale factor
The ratio by which a figure is enlarged or reduced.

scatter graph
A plot of one variable against another.

scatter plot
A graph of paired values of two variables, one point per individual.

scientific notation
A number written as $a \times 10^n$ with $1 \leq a < 10$.

sequence
An ordered list of values generated by a rule.

significant figures
The digits of a number that carry information about its precision.

similar
Having the same shape but not necessarily the same size; the symbol is $\sim$.

simultaneous equations
Two or more equations that must be satisfied at the same time.

simultaneous solution
A pair of values satisfying two equations at once.

sine
Opposite divided by hypotenuse.

slant height
The distance along a cone's surface from rim to apex.

slope
See gradient.

solution set
All the values that satisfy a statement.

subject
The variable that stands alone on one side of a formula.

substitution
Replacing a variable by a number or expression; also a method of solving simultaneous equations.

surd
An irrational root, such as $\sqrt{2}$, left in exact form.

surds
Irrational roots left in exact form.

surface area
The total area of a solid's faces, in square units.

symmetry
A property of a figure that is unchanged by a reflection, rotation or translation.

tangent
Opposite divided by adjacent.

term
A product of numbers and variables, separated by $+$ or $-$ signs.

terminating decimal
A decimal that ends after finitely many digits.

tessellation
A tiling of the plane with no gaps and no overlaps.

tolerance
The permitted variation in a manufactured measurement.

transformation
A rule that moves every point of a figure.

translation
A transformation that moves every point the same distance in the same direction.

tree diagram
A branching diagram showing successive events.

trigonometric ratio
For an acute angle in a right triangle, the ratio of two sides: sine, cosine or tangent.

two-way table
A table that counts individuals by two categorical variables.

variable
A letter standing for a number, or for a quantity that can change.

vector
A quantity with magnitude and direction; drawn as an arrow.

volume
The space a solid occupies, in cubic units.

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Sources, credits and acknowledgements

Mathematical sources

All mathematics in this book is standard MYP Year 3 material and was developed from the curriculum framework, not from any published coursebook. The sequence of topics, the worked examples, the exercises, the investigations and the applications were written for this edition. Where a classical result is cited (the irrationality of the square root of 2, the Babylonian square-root method, Pythagoras' theorem, the sieve of Eratosthenes), the result is in the public domain of mathematics and is attributed in the text to its historical source.

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Data

Data sets used in exercises are either constructed for the purpose of the question or drawn from published figures cited at the point of use. Constructed data are labelled as such. No data set is presented as a real measurement when it is not.

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What the course asks of you

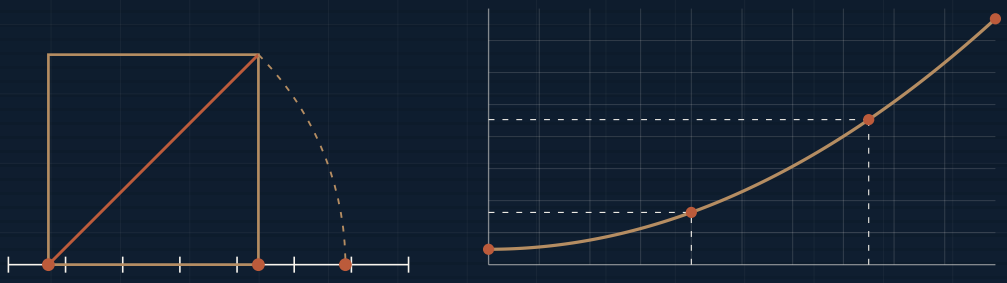
Each chapter opens with a question you cannot answer yet, gives you the evidence to answer it, and then makes you explain what you found. The mathematics is the mathematics it has always been; what changes is that you are asked to notice why it works, and to say so clearly.

TEN CHAPTERS, FIVE PARTS

Number and algebraic structure
Relationships and equations
Geometry, proof and space
Data, chance and evidence

FOUR CRITERIA, ONE HABIT

Knowing and understanding
Investigating patterns
Communicating
Applying mathematics in real-life contexts



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